

Infinitesimal extensions

SGA 3, Exposé III

English draft

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About this draft

This is an unofficial English translation of Exposé III, “Infinitesimal extensions” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Exp3-14oct24.pdf` (78 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé III. Infinitesimal extensions

by M. Demazure

⁽⁰⁾In this exposé, one places oneself in the following general situation. One has a scheme S and a coherent nilpotent ideal $\mathcal{I}$ on S . One denotes by S_n the closed subscheme of S defined by the ideal $\mathcal{I}^{n+1}$ ($n \geq 0$). In particular S_0 is defined by $\mathcal{I}$. As $\mathcal{I}$ is nilpotent, S_n equals S for n large enough, and the S_i have the same underlying topological space. A typical example of this situation is the following: S is the spectrum of a local artinian ring A , $\mathcal{I}$ is the ideal defined by the radical of A , hence S_0 is the spectrum of the residue field of A .

In the preceding situation, one is given a certain number of data above S_0 , and one seeks above S data which lift them, that is to say, which give them back by base change from S to S_0 . This is done step by step, by means of the S_n . At each step, one proposes to define the obstructions encountered and to classify, when they exist, the solutions obtained.

The passage from S_n to S_{n+1} may be generalized as follows: one has a scheme S , two ideals $\mathcal{I}$ and $\mathcal{J}$ with $\mathcal{I} \supset \mathcal{J}$, and $\mathcal{I} \cdot \mathcal{J} = 0$ (in the preceding case, S , $\mathcal{I}$ and $\mathcal{J}$ are respectively S_{n+1} , $\mathcal{I}/\mathcal{I}^{n+2}$, $\mathcal{I}^{n+1}/\mathcal{I}^{n+2}$). One denotes by S_0 (resp. $S_{\mathcal{J}}$) the closed subscheme of S defined by $\mathcal{I}$ (resp. $\mathcal{J}$), and one poses a problem of extension from $S_{\mathcal{J}}$ to S .

In SGA 1 III were treated the problems of extension of morphisms of schemes and of extension of schemes. We shall pose here the problems of extension of morphisms of groups, of extension of group structures and of extension of subgroups.

We have gathered in an n° 0 the results of SGA 1 III which will be useful to us, so as to put them in the form most practical for our purpose, and so as to spare the reader from having constantly to refer to SGA 1 III.⁽¹⁾ The n° 1 gathers computations of cohomology of groups, useful in the sequel, and which have nothing to do with the theory of schemes. Numbers 2 and 3 treat respectively of *the extension of morphisms of groups and of the extension of group structures*. In n° 4, we have briefly recalled the proof of a result stated in TDTE IV concerning the extension of subschemes, and applied this result to the problem of extension of subgroups. *For the sequel of the Seminar, only the result of n° 2, concerning the extension of morphisms of groups, will be indispensable.*⁽²⁾

The idea of reducing the problems of infinitesimal extensions to the usual cohomology computations in group extensions was suggested by J. Giraud during the oral exposé (whose computations were distinctly more complicated and less transparent). Unfortunately, it seems that this method applies well only to the first two problems studied, and we have been unable to avoid rather painful computations in the case of extensions of subgroups.

To simplify the language, we shall call Y -functor, resp. Y -scheme, etc., a functor, resp. scheme, etc., equipped with a morphism into the functor Y , thus extending the definitions of Exposé I (which concerned only the case of a representable Y).

0. Reminders from SGA 1 III and various remarks

Let us first state a general definition.

⁽⁰⁾**N.D.E.** Version of 14/10/2024

⁽¹⁾**N.D.E.** The reader may prefer to begin by reading section 1, which is easier, and which can serve as motivation and as a guide for the results obtained in section 0.

⁽²⁾**N.D.E.** However, 3.10 is used in XXIV, 1.13. On the other hand, 4.37 is used in the proof of IX, 3.2 bis and 3.6 bis, but these also follow from results of Exp. X, which do not use 4.37.

Definition 0.1. Let $\mathcal{C}$ be a category, X an object of $\widehat{\mathcal{C}}$, G a $\widehat{\mathcal{C}}$ -group operating on X . One says that X is formally principal homogeneous⁽³⁾ under G if the following equivalent conditions are satisfied:

- (i) for each object S of $\mathcal{C}$, the set $X(S)$ is empty or principal homogeneous under $G(S)$;
- (ii) the morphism of functors $G \times X \longrightarrow X \times X$ defined set-theoretically by $(g, x) \mapsto (gx, x)$ is an isomorphism.

This done, we are going to put the results of SGA 1 III, § 5⁽⁴⁾ in the form which will be the most useful to us. We shall employ the following general notations throughout this number. One has a scheme S and on S two quasi-coherent ideals $\mathcal{I}$ and $\mathcal{J}$ such that

$$\mathcal{I} \supset \mathcal{J} \quad \text{and} \quad \mathcal{I} \cdot \mathcal{J} = 0.$$

One will therefore have in particular $\mathcal{J}^2 = 0$. One will denote by S_0 (resp. $S_{\mathcal{J}}$) the closed subscheme of S defined by the ideal $\mathcal{I}$ (resp. $\mathcal{J}$). For every S -functor X , one will systematically denote by X_0 and $X_{\mathcal{J}}$ the functors obtained by base change from S to S_0 and $S_{\mathcal{J}}$. The same notations for a morphism.

Definition 0.1.1. ⁽⁵⁾Let X be an S -functor. Let us define a functor X^+ above S by the formula

$$\mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_{S_{\mathcal{J}}}(Y_{\mathcal{J}}, X_{\mathcal{J}}) = \mathrm{Hom}_S(Y_{\mathcal{J}}, X)$$

for a variable S -scheme Y . In the notations of Exp. II, 1, one has

$$X^+ \simeq \coprod_{S_{\mathcal{J}}/S} X_{\mathcal{J}}.$$

The identity morphism of $X_{\mathcal{J}}$ defines by construction an S -morphism

$$p_X: X \longrightarrow X^+.$$

⁽⁶⁾Explicitly, for every S -scheme Y , the map

$$p_X(Y): \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+) = \mathrm{Hom}_S(Y_{\mathcal{J}}, X)$$

is the map induced by the morphism $Y_{\mathcal{J}} \longrightarrow Y$.

Remark 0.1.2. Let us now remark that if X is an S -functor in groups, $X_{\mathcal{J}}$ is an $S_{\mathcal{J}}$ -functor in groups; then the defining formula of X^+ equips it with a structure of S -functor in groups, and the description of p_X above shows that $p_X: X \longrightarrow X^+$ is a morphism of S -functors in groups.

Remark 0.1.3. On the other hand, for every S -functor in groups Y , one has

$$\mathrm{Hom}_{S\text{-gr.}}(Y, X^+) = \mathrm{Hom}_{S_{\mathcal{J}}\text{-gr.}}(Y_{\mathcal{J}}, X_{\mathcal{J}}).$$

Indeed, let $f \in \mathrm{Hom}_S(Y, X^+)$, corresponding to $f_{\mathcal{J}} \in \mathrm{Hom}_{S_{\mathcal{J}}}(Y_{\mathcal{J}}, X_{\mathcal{J}})$; the condition for $f \in \mathrm{Hom}_{S\text{-gr.}}(Y, X^+)$ is that, for every $T \longrightarrow S$ and $y, y' \in Y(T)$, one has $f(y \cdot y') = f(y) \cdot f(y')$, and this is equivalent to

$$f_{\mathcal{J}}(y_{\mathcal{J}}) \cdot f_{\mathcal{J}}(y'_{\mathcal{J}}) = f_{\mathcal{J}}((y \cdot y')_{\mathcal{J}});$$

as $(y \cdot y')_{\mathcal{J}} = y_{\mathcal{J}} \cdot y'_{\mathcal{J}}$ (since $Y(T) \longrightarrow Y(T_{\mathcal{J}}) = Y_{\mathcal{J}}(T_{\mathcal{J}})$ is a morphism of groups), this is the condition for $f_{\mathcal{J}}$ to be a morphism of groups. Applying this to $Y = X$, one finds again that p_X , which corresponds to $\mathrm{id}_{X_{\mathcal{J}}}$, is a morphism of S -functors in groups.

⁽³⁾**N.D.E.** One also says “pseudo-torsor”, cf. EGA IV₄, 16.5.15. On the other hand, the more general notion of formally homogeneous object (not necessarily principal homogeneous) is defined in Exp. IV, 6.7.1.

⁽⁴⁾**N.D.E.** See also EGA IV₄, 16.5.14–18.

⁽⁵⁾**N.D.E.** One has added the numbering 0.1.1, ..., 0.1.13, so as to bring out the definitions and results which are found there.

⁽⁶⁾**N.D.E.** One has added the following sentence, and spelled out the two remarks which follow.

Let us now return to the general case, but suppose that X is an S -scheme. An S -morphism of a variable S -scheme Y into X^+ being by definition an $S_{\mathcal{J}}$ -morphism $g_{\mathcal{J}}$ of $Y_{\mathcal{J}}$ into $X_{\mathcal{J}}$, one is going to define an X^+ -functor in abelian groups L_X as follows.

Scholium 0.1.4. ⁽⁷⁾If $\pi: Y \rightarrow S$ is a morphism of schemes and $\mathcal{M}$ an $\mathcal{O}_S$ -module, one denotes by $\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_Y$ the inverse image $\pi^*(\mathcal{M})$. If $\mathcal{J}$ is an ideal of $\mathcal{O}_S$, one denotes by $\mathcal{J}\mathcal{O}_Y$ the ideal of $\mathcal{O}_Y$, image of the morphism

$$\pi^*(\mathcal{J}) = \mathcal{J} \otimes_{\mathcal{O}_S} \mathcal{O}_Y \rightarrow \mathcal{O}_Y.$$

Let us note that, for every morphism of S -schemes $f: Z \rightarrow Y$, one has an epimorphism of $\mathcal{O}_Z$ -modules

$$(0.1.4) \quad f^*(\mathcal{J}\mathcal{O}_Y) \rightarrow \mathcal{J}\mathcal{O}_Z,$$

as follows from the commutative diagram below:

$$\begin{array}{ccc} \mathcal{J} \otimes_{\mathcal{O}_S} \mathcal{O}_Y \otimes_{\mathcal{O}_Y} \mathcal{O}_Z & & \mathcal{J} \otimes_{\mathcal{O}_S} \mathcal{O}_Z \\ \downarrow & & \downarrow \\ f^*(\mathcal{J}\mathcal{O}_Y) = (\mathcal{J}\mathcal{O}_Y) \otimes_{\mathcal{O}_Y} \mathcal{O}_Z & \longrightarrow & \mathcal{J}\mathcal{O}_Z. \end{array}$$

Definition 0.1.5. ⁽⁸⁾Let X be an S -scheme. For every X^+ -scheme Y , given by a morphism $g_{\mathcal{J}}: Y_{\mathcal{J}} \rightarrow X_{\mathcal{J}}$, one sets

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y),$$

where Ω_{X_0/S_0}^1 denotes the module of relative differentials of X_0 with respect to S_0 (cf. SGA 1, I.1 or EGA IV₄, 16.3), and where one regards $\mathcal{J}\mathcal{O}_Y$ as an $\mathcal{O}_{Y_0}$ -module thanks to the fact that it is annihilated by $\mathcal{I}$.

Then, L_X is an X^+ -functor in abelian groups.⁽⁹⁾ Indeed, for every X^+ -morphism $f: Z \rightarrow Y$, the functor f_0^* and the morphism $f_0^*(\mathcal{J}\mathcal{O}_Y) \rightarrow \mathcal{J}\mathcal{O}_Z$ of (0.1.4) induce a natural morphism of abelian groups $L_X(f)$:

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) &\longrightarrow \mathrm{Hom}_{\mathcal{O}_{Z_0}}(f_0^*g_0^*(\Omega_{X_0/S_0}^1), f_0^*(\mathcal{J}\mathcal{O}_Y)) \\ &\longrightarrow \mathrm{Hom}_{\mathcal{O}_{Z_0}}(f_0^*g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Z). \end{aligned}$$

Finally, let us remark that $L_X(f)$ is described in a simple way locally, as follows. Let us first note that Y and $Y_{\mathcal{J}}$ have the same underlying topological space, and that the same holds for V and $V_{\mathcal{J}}$, if V is an open of Y . Let then $U = \mathrm{Spec}(A)$ be an affine open of X above an affine open $\mathrm{Spec}(\Lambda)$ of S , $V = \mathrm{Spec}(B)$ an affine open of Y such that $g_{\mathcal{J}}(V_{\mathcal{J}}) \subset U$, and $W = \mathrm{Spec}(C)$ an affine open of $f^{-1}(V)$. Let J and I be the ideals of Λ corresponding to $\mathcal{J}$ and $\mathcal{I}$. Then f (resp. $g_{\mathcal{J}}$) induces a morphism of Λ -algebras $\theta: B \rightarrow C$ (resp. $\phi: A \rightarrow B/IB$), and one evidently has $\theta(JB) \subset JC$. On the other hand, $m \in \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X/S}^1), \mathcal{J}\mathcal{O}_Y)$ induces an element D of

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_{V_0}}(g_0^*(\Omega_{U/S}^1), \mathcal{J}\mathcal{O}_V) &= \mathrm{Hom}_{B/IB}(\Omega_{A/\Lambda}^1 \otimes_A B/IB, JB) \\ &= \mathrm{Der}_{\Lambda}(A, JB), \end{aligned}$$

⁽⁷⁾**N.D.E.** One has added this scholium.

⁽⁸⁾**N.D.E.** The introduction of the functor L_X leads to an extension “functorial in Y ” (and in X) of SGA 1, III, Prop. 5.1; see below 0.1.8 and 0.1.9, as well as 0.1.10 and 0.2. Moreover, when X is an S -functor in groups, L_X gives rise, under certain hypotheses, to an S -functor in groups L'_X and to an exact sequence $1 \rightarrow L'_X \rightarrow X \rightarrow X^+$ (cf. 0.4, 0.9 and 0.11), which play an essential role in this exposé (cf. Theorems 2.1, 3.5 and 4.21).

⁽⁹⁾**N.D.E.** One has added what follows.

and the image of $L_X(f)(m)$ in

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_{W_0}}(f_0^* g_0^*(\Omega_{U/S}^1), \mathcal{J}\mathcal{O}_Z) &= \mathrm{Hom}_{C/IC}(\Omega_{A/\Lambda}^1 \otimes_A C/IC, JC) \\ &= \mathrm{Der}_\Lambda(A, JC) \end{aligned}$$

is none other than $\theta \circ D$.

Remark 0.1.6. ⁽¹⁰⁾Let $f: X \rightarrow W$ be an S -morphism. It induces an S -morphism $f^+: X^+ \rightarrow W^+$ defined as follows: if g is an element of $\mathrm{Hom}_S(Y, X^+)$, corresponding to an S -morphism $g_{\mathcal{J}}: Y_{\mathcal{J}} \rightarrow X$, then $f^+(g)$ is the element $f \circ g_{\mathcal{J}}$ of $\mathrm{Hom}_S(Y, W^+)$. It is clear that the following diagram is commutative:

$$\begin{array}{ccc} X & \xrightarrow{f} & W \\ p_X \downarrow & & \downarrow p_W \\ X^+ & \xrightarrow{f^+} & W^+. \end{array}$$

Reminders 0.1.7. ⁽¹¹⁾In this paragraph, given an S -scheme X , one “recalls” certain functorial properties of the module of differentials $\Omega_{X/S}^1$ and of the first infinitesimal neighborhood of the diagonal, $\Delta_{X/S}^{(1)}$, properties which follow easily from EGA IV₄, §§ 16.1–16.4, but which do not appear there explicitly.

a) Let us begin by recalling the following facts (cf. EGA II, §§ 1.2–1.5). Let $g: Y \rightarrow X$ be a morphism of schemes, $\pi: X' \rightarrow X$ an *affine* X -scheme, $\mathcal{B}$ the quasi-coherent $\mathcal{O}_X$ -algebra $\pi_*(\mathcal{O}_{X'})$; then the Y -scheme $Y \times_X X'$ is affine and corresponds to the quasi-coherent $\mathcal{O}_Y$ -algebra $g^*(\mathcal{B})$, and one has a commutative diagram of bijections:

$$\begin{array}{ccc} \mathrm{Hom}_X(Y, X') & \xrightarrow{\sim} & \mathrm{Hom}_Y(Y, Y \times_X X') \\ \wr \downarrow & & \downarrow \wr \\ \mathrm{Hom}_{\mathcal{O}_X\text{-alg.}}(\mathcal{B}, g_*(\mathcal{O}_Y)) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(g^*(\mathcal{B}), \mathcal{O}_Y). \end{array}$$

Moreover, these bijections are functorial in the pair (X, X') , i.e., if W' is an affine W -scheme, corresponding to the quasi-coherent $\mathcal{O}_W$ -algebra $\mathcal{A}$, if one has a commutative diagram of morphisms of schemes

$$\begin{array}{ccccc} & & X' & \xrightarrow{f'} & W' \\ & \nearrow g' & & & \\ Y & \xrightarrow{g} & X & \xrightarrow{f} & W \end{array}$$

and if one denotes by $\phi: \mathcal{A} \rightarrow f_*(\mathcal{B})$ and $\phi^\sharp: f^*(\mathcal{A}) \rightarrow \mathcal{B}$ (resp. $\theta: \mathcal{B} \rightarrow g_*(\mathcal{O}_Y)$ and $\theta^\sharp: g^*(\mathcal{B}) \rightarrow \mathcal{O}_Y$) the algebra morphisms associated with f' (resp. with a variable X -morphism $g': Y \rightarrow X'$), then one has the following commutative diagram (where Y is

⁽¹⁰⁾**N.D.E.** One has spelled out this remark.

⁽¹¹⁾**N.D.E.** One has added this paragraph (see also [DG70], § I.4, n^{os} 1–2).

viewed as a W -scheme via $f \circ g$):

$$\begin{array}{ccc}
 \mathrm{Hom}_X(Y, X') & \xrightarrow{g' \mapsto f' \circ g'} & \mathrm{Hom}_W(Y, W') \\
 \downarrow \wr & & \searrow \sim \\
 & & \mathrm{Hom}_{\mathcal{O}_W\text{-alg.}}(\mathcal{A}, f_* g_*(\mathcal{O}_Y)) \\
 \mathrm{Hom}_{\mathcal{O}_X\text{-alg.}}(\mathcal{B}, g_*(\mathcal{O}_Y)) & \xrightarrow{\theta \mapsto \theta \circ \phi^\#} & \mathrm{Hom}_{\mathcal{O}_X\text{-alg.}}(f^*(\mathcal{A}), g_*(\mathcal{O}_Y)) \\
 \downarrow \wr & & \downarrow \wr \\
 \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(g^*(\mathcal{B}), \mathcal{O}_Y) & \xrightarrow{\theta^\# \mapsto \theta^\# \circ g^*(\phi^\#)} & \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(g^* f^*(\mathcal{A}), \mathcal{O}_Y).
 \end{array}$$

$\theta \mapsto f_*(\theta) \circ \phi$ (arrow from $\mathrm{Hom}_{\mathcal{O}_X\text{-alg.}}(f^*(\mathcal{A}), g_*(\mathcal{O}_Y))$ to $\mathrm{Hom}_{\mathcal{O}_W\text{-alg.}}(\mathcal{A}, f_* g_*(\mathcal{O}_Y))$)

b) Let now X be an S -scheme. Let $\Omega_{X/S}^1$ be the module of differentials of X over S , and $\Delta_{X/S}^{(1)}$ the first infinitesimal neighborhood of the diagonal immersion $\delta_X: X \rightarrow X \times_S X$; it is a subscheme of $X \times_S X$, of which the diagonal $\Delta_{X/S}$ is a closed subscheme. One denotes by pr_X^i ($i = 1, 2$) the two projections of $X \times_S X$, and by π_X the restriction of pr_X^1 to $\Delta_{X/S}^{(1)}$.

On the one hand, every morphism $f: X \rightarrow W$ of S -schemes induces an S -morphism $\Delta^{(1)} f: \Delta_{X/S}^{(1)} \rightarrow \Delta_{W/S}^{(1)}$ such that the following diagram is commutative:

$$\begin{array}{ccccccc}
 X & \xrightarrow{\delta_X} & \Delta_{X/S}^{(1)} & \longrightarrow & X \times_S X & \xrightarrow{\mathrm{pr}_X^i} & X \\
 f \downarrow & & \Delta^{(1)} f \downarrow & & f \times f \downarrow & & \downarrow f \\
 W & \xrightarrow{\delta_W} & \Delta_{W/S}^{(1)} & \longrightarrow & W \times_S W & \xrightarrow{\mathrm{pr}_W^i} & W.
 \end{array}$$

On the other hand, $\Delta_{X/S}^{(1)}$ is, via the projection π_X , an affine X -scheme, spectrum of the augmented quasi-coherent $\mathcal{O}_X$ -algebra

$$\mathcal{P}_{X/S}^1 = \mathcal{O}_X \oplus \Omega_{X/S}^1,$$

where $\Omega_{X/S}^1$ is an ideal of square zero; the augmentation is the morphism of $\mathcal{O}_X$ -algebras $\varepsilon_X: \mathcal{P}_{X/S}^1 \rightarrow \mathcal{O}_X$ which vanishes on $\Omega_{X/S}^1$ and which corresponds to the closed immersion $\delta_X: X \hookrightarrow \Delta_{X/S}^{(1)}$. Then, every morphism of S -schemes $f: X \rightarrow W$ induces a morphism of augmented $\mathcal{O}_X$ -algebras

$$f^*(\mathcal{P}_{W/S}^1) = \mathcal{O}_X \oplus f^*(\Omega_{W/S}^1) \rightarrow \mathcal{P}_{X/S}^1 = \mathcal{O}_X \oplus \Omega_{X/S}^1,$$

i.e. , in an equivalent way, a morphism of $\mathcal{O}_X$ -modules

$$f_{X/W/S}: f^*(\Omega_{W/S}^1) \rightarrow \Omega_{X/S}^1,$$

cf. EGA IV₄ (16.4.3.6) (and (16.4.18.2) for the notation $f_{X/W/S}$).

As $\pi_X: \Delta_{X/S}^{(1)} \rightarrow X$ is affine, then, by a), $\Delta^{(1)} f$ is entirely determined by $f_{X/W/S}$ and, for every X -scheme $g: Y \rightarrow X$, the set

$$\mathrm{Hom}_X(Y, \Delta_{X/S}^{(1)}) \simeq \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(\mathcal{O}_Y \oplus g^*(\Omega_{X/S}^1), \mathcal{O}_Y)$$

is identified with a subset of $\mathrm{Hom}_{\mathcal{O}_Y}(g^*(\Omega_{X/S}^1), \mathcal{O}_Y)$, namely the subset

$$\mathrm{Hom}_{\mathcal{O}_Y}^\square(g^*(\Omega_{X/S}^1), \mathcal{O}_Y)$$

formed of the $\mathcal{O}_Y$ -morphisms $\psi: g^*(\Omega_{X/S}^1) \rightarrow \mathcal{O}_Y$ such that $\text{Im}(\psi)$ is an ideal of $\mathcal{O}_Y$ of square zero.⁽¹²⁾

Consequently, applying a) to the diagram

$$\begin{array}{ccccc} & & \Delta_{X/S}^{(1)} & \xrightarrow{\Delta^{(1)}f} & \Delta_{W/S}^{(1)} \\ & \nearrow g' & & & \\ Y & \xrightarrow{g} & X & \xrightarrow{f} & W \end{array}$$

and taking into account the fact that $\Delta^{(1)}f$ is the restriction to $\Delta_{X/S}^{(1)}$ of $f \times f$, one obtains the following commutative diagram, functorial in the X -scheme $Y \xrightarrow{g} X$:

$$(0.1.7 \text{ (*)}) \quad \begin{array}{ccc} \text{Hom}_X(Y, X \times_S X) & \xrightarrow{(g, g') \mapsto (f \circ g, f \circ g')} & \text{Hom}_W(Y, \Delta_{W/S}^{(1)}) \\ \uparrow & & \uparrow \\ \text{Hom}_X(Y, \Delta_{X/S}^{(1)}) & \xrightarrow{g' \mapsto (\Delta^{(1)}f) \circ g'} & \text{Hom}_W(Y, \Delta_{W/S}^{(1)}) \\ \uparrow \simeq & & \uparrow \simeq \\ \text{Hom}_{\mathcal{O}_Y}^\square(g^*(\Omega_{X/S}^1), \mathcal{O}_Y) & \xrightarrow{\psi \mapsto \psi \circ g^*(f_{X/W/S})} & \text{Hom}_{\mathcal{O}_Y}^\square(g^*f^*(\Omega_{W/S}^1), \mathcal{O}_Y) \end{array}$$

Remark 0.1.7.1. Let us end this paragraph with the following remark, which will be useful later (cf. 0.1.10). If one denotes by $L_X^\square$ the X -functor which, to every X -scheme $g: Y \rightarrow X$, associates $\text{Hom}_{\mathcal{O}_Y}^\square(g^*(\Omega_{X/S}^1), \mathcal{O}_Y)$, and by $L_f^\square: L_X^\square \rightarrow L_W^\square$ the morphism of functors defined above (which, to every $\psi \in L_X^\square(Y)$, associates $\psi \circ g^*(f_{X/W/S})$), what precedes shows that one has a commutative diagram of functors:

$$\begin{array}{ccccc} X \times_X L_X^\square & \xleftarrow{\sim} & \Delta_{X/S}^{(1)} & \hookrightarrow & X \times_S X \\ f \times L_f^\square \downarrow & & \Delta^{(1)}f \downarrow & & \downarrow f \times f \\ W \times_W L_W^\square & \xleftarrow{\sim} & \Delta_{W/S}^{(1)} & \hookrightarrow & W \times_S W. \end{array}$$

Theorem 0.1.8. (SGA 1, III 5.1)⁽¹³⁾ Let Y, X be two S -schemes, $\mathfrak{J}$ a quasi-coherent ideal of $\mathcal{O}_Y$ of square zero, $Y_{\mathfrak{J}}$ the closed subscheme of Y defined by $\mathfrak{J}$, and $g_{\mathfrak{J}}: Y_{\mathfrak{J}} \rightarrow X$ an S -morphism.

a) The set $P(g_{\mathfrak{J}})$ of S -morphisms $g: Y \rightarrow X$ which extend $g_{\mathfrak{J}}$ is either empty, or principal homogeneous under the abelian group

$$\text{Hom}_{\mathcal{O}_{Y_{\mathfrak{J}}}}(g_{\mathfrak{J}}^*(\Omega_{X/S}^1), \mathfrak{J}).$$

b) If $i: Y_0 \hookrightarrow Y_{\mathfrak{J}}$ is the closed immersion defined by a quasi-coherent ideal $\mathfrak{I} \supset \mathfrak{J}$ such that $\mathfrak{I}\mathfrak{J} = 0$, and if $g_0 = g_{\mathfrak{J}} \circ i$, the preceding abelian group is isomorphic to

$$\text{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X/S}^1), \mathfrak{I}).$$

⁽¹²⁾**N.D.E.** In the proof of SGA 1, III 5.1, one must accordingly correct the sentence “Now the algebra homomorphisms $\mathcal{A} \rightarrow \mathcal{O}_Y$ correspond . . .” (the sequel of *loc. cit.* being unchanged).

⁽¹³⁾**N.D.E.** One has reproduced here the statement and the proof of SGA 1, III 5.1, which one needs in order to prove Proposition 0.2 below (see already Corollary 0.1.9 which follows). On the other hand, $\mathfrak{J}$ (resp. $\mathfrak{I}$) is another calligraphy of the letter J (resp. I); returning to the earlier notations ($\mathcal{J} \subset \mathcal{I}$ ideals of $\mathcal{O}_S$ such that $\mathcal{I}\mathcal{J} = 0$), one will apply this to $\mathfrak{J} = \mathcal{J}\mathcal{O}_Y$ (resp. $\mathfrak{I} = \mathcal{I}\mathcal{O}_Y$).

PROOF. (b) follows at once from (a). Indeed, $\mathfrak{J}$, being annihilated by $\mathfrak{J}$, may be regarded as an $\mathcal{O}_{Y_0}$ -module, whence, by adjunction,

$$\mathrm{Hom}_{\mathcal{O}_{Y_3}}(g_3^*(\Omega_{X/S}^1), \mathfrak{J}) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(i^*g_3^*(\Omega_{X/S}^1), \mathfrak{J}).$$

To prove (a), one may suppose $P(g_3) \neq \emptyset$, i.e. that there exists an S -morphism $g: Y \rightarrow X$ extending g_3 . Let us denote by j the immersion $Y_3 \hookrightarrow Y$. Then $P(g_3)$ is the set of S -morphisms $g': Y \rightarrow X$ such that $g' \circ j = g_3$. The datum of such a g' is equivalent to the datum of an S -morphism

$$h: Y \rightarrow X \times_S X$$

such that $\mathrm{pr}_1 \circ h = g$ and $h_3 = \delta \circ g_3$, where $h_3 = h \circ j$ and δ is the diagonal immersion $X \hookrightarrow X \times_S X$:

$$\begin{array}{ccc} X \times_S X & \xleftarrow{h_3 = \delta \circ g_3} & Y_3 \\ \mathrm{pr}_1 \downarrow & \swarrow h & \downarrow j \\ X & \xleftarrow{g} & Y. \end{array}$$

As h_3 factors through δ and Y is in the first infinitesimal neighborhood of the immersion $j: Y_3 \rightarrow Y$ (since $\mathfrak{J}^2 = 0$), then, by functoriality (cf. EGA IV₄, 16.2.2 (i)), the sought-for h factor, in a unique way, through $\Delta_{X/S}^{(1)}$ (cf. 0.1.7). Let us set

$$Y' = \Delta_{X/S}^{(1)} \times_X Y \quad \text{and} \quad Y'_3 = Y' \times_Y Y_3 = \Delta_{X/S}^{(1)} \times_X Y_3.$$

Then the sought-for h are in bijection with the sections u of $Y' \rightarrow Y$ which extend the section $u_3 = (\delta \circ g_3, \mathrm{id})$ of $Y'_3 \rightarrow Y_3$. On the other hand, Y' (resp. Y'_3) is a scheme *affine over* Y (resp. Y_3), corresponding to the quasi-coherent algebra

$$\mathcal{A} = \mathcal{O}_Y \oplus g^*(\Omega_{X/S}^1), \quad \text{resp.} \quad \mathcal{A}_3 = \mathcal{A} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_3} = \mathcal{O}_{Y_3} \oplus g_3^*(\Omega_{X/S}^1).$$

Let us denote by $\varepsilon: \mathcal{A} \rightarrow \mathcal{O}_Y$ the canonical augmentation of $\mathcal{A}$ (i.e. the morphism of $\mathcal{O}_Y$ -algebras $\mathcal{A} \rightarrow \mathcal{O}_Y$ which vanishes on $g^*(\Omega_{X/S}^1)$), and let us define likewise $\varepsilon_3: \mathcal{A}_3 \rightarrow \mathcal{O}_{Y_3}$. Then

$$\Gamma(Y'/Y) \simeq \mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(\mathcal{A}, \mathcal{O}_Y), \quad \Gamma(Y'_3/Y_3) \simeq \mathrm{Hom}_{\mathcal{O}_{Y_3}\text{-alg.}}(\mathcal{A}_3, \mathcal{O}_{Y_3}),$$

and, via these isomorphisms, the section $u = (\delta \circ g, \mathrm{id})$ (resp. u_3) corresponds to ε (resp. ε_3).

Consequently, $P(g_3)$ is in bijection with the set of morphisms of algebras $\mathcal{A} \rightarrow \mathcal{O}_Y$ which reduce according to ε_3 , and via this bijection, g corresponds to ε .

Set $\mathcal{M} = g^*(\Omega_{X/S}^1)$. Then $\mathrm{Hom}_{\mathcal{O}_Y\text{-alg.}}(\mathcal{A}, \mathcal{O}_Y)$ is identified with the set of $\mathcal{O}_Y$ -morphisms $\psi: \mathcal{M} \rightarrow \mathcal{O}_Y$ such that $\mathrm{Im}(\psi)$ is an ideal of square zero, and one is interested in those which induce the zero morphism $\mathcal{M} \rightarrow \mathcal{O}_{Y_3} = \mathcal{O}_Y/\mathfrak{J}$, i.e. which map $\mathcal{M}$ into $\mathfrak{J}$. Conversely, since $\mathfrak{J}^2 = 0$, every $\mathcal{O}_Y$ -morphism $\phi: \mathcal{M} \rightarrow \mathfrak{J}$ comes from a (unique) morphism of algebras $\mathcal{A} \rightarrow \mathcal{O}_Y$, reducing according to ε_3 . Finally, one has

$$\mathrm{Hom}_{\mathcal{O}_Y}(g^*(\Omega_{X/S}^1), \mathfrak{J}) = \mathrm{Hom}_{\mathcal{O}_{Y_3}}(g_3^*(\Omega_{X/S}^1), \mathfrak{J})$$

since $\mathfrak{J}^2 = 0$ (cf. the proof of (b) already seen). One therefore obtains a bijection

$$P(g_3) \simeq \mathrm{Hom}_{\mathcal{O}_{Y_3}}(g_3^*(\Omega_{X/S}^1), \mathfrak{J})$$

by which g corresponds to the zero morphism.

For every $m \in \mathrm{Hom}_{\mathcal{O}_{Y_3}}(g_3^*(\Omega_{X/S}^1), \mathfrak{J})$, let us denote by $m \cdot g$ the element of $P(g_3)$ associated with g and m by the preceding bijection. One has already seen that $0 \cdot g = g$; it remains to see that

$$(0.1.8 \text{ (*)}) \quad m' \cdot (m \cdot g) = (m + m') \cdot g.$$

This is verified locally.⁽¹⁴⁾ Indeed, the two preceding morphisms $Y \rightarrow X$ induce the same continuous map as g between the underlying topological spaces; it therefore suffices to check that for every affine open $U = \operatorname{Spec}(A)$ of X lying over an affine open $\operatorname{Spec}(\Lambda)$ of S , and every affine open $V = \operatorname{Spec}(B)$ of $g^{-1}(U)$, they induce the same morphism of Λ -algebras $A \rightarrow B$.

Let $J = \Gamma(V, \mathfrak{J})$ and let ϕ, ψ and η be the morphisms $A \rightarrow B$ induced by $g, m \cdot g$ and $m' \cdot (m \cdot g)$ respectively; they coincide modulo J . One can write in a unique way $\psi = \phi + D$ (resp. $\eta = \psi + D'$), where D (resp. D') is an element of

$$\operatorname{Dér}_\phi(A, J) = \{\delta \in \operatorname{Hom}_\Lambda(A, J) \mid \delta(ab) = \phi(a)\delta(b) + \phi(b)\delta(a)\}$$

(resp. $\operatorname{Dér}_\psi(A, J)$). But $\operatorname{Dér}_\phi(A, J) = \operatorname{Dér}_\psi(A, J)$ since $J^2 = 0$, and both are identified with

$$\operatorname{Hom}_{B/J}(\Omega_{A/\Lambda}^1 \otimes_A B/J, J),$$

and via this identification D corresponds to m and D' to m' . Then $\eta = \phi + D + D'$ and $D + D'$ corresponds to $m + m'$, whence the equality (*). $\square$

Corollary 0.1.9. ⁽¹⁵⁾ *Let X be an S -scheme; let us take up again the notations of 0.1.5. Then X is equipped with a (left) operation of the abelian X^+ -group L_X , which makes X into a formally principal homogeneous object under L_X over X^+ , i.e. one has an isomorphism of X^+ -functors:*

$$L_X \times_{X^+} X \xrightarrow{\sim} X \times_{X^+} X$$

(defined set-theoretically by $(m, x) \mapsto (x, m \cdot x)$).

PROOF. Let i_0 be the immersion $X_0 \hookrightarrow X$. Let us first note that, since $X_0 = X \times_S S_0$, one has $i_0^*(\Omega_{X/S}^1) \simeq \Omega_{X_0/S_0}^1$ (cf. EGA IV, 16.4.5).

Let Y be an X^+ -scheme, given by an S -morphism $g_{\mathcal{J}}: Y_{\mathcal{J}} \rightarrow X$, and let $g_0: Y_0 \rightarrow X_0$ be the morphism obtained by base change. By 0.1.8, if $\operatorname{Hom}_{X^+}(Y, X)$ is non-empty, it is a principal homogeneous set under the group

$$\operatorname{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X/S}^1), \mathcal{J}\mathcal{O}_Y),$$

which is identified with $\operatorname{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) = L_X(Y)$. One therefore has a bijection

$$L_X(Y) \times \operatorname{Hom}_{X^+}(Y, X) \xrightarrow{\sim} \operatorname{Hom}_{X^+}(Y, X \times_{X^+} X)$$

given by $(m, g) \mapsto (g, m \cdot g)$. Let us show that this is “functorial in Y ”.

Let $f: Z \rightarrow Y$ be a morphism of S -schemes. The point is to show that the diagram below is commutative:

$$\begin{array}{ccc} L_X(Y) \times \operatorname{Hom}_{X^+}(Y, X) & \xrightarrow{\sim} & \operatorname{Hom}_{X^+}(Y, X \times_{X^+} X) \\ \downarrow L_X(f) \times f & & \downarrow f \times f \\ L_X(Z) \times \operatorname{Hom}_{X^+}(Z, X) & \xrightarrow{\sim} & \operatorname{Hom}_{X^+}(Z, X \times_{X^+} X). \end{array}$$

If $\operatorname{Hom}_{X^+}(Y, X) = \emptyset$, there is nothing to show. It therefore suffices to see that, for every S -morphism $g: Y \rightarrow X$ extending $g_{\mathcal{J}}$ and every $m \in L_X(Y)$, one has:

$$(0.1.9 \text{ (*)}) \quad (m \cdot g) \circ f = L_X(f)(m) \cdot (g \circ f).$$

These two S -morphisms $Z \rightarrow X$ coincide on $Z_{\mathcal{J}}$ with $g_{\mathcal{J}} \circ f_{\mathcal{J}}$; in particular, they induce the same continuous map as $g \circ f$ between the underlying topological spaces. Consequently, it suffices to see that, if $z \in Z$, $y = f(z)$, $x = g(y)$, and if A, B, C denote respectively the local rings $\mathcal{O}_{X,x}, \mathcal{O}_{Y,y}, \mathcal{O}_{Z,z}$, then the morphisms $A \rightarrow C$ induced by $(m \cdot g) \circ f$ and

⁽¹⁴⁾**N.D.E.** What precedes is reproduced almost word for word from SGA 1, III 5.1; what follows has been spelled out.

⁽¹⁵⁾**N.D.E.** One has added this corollary of SGA 1, III 5.1, which proves point (i) of proposition 0.2 further on.

$L_X(f)(m) \cdot (g \circ f)$ coincide. Let us denote by s the image of x in S , $\Lambda = \mathcal{O}_{S,s}$, and let J and I be the ideals of Λ corresponding to $\mathcal{J}$ and $\mathcal{I}$, and let $\phi, \psi: A \rightarrow B$ and $\theta: B \rightarrow C$ be the morphisms of Λ -algebras induced by $g, m \cdot g$ and f . Then m induces an element D of

$$\mathrm{Hom}_{B/IB}(\Omega_{A/\Lambda}^1 \otimes_A B/IB, JB) = \mathrm{Dér}_\Lambda(A, JB)$$

and one has $\psi = \phi + D$; hence $(m \cdot g) \circ f$ corresponds to $\theta \circ \psi = \theta \circ \phi + \theta \circ D$. Now, one has seen in 0.1.5 that $\theta \circ D$ is the image of $L_X(f)(m)$ in

$$\mathrm{Hom}_{C/IC}(\Omega_{A/\Lambda}^1 \otimes_A C/IC, JC) = \mathrm{Dér}_\Lambda(A, JC);$$

consequently, $\theta \circ \phi + \theta \circ D$ is the image of $L_X(f)(m) \cdot (g \circ f)$. This proves the equality (0.1.9 (*)). $\square$

Corollary 0.1.10. ⁽¹⁶⁾a) L_X depends functorially on X : for every S -morphism $f: X \rightarrow W$, there exists an S -morphism $L_f: L_X \rightarrow L_W$ which is a morphism of abelian groups “over f^+ ”, i.e. the diagram

$$\begin{array}{ccc} L_X & \xrightarrow{L_f} & L_W \\ \downarrow & & \downarrow \\ X^+ & \xrightarrow{f^+} & W^+ \end{array}$$

is commutative and, for every $Y \rightarrow X^+$,

$$L_f(Y): \mathrm{Hom}_{X^+}(Y, L_X) \rightarrow \mathrm{Hom}_{W^+}(Y, L_W)$$

(where Y is over W^+ via f^+) is a morphism of abelian groups.

b) Moreover, the following diagram is commutative:

$$\begin{array}{ccc} L_X \times_{X^+} X & \xrightarrow{\sim} & X \times_{X^+} X \\ \downarrow L_f \times f & & \downarrow f \times f \\ L_W \times_{W^+} W & \xrightarrow{\sim} & W \times_{W^+} W. \end{array}$$

PROOF. a) L_f is induced by the morphism of $\mathcal{O}_{X_0}$ -modules $f_{X_0/W_0/S_0}: f_0^*(\Omega_{W_0/S_0}^1) \rightarrow \Omega_{X_0/S_0}^1$ (cf. 0.1.7 b)): for every X^+ -scheme Y , given by an S -morphism $g_{\mathcal{J}}: Y_{\mathcal{J}} \rightarrow X$, one has a commutative diagram, functorial in Y :

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) & \xrightarrow{L_f(Y)} & \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*f_0^*(\Omega_{W_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) \\ \downarrow & & \downarrow \\ \{g_{\mathcal{J}}\} & \xrightarrow{\quad} & \{f_{\mathcal{J}} \circ g_{\mathcal{J}}\} \end{array}$$

where $L_f(Y)$ is the map $\psi \mapsto \psi \circ g_0^*(f_{X_0/W_0/S_0})$, which is indeed a morphism of abelian groups. ⁽¹⁷⁾

Let us prove (b). Let Y be an X^+ -scheme; if $\mathrm{Hom}_{X^+}(Y, X) = \emptyset$ there is nothing to show. Let therefore $g \in \mathrm{Hom}_{X^+}(Y, X)$; one must see that for every $m \in L_X(Y)$, one has:

$$(0.1.10 \text{ (*)}) \quad f \circ (m \cdot g) = L_f(Y)(m) \cdot (f \circ g).$$

Now, g being fixed, $\mathrm{Hom}_X(Y, X \times_{X^+} X)$ is a subset of $\mathrm{Hom}_X(Y, \Delta_{X/S}^{(1)})$ and

$$L_X(Y) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) = \mathrm{Hom}_{\mathcal{O}_Y}(g^*(\Omega_{X/S}^1), \mathcal{J}\mathcal{O}_Y)$$

⁽¹⁶⁾**N.D.E.** One has spelled out the “functoriality in X ” of L_X , in particular point b) below.

⁽¹⁷⁾**N.D.E.** One will note that L_f depends only on $f_0: X_0 \rightarrow W_0$.

a subset of $L_X^\square(Y)$ (cf. 0.1.7); finally $L_f(Y)$ is the restriction to $L_X(Y)$ of the map $L_f^\square(Y)$. Moreover, the bijection

$$L_X(Y) \xrightarrow{\sim} \text{Hom}_X(Y, X \times_{X^+} X), \quad m \mapsto (g, m \cdot g)$$

is (the inverse of) the restriction to $L_X(Y) \subset L_X^\square(Y)$ of the bijection $\text{Hom}_X(Y, \Delta_{X/S}^{(1)}) \xrightarrow{\sim} \{g\} \times L_X^\square(Y)$ considered in 0.1.7.1. Consequently, the equality (0.1.10 (*)) follows from (0.1.7 (*)); indeed, if one denotes by g' the X -morphism $Y \rightarrow \Delta_{X/S}^{(1)}$ defined by $(g, m \cdot g)$, then the element of $L_W(Y)$ corresponding to $(f \circ g, f \circ g')$ is $L_f^\square(Y)(m) = L_f(Y)(m)$, i.e. one indeed has

$$L_f(m) \cdot (f \circ g) = f \circ (m \cdot g).$$

□

Lemma 0.1.11. *Let X, X' be two S -schemes. One has a commutative diagram:*

$$\begin{array}{ccc} L_X \times_S L_{X'} & \xrightarrow{\sim} & L_{X \times_S X'} \\ \downarrow & & \downarrow \\ X^+ \times_S X'^+ & \xrightarrow{\sim} & (X \times_S X')^+. \end{array}$$

PROOF. ⁽¹⁸⁾First, for every S -scheme Y , $\text{Hom}_S(Y, X^+ \times_S X'^+)$ equals $\text{Hom}_S(Y, X^+) \times \text{Hom}_S(Y, X'^+)$ and the latter is isomorphic to

$$\text{Hom}_S(Y_{\mathcal{J}}, X) \times \text{Hom}_S(Y_{\mathcal{J}}, X') = \text{Hom}_S(Y, (X \times_S X')^+);$$

this proves that $X^+ \times_S X'^+ \simeq (X \times_S X')^+$.

Next, let Y be a scheme over $X^+ \times_S X'^+$ via a morphism $h: Y_{\mathcal{J}} \rightarrow X \times_S X'$; set $f = p \circ h$ and $g = q \circ h$, where one has denoted by p, q the projections of $X \times_S X'$ to X and X' . Since

$$\Omega_{(X_0 \times_{S_0} X'_0)/S_0}^1 \simeq p_0^*(\Omega_{X_0/S_0}^1) \oplus q_0^*(\Omega_{X'_0/S_0}^1)$$

(cf. EGA IV₄, 16.4.23), one obtains a natural isomorphism:

$$\begin{aligned} & \text{Hom}_{\mathcal{O}_{Y_0}}(f_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) \times \text{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X'_0/S_0}^1), \mathcal{J}\mathcal{O}_Y) \\ & \simeq \text{Hom}_{\mathcal{O}_{Y_0}}(h_0^*(\Omega_{(X_0 \times_{S_0} X'_0)/S_0}^1), \mathcal{J}\mathcal{O}_Y), \end{aligned}$$

i.e. $L_X(Y) \times L_{X'}(Y) \simeq L_{X \times_S X'}(Y)$. □

Remark 0.1.12. ⁽¹⁹⁾Let $\mathcal{C}$ be a category stable under fibered products, S an object of $\mathcal{C}$, T_1, T_2 two objects over S and, for $i = 1, 2$, L_i and X_i two objects over T_i :

$$\begin{array}{ccccc} L_1 & & X_1 & & L_2 & & X_2 \\ & \searrow & \swarrow & & \searrow & \swarrow & \\ & T_1 & & & T_2 & & \\ & & \searrow & & \swarrow & & \\ & & S & & & & \end{array}$$

Then one has a natural isomorphism:

$$(L_1 \times_{T_1} X_1) \times_S (L_2 \times_{T_2} X_2) \simeq (L_1 \times_S L_2) \times_{T_1 \times_S T_2} (X_1 \times_S X_2).$$

Consequently, one deduces from the preceding lemma the:

⁽¹⁸⁾**N.D.E.** One has spelled out the proof.

⁽¹⁹⁾**N.D.E.** One has added this remark, as well as the corollary which follows.

Corollary 0.1.13. *Let X_1, X_2 be two S -schemes. One has a commutative diagram of isomorphisms:*

$$\begin{array}{ccc} L_{X_1 \times_S X_2} \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) & & \\ \downarrow (0.1.11) \simeq & \searrow \simeq & \\ (L_{X_1} \times_S L_{X_2}) \times_{(X_1^+ \times_S X_2^+)} (X_1 \times_S X_2) & \xrightarrow[(0.1.12)]{\simeq} & (L_{X_1} \times_{X_1^+} X_1) \times_S (L_{X_2} \times_{X_2^+} X_2). \end{array}$$

We can now state:

Proposition 0.2. *For every S -scheme X , one can define a (left) operation of the abelian X^+ -group L_X on the X^+ -object X , such that:*

- (i) *this operation makes X into a formally principal homogeneous object under L_X over X^+ , i.e. the morphism*

$$L_X \times_{X^+} X \longrightarrow X \times_{X^+} X$$

is an isomorphism of X^+ -functors;

- (ii) *this operation is functorial in the S -scheme X , i.e. for every S -morphism $f: X \longrightarrow W$, the following diagram is commutative:*

$$\begin{array}{ccc} L_X \times_{X^+} X & \longrightarrow & X \\ L_f \times f \downarrow & & \downarrow f \\ L_W \times_{W^+} W & \longrightarrow & W; \end{array}$$

- (iii) *this operation “commutes with fibered product”, i.e. for all S -schemes X_1 and X_2 , the following diagram is commutative:*

$$\begin{array}{ccc} L_{X_1 \times_S X_2} \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) & \longrightarrow & X_1 \times_S X_2 \\ \downarrow \simeq & \searrow \simeq & \uparrow \\ (L_{X_1} \times_S L_{X_2}) \times_{(X_1 \times_S X_2)^+} (X_1 \times_S X_2) & \xrightarrow{\simeq} & (L_{X_1} \times_{X_1^+} X_1) \times_S (L_{X_2} \times_{X_2^+} X_2). \end{array}$$

PROOF. ⁽²⁰⁾(i) and (ii) follow respectively from corollaries 0.1.9 and 0.1.10. To prove (iii), set $P(X) = L_X \times_{X^+} X$, for every S -scheme X . Then, by (ii) applied to the projections $p_i: X_1 \times_S X_2 \longrightarrow X_i$, one obtains commutative squares

$$\begin{array}{ccc} P(X_1 \times_S X_2) & \longrightarrow & X_1 \times_S X_2 \\ L_{p_i} \times p_i \downarrow & & \downarrow p_i \\ P(X_i) & \longrightarrow & X_i \end{array}$$

for $i = 1, 2$, and hence a commutative square:

$$\begin{array}{ccc} P(X_1 \times_S X_2) & \longrightarrow & X_1 \times_S X_2 \\ \downarrow & & \\ P(X_1) \times_S P(X_2) & \longrightarrow & X_1 \times_S X_2. \end{array}$$

Combining this with corollary 0.1.13, one obtains that the vertical arrow is an isomorphism, and that one has the commutative diagram announced in (iii). $\square$

⁽²⁰⁾**N.D.E.** One has modified and spelled out the original, taking into account the additions made previously; these incorporate, spelling it out, the content of page 89 of the original.

Remark 0.3. Suppose the X^+ -scheme Y *flat* over S (cf. SGA 1, IV). One can then write

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^*(\Omega_{X_0/S_0}^1), \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}).$$

Remark 0.4. Let us denote by $\pi_0: X_0 \rightarrow S_0$ the structural morphism and suppose that there exists an $\mathcal{O}_{S_0}$ -module ω_{X_0/S_0}^1 such that $\Omega_{X_0/S_0}^1 \simeq \pi_0^*(\omega_{X_0/S_0}^1)$ (the case will present itself in particular when X_0 is an S_0 -group, cf. II, 4.11). If one defines a functor L'_X over S by the formula

$$(0.4.1) \quad \mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}, \mathcal{J} \mathcal{O}_Y),$$

one then has $\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_S(Y, L'_X)$ for every X^+ -scheme Y , i.e.

$$L_X = L'_X \times_S X^+.$$

⁽²¹⁾Then, since $L_X \times_{X^+} X = L'_X \times_S X$, the operation of L_X on X induces an operation of L'_X on X , and this operation respects the morphism $p_X: X \rightarrow X^+$; indeed, if Y is an S -scheme, $h: Y \rightarrow X$ an S -morphism and m an element of $L'_X(Y)$, then h and $m \cdot h$ have the same restriction to $Y_{\mathcal{J}}$, i.e. $p_X(m \cdot h) = p_X(h)$.

Remark 0.5. Let us keep the hypotheses and notations of 0.4 and suppose moreover that Y is an S -scheme *flat* over S . One then has

$$\mathrm{Hom}_{X^+}(Y, L_X) = \mathrm{Hom}_S(Y, L'_X) = \mathrm{Hom}_{S_0}(Y_0, L_{0X}),$$

where the S_0 -functor in abelian groups L_{0X} is defined by the following identity (with respect to the variable S_0 -scheme T):

$$(0.5.1) \quad \mathrm{Hom}_{S_0}(T, L_{0X}) = \mathrm{Hom}_{\mathcal{O}_T}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T).$$

In the notations of II, 1, one has therefore shown that the functors L'_X and $\prod_{S_0/S} L_{0X}$ have the same restriction to the full subcategory of $(\mathbf{Sch})_S$ whose objects are the S -schemes Y *flat* over S .

Remark 0.6. Let us keep the hypotheses and notations of 0.5⁽²²⁾ and suppose moreover that there exists a section ε_0 of $\pi_0: X_0 \rightarrow S_0$; one then has $\omega_{X_0/S_0}^1 \simeq \varepsilon_0^*(\Omega_{X_0/S_0}^1)$. First, one has (independently of the preceding hypothesis):

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \Gamma(T, \mathcal{H}\mathrm{om}_{\mathcal{O}_T}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T)).$$

Suppose now that ω_{X_0/S_0}^1 admits a finite presentation (cf. EGA 0_I, 5.2.5), which will in particular be the case if X_0 is *locally of finite presentation* over S_0 (cf. EGA IV₄, 16.4.22). Then, if T is *flat* over S_0 , it follows from EGA 0_I, 6.7.6 that

$$\mathcal{H}\mathrm{om}_{\mathcal{O}_T}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T) \simeq \mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T,$$

whence

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \Gamma(T, \mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T).$$

Introducing the notation $\mathbf{W}(\)$ of I, 4.6.1, one has therefore proved that for every S_0 -scheme T *flat* over S_0 , one has

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \mathrm{Hom}_{S_0}(T, \mathbf{W}(\mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}))).$$

In summary, if ω_{X_0/S_0}^1 *admits a finite presentation*, and if one restricts oneself to the category of S -schemes *flat* over S , one has

$$(0.6.1) \quad L'_X = \prod_{S_0/S} \mathbf{W}(\mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J})),$$

and $\mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J})$ is a *quasi-coherent* $\mathcal{O}_{S_0}$ -module, by EGA I, 9.1.1.

⁽²¹⁾**N.D.E.** One has spelled out what follows.

⁽²²⁾**N.D.E.** One has added the hypothesis which follows, implicit in the original.

Let us remark finally that if ω_{X_0/S_0}^1 is moreover locally free (of finite rank), for example if X_0 is *smooth* over S_0 (in which case it is automatically locally of finite presentation over S_0), one has

$$(0.6.2) \quad \mathcal{H}om_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) \simeq \underline{\mathrm{Lie}}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J},$$

where one denotes by abuse of language (X_0 not necessarily being an S_0 -group) $\underline{\mathrm{Lie}}(X_0/S_0)$ the dual of the $\mathcal{O}_{S_0}$ -module ω_{X_0/S_0}^1 .⁽²³⁾

Proposition 0.2 (and its proof) has two important corollaries.⁽²⁴⁾

Corollary 0.7. *Let X be an S -scheme.*

- a) *Every S -endomorphism of X inducing the identity on $X_{\mathcal{J}}$ is an automorphism.*
- b) *One has an exact sequence of groups:*

$$0 \longrightarrow \mathrm{Hom}_{\mathcal{O}_{X_0}}(\Omega_{X_0/S_0}^1, \mathcal{J}\mathcal{O}_X) \xrightarrow{i} \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_{\mathcal{J}}}(X_{\mathcal{J}}).$$

- c) *Moreover, if one lets $\mathrm{Aut}_S(X)$ operate on the first group by transport of structure, one has, for all $u \in \mathrm{Aut}_S(X)$ and $m \in \mathrm{Hom}_{\mathcal{O}_{X_0}}(\Omega_{X_0/S_0}^1, \mathcal{J}\mathcal{O}_X)$:*

$$i(um) = u i(m) u^{-1}.$$

PROOF. By 0.2 (i), $\mathrm{Hom}_{X^+}(X, X)$ is a principal homogeneous set under $\mathrm{Hom}_{X^+}(X, L_X)$, for it is certainly non-empty; indeed, it contains a marked point: the identity automorphism of X .⁽²⁵⁾ Consequently, the map $m \mapsto m \cdot \mathrm{id}_X$ induces a *bijection*

$$\mathrm{Hom}_{\mathcal{O}_{X_0}}(\Omega_{X_0/S_0}^1, \mathcal{J}\mathcal{O}_X) = L_X(X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X, X).$$

Let $m \in L_X(X)$ and let $f = m' \cdot \mathrm{id}_X$ be an element of $\mathrm{Hom}_{X^+}(X, X)$. Applying 0.2 (ii) to f , one obtains that:

$$f \circ (m \cdot \mathrm{id}_X) = L_f(X)(m) \cdot f = L_f(X)(m) \cdot (m' \cdot \mathrm{id}_X).$$

On the other hand, as f is an X^+ -endomorphism of X , one has $f_{\mathcal{J}} = \mathrm{id}_{X_{\mathcal{J}}}$ and hence $f_0 = \mathrm{id}_{X_0}$; as L_f depends only on f_0 (cf. N.D.E. (17) in 0.1.10), one therefore has $L_f(X)(m) = m$. Consequently, the equality above is rewritten:

$$(m' \cdot \mathrm{id}_X) \circ (m \cdot \mathrm{id}_X) = m \cdot (m' \cdot \mathrm{id}_X) = (m + m') \cdot \mathrm{id}_X.$$

This shows that the bijection $m \mapsto m \cdot \mathrm{id}_X$ transforms the group law of $\mathrm{Hom}_{X^+}(X, L_X)$ into the composition law of the X^+ -endomorphisms of X .

It follows first that every element of $\mathrm{Hom}_{X^+}(X, X)$ is invertible, which is the first assertion of the statement, and then that one has an exact sequence

$$0 \longrightarrow \mathrm{Hom}_{X^+}(X, L_X) \xrightarrow{i} \mathrm{Aut}_S(X) \longrightarrow \mathrm{Aut}_{S_{\mathcal{J}}}(X_{\mathcal{J}}),$$

which is the second.

Let us remark now that the morphism i defined above is functorial in X for isomorphisms, for it is defined in structural terms from the operation of L_X on X over X^+ , itself functorial in X by assertion (ii) of proposition 0.2.⁽²⁶⁾ Therefore every automorphism u of X over S induces by transport of structure isomorphisms

$$h: \mathrm{Hom}_{X^+}(X, L_X) \xrightarrow{\sim} \mathrm{Hom}_{X^+}(X, L_X)$$

⁽²³⁾**N.D.E.** This is justified by II, 3.3 and 4.11. Indeed, the functor $L_{X_0/S_0}^{\varepsilon_0}$, “tangent space to X_0 over S_0 at the point ε_0 ”, is represented by the vector fibration $\mathbf{V}(\varepsilon_0^*(\Omega_{X_0/S_0}^1)) = \mathbf{V}(\omega_{X_0/S_0}^1)$ over S_0 , whose sheaf of sections is the dual module $\mathcal{H}om_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{O}_{S_0})$; the latter is $\underline{\mathrm{Lie}}(X_0/S_0)$ when X_0 is an S_0 -group and ε_0 the unit section.

⁽²⁴⁾**N.D.E.** One has added to 0.7 the corollary 0.7.bis, which was used implicitly in the proof of 0.8. Let us point out here that the numbers 0.8 to 0.12, as well as 0.17, which play an important technical role in the sequel of this exposé, are consequences of 0.7 and 0.7.bis.

⁽²⁵⁾**N.D.E.** One has set out what follows in detail.

⁽²⁶⁾**N.D.E.** One has set out what follows in detail.

and $f: \text{Aut}_S(X) \xrightarrow{\sim} \text{Aut}_S(X)$ such that the following diagram is commutative:

$$\begin{array}{ccc} \text{Hom}_{X^+}(X, L_X) & \xrightarrow{i} & \text{Aut}_S(X) \\ h \downarrow & & \downarrow f \\ \text{Hom}_{X^+}(X, L_X) & \xrightarrow{i} & \text{Aut}_S(X) \end{array}$$

i.e. such that $f \circ i = i \circ h$. On the other hand, f is given by the commutative diagram:

$$\begin{array}{ccc} X & \xrightarrow{a} & X \\ u \downarrow & & \downarrow u \\ X & \xrightarrow{f(a)} & X, \end{array}$$

that is, $f(a) = u \circ a \circ u^{-1}$, for every $a \in \text{Aut}_S(X)$. Writing $i(h(m)) = f(i(m))$, one finds the desired formula. $\square$

Corollary 0.7.bis. *Let X be an S -scheme such that $X_{\mathcal{J}}$ is an $S_{\mathcal{J}}$ -monoid. Then L_X is equipped with a structure of S -monoid, one has a split exact sequence of S -monoids:*

$$1 \longrightarrow L'_X \xrightarrow{i} L_X \xrightleftharpoons[p]{p} X^+ \longrightarrow 1$$

and the monoid law induced on L'_X coincides with its structure of abelian group. In particular, if $X_{\mathcal{J}}$ is an $S_{\mathcal{J}}$ -group, then L_X is an S -group and is the semidirect product of X^+ and of L'_X .

PROOF. Indeed, as $X_{\mathcal{J}}$ is an $S_{\mathcal{J}}$ -monoid, then $X^+ = \prod_{S_{\mathcal{J}}/S} X_{\mathcal{J}}$ is an S -monoid (indeed, one has $X^+(Y) = X_{\mathcal{J}}(Y_{\mathcal{J}})$ for every $Y \rightarrow S$). For every S -scheme Y , let us denote by $\tilde{Y}_{\mathcal{J}}$ the affine $Y_{\mathcal{J}}$ -scheme corresponding to the quasi-coherent $\mathcal{O}_{Y_{\mathcal{J}}}$ -algebra $\mathcal{O}_{Y_{\mathcal{J}}} \oplus \mathcal{J}\mathcal{O}_Y$ (i.e. the graded algebra associated with the filtration $\mathcal{O}_Y \supset \mathcal{J}\mathcal{O}_Y$). Then $L_X(Y)$ is identified with $X_{\mathcal{J}}(\tilde{Y}_{\mathcal{J}})$ and $L'_X(Y)$ with the kernel of the morphism $p: X_{\mathcal{J}}(\tilde{Y}_{\mathcal{J}}) \rightarrow X_{\mathcal{J}}(Y_{\mathcal{J}})$ induced by the “zero section” $Y_{\mathcal{J}} \rightarrow \tilde{Y}_{\mathcal{J}}$ (i.e. by the morphism of $\mathcal{O}_{Y_{\mathcal{J}}}$ -algebras $\mathcal{O}_{\tilde{Y}_{\mathcal{J}}} \rightarrow \mathcal{O}_{Y_{\mathcal{J}}}$ vanishing on the ideal $\mathcal{J}\mathcal{O}_Y$). One therefore has, for every $Y \rightarrow S$, a split exact sequence of monoids, functorial in Y :

$$1 \longrightarrow L'_X(Y) \xrightarrow{i} L_X(Y) \xrightleftharpoons[p]{p} X^+(Y) \longrightarrow 1.$$

It remains to see that the monoid law induced on L'_X coincides with its structure of abelian group. Let us denote by μ the monoid law of L_X and by e its unit section; one must show that for all $m, m' \in L'_X(Y)$, one has

$$\mu(m \cdot e, m' \cdot e) = (m + m') \cdot e.$$

This can be seen in one or the other of the following ways. On the one hand, one can take up again the proof of the equality (0.1.10 (*)) upon replacing the morphism $f: X \rightarrow W$ which figures there by the morphism $\mu: L_X \times_S L_X \rightarrow L_X$. Identifying $X^+(Y) = X_{\mathcal{J}}(Y_{\mathcal{J}})$ with its image by s in $L_X(Y) = X_{\mathcal{J}}(\tilde{Y}_{\mathcal{J}})$ one obtains that, for all $g, g' \in X_{\mathcal{J}}(Y_{\mathcal{J}})$ and $m, m' \in L'_X(Y)$, one has

$$(\star) \quad \mu(m \cdot g, m' \cdot g') = L_{\mu}^{(g, g')}(m, m') \cdot \mu(g, g'),$$

where $L_{\mu}^{(g, g')}$ denotes the derived morphism of μ at the point (g, g') (i.e. $\tilde{Y}_{\mathcal{J}}$ lies over $L_X \times_S L_X$ via (g, g')). In particular, one has $\mu(m \cdot e, m' \cdot e) = L_{\mu}^{(e, e)}(m, m') \cdot e$; but $L_{\mu}^{(e, e)}(m, m') = L_{\ell_e}(m') + L_{r_e}(m)$, where ℓ_e (resp. r_e) denotes the left translation (resp. the right translation) by e , which is the identity map of $X_{\mathcal{J}}$, whence $L_{\mu}^{(e, e)}(m, m') = m + m'$.

Or else, one can proceed as follows (cf. the proof of [DG70], § II.4, Th. 3.5). By lemma 0.1.11, the formation of X^+ and of L_X “commutes with the product” and the same therefore holds for L'_X ; it follows that the morphism $\mu': L'_X \times L'_X \rightarrow L'_X$, induced by μ , is

a homomorphism for the structure of abelian groups. One then deduces from lemma 3.10 of Exp. II that μ' coincides with the abelian group law. $\square$

0.8. — ⁽²⁷⁾ Suppose now that X is an S -scheme such that $X_{\mathcal{J}}$ is an $S_{\mathcal{J}}$ -group. Suppose that there exists an S -morphism

$$P: X \times_S X \longrightarrow X$$

such that the morphism obtained by base change

$$P_{\mathcal{J}}: X_{\mathcal{J}} \times_{S_{\mathcal{J}}} X_{\mathcal{J}} \longrightarrow X_{\mathcal{J}}$$

is the group law of $X_{\mathcal{J}}$. (An important particular case of the preceding situation will be the case where X is an S -group and where one takes for P its group law.) One deduces from this a morphism

$$L_P: L_X \times_S L_X \simeq L_{X \times_S X} \longrightarrow L_X$$

which, in fact, does not depend on P , for it is computed by means of the group law $P_{\mathcal{J}}$ of $X_{\mathcal{J}}$ as we are going to see now.⁽²⁸⁾ Indeed, by (ii) and (iii) of 0.2, for every $Y \longrightarrow S$ and $x, x' \in X(Y)$, $m, m' \in L'_X(Y)$, one has

$$P(m \cdot x, m' \cdot x') = P((m, m') \cdot (x, x')) = L_P^{(x, x')}(m, m') \cdot \mu(g, g')$$

where g (resp. g') is the image of x (resp. x') in $X^+(Y)$. Moreover (cf. the proof of 0.10), $L_P^{(x, x')}$ equals $L_{\mu}^{(g, g')}$ and, by 0.7.bis ($\star$), the latter is the element of $L'_X(Y)$ defined by the following equality in $L_X(Y)$:

$$L_{\mu}^{(g, g')}(m, m') \cdot \mu(g, g') = \mu(m \cdot g, m' \cdot g'),$$

that is, if one denotes by $\times$ (instead of μ) the group law of L_X and by Ad “the adjoint operation” of X^+ on L'_X (which factors through X_0 and which is induced by the adjoint operation of X_0 on ω_{X_0/S_0}^1), one obtains that

$$L_{\mu}^{(g, g')}(m, m') \times g \times g' = m \times g \times m' \times g' = (m \times \text{Ad}(g)(m')) \times g \times g'$$

whence finally $L_P^{(x, x')}(m, m') = m \times \text{Ad}(g)(m')$. One therefore obtains the following:

Proposition 0.8. *Let $P: X \times_S X \longrightarrow X$ be an S -morphism such that $P_{\mathcal{J}}$ equips $X_{\mathcal{J}}$ with a structure of $S_{\mathcal{J}}$ -group. Let us denote by $\times$ the group law of L'_X and by $(m, x) \mapsto m \cdot x$ the morphism $L'_X \times_S X \longrightarrow X$ defining the action of L'_X on X , and let $\text{Ad}: X^+ \longrightarrow \text{Aut}_{S\text{-gr.}}(L'_X)$ be “the adjoint operation” of X^+ on L'_X (which is induced by the adjoint operation of X_0 on ω_{X_0/S_0}^1). Then, for every $S' \longrightarrow S$ and $x, x' \in X(S')$, $m, m' \in L'_X(S')$, one has:*

$$(0.8.1) \quad P(m \cdot x, m' \cdot x') = (m \times \text{Ad } p_X(x)(m')) \cdot P(x, x').$$

If X is an S -group, one will denote by $*$ its law, by e its unit section, and by i the S -morphism defined by:

$$i(m) = m \cdot e,$$

for every $S' \longrightarrow S$ and $m \in L'_X(S')$.

Corollary 0.9. *Let X be an S -group. Then X^+ is equipped naturally with a structure of S -group, and p_X is a morphism of S -groups. Moreover, the S -morphism*

$$i: L'_X \longrightarrow X, \quad m \mapsto m \cdot e$$

is an isomorphism of S -groups of L'_X onto $\text{Ker}(p_X)$, and one has, for all $S' \longrightarrow S$, $x' \in X(S')$, $m \in L'_X(S')$:

$$(0.9.1) \quad m \cdot x' = (m \cdot e) * x' = i(m) * x'.$$

⁽²⁷⁾**N.D.E.** One has added here the number 0.8 in order to mark the return to the original.

⁽²⁸⁾**N.D.E.** One has set out the original in detail in what follows.

The first two assertions have already been proved in 0.1.2. As X is formally principal homogeneous over X^+ under $L_X = L'_X \times_S X^+$, the morphism i is indeed an isomorphism of S -functors of L'_X onto the kernel of p_X . The fact that i is a morphism of groups and the formula (0.9.1) result from the formula (0.8.1) applied respectively to $x = x' = e$, and to $x = e, m' = 1$.

Corollary 0.10. *Let X be an S -group. With the preceding notations, for every $S' \rightarrow S$ and all $x \in X(S')$ and $m' \in L'_X(S')$, one has*

$$(0.10.1) \quad x * i(m') * x^{-1} = i(\text{Ad } p_X(x)(m')).$$

This results from the equality $i(m') * x^{-1} = m' \cdot x^{-1}$ and from (0.8.1) applied to $m = 1$ and $x' = x^{-1}$.

When X is an S -group, we have therefore determined explicitly the kernel of $X \rightarrow X^+$ and the operation of the inner automorphisms of X on this kernel. We are now going to see that one can do likewise for certain S -functors in groups, not necessarily representable. One case will be useful to us, that of the functors Aut (I 1.7). Let us state at once:

Proposition 0.11. *Let E be an S -scheme. Set $X = \text{Aut}_S(E)$. The kernel of the morphism of S -functors in groups*

$$p_X: X \rightarrow X^+$$

is identified canonically with the commutative S -functor in groups L'_X defined by

$$\text{Hom}_S(Y, L'_X) = \text{Hom}_{\mathcal{O}_{E_0 \times_{S_0} Y_0}}(\Omega_{E_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}, \mathcal{J}\mathcal{O}_{E \times_S Y}),$$

where Y denotes a variable S -scheme.

Indeed, if Y is a variable S -scheme, one has $\text{Hom}_S(Y, X) = \text{Aut}_Y(E \times_S Y)$, and

$$\text{Hom}_S(Y, X^+) = \text{Hom}_S(Y_{\mathcal{J}}, X) = \text{Aut}_{Y_{\mathcal{J}}}(E \times_S Y_{\mathcal{J}}) = \text{Aut}_{Y_{\mathcal{J}}}((E \times_S Y) \times_Y Y_{\mathcal{J}}).$$

By applying 0.7 b) to the Y -scheme $E \times_S Y$, one obtains an isomorphism of groups:

$$\text{Hom}_S(Y, L'_X) \simeq \text{Ker}(\text{Hom}_S(Y, X) \rightarrow \text{Hom}_S(Y, X^+)),$$

an isomorphism which one easily verifies to be functorial in the S -scheme Y . One therefore obtains an isomorphism of S -groups

$$L'_X \simeq \text{Ker}(X \rightarrow X^+),$$

which completes the proof of proposition 0.11.

Corollary 0.12. ⁽²⁹⁾ *One keeps the notations of 0.11: E is an S -scheme and $X = \text{Aut}_S(E)$. One has a natural operation f of X on L'_X defined in the following manner. For every S -scheme Y , one has*

$$\text{Hom}_S(Y, X) = \text{Aut}_Y(E \times_S Y)$$

and

$$\text{Hom}_S(Y, L'_X) = \text{Hom}_{\mathcal{O}_{E_0 \times_{S_0} Y_0}}(\Omega_{E_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0/Y_0}, \mathcal{J}\mathcal{O}_{E \times_S Y})$$

(N.B. $\Omega_{E_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0} \simeq \Omega_{E_0 \times_{S_0} Y_0/Y_0}^1$, cf. EGA IV₄, 16.4.5); the first group operates on the second by transport of structure and this operation is indeed functorial in Y . One then has the formula:

$$(0.12.1) \quad x i(m) x^{-1} = i(f(x)m),$$

for every $Y \rightarrow S$ and all $x \in \text{Hom}_S(Y, X)$, $m \in \text{Hom}_S(Y, L'_X)$.

Indeed, this results from 0.7 c) applied to the Y -scheme $E \times_S Y$.

⁽²⁹⁾**N.D.E.** One has changed “Remark” into “Corollary”.

Recall 0.13. The direct image of a quasi-coherent module by a morphism of finite presentation is quasi-coherent. Under the same conditions, the formation of the direct image commutes with *flat* base change: in the situation

$$\begin{array}{ccc} T & \xleftarrow{g'} & T' = T \times_S S' \\ f \downarrow & & \downarrow f' \\ S & \xleftarrow{g} & S', \end{array}$$

if one supposes f (and hence f') of finite presentation and g (and hence g') *flat*, one has for every quasi-coherent $\mathcal{O}_T$ -module $\mathcal{F}$

$$f_*(\mathcal{F}) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} = f'_*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}),$$

or, in a more aesthetic manner

$$g^*(f_*(\mathcal{F})) = f'_*(g'^*(\mathcal{F})).$$

These two facts are more generally valid for a morphism f quasi-compact and quasi-separated, cf. EGA I, 9.2.1 and EGA III₁, 1.4.15 in the quasi-compact separated case (taking into account EGA III₂, Err_{III} 25) and EGA IV₁, 1.7.4 and 1.7.21.

Remark 0.14. ⁽³⁰⁾ Let us take up again the notations of 0.11: let E be an S -scheme, $X = \text{Aut}_S(E)$ and L'_X the commutative S -functor in groups defined by:

$$\begin{aligned} L'_X(Y) &= \text{Hom}_{\mathcal{O}_{E_0 \times_{S_0} Y_0}}(\Omega_{E_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}, \mathcal{JO}_{E \times_S Y}) \\ &= \text{Hom}_{\mathcal{O}_{E \times_S Y}}(\Omega_{E/S}^1 \otimes_{\mathcal{O}_S} \mathcal{O}_Y, \mathcal{JO}_{E \times_S Y}) \\ &= \Gamma(E \times_S Y, \mathcal{H}om_{\mathcal{O}_{E \times_S Y}}(\Omega_{E/S}^1 \otimes_{\mathcal{O}_S} \mathcal{O}_Y, \mathcal{JO}_{E \times_S Y})). \end{aligned}$$

Suppose Y is flat over S ; then one has isomorphisms:

$$\mathcal{JO}_{E \times_S Y} \xleftarrow{\sim} (J\mathcal{O}_E) \otimes_{\mathcal{O}_S} \mathcal{O}_Y \xrightarrow{\sim} (J\mathcal{O}_E) \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}.$$

Suppose moreover that E is of finite presentation over S ; then $\Omega_{E/S}^1$ is an $\mathcal{O}_E$ -module of finite presentation (cf. EGA IV₄, 16.4.22) and hence, by EGA 0_I, 6.7.6, one has:

$$\mathcal{H}om_{\mathcal{O}_{E \times_S Y}}(\Omega_{E/S}^1 \otimes_{\mathcal{O}_S} \mathcal{O}_Y, (J\mathcal{O}_E) \otimes_{\mathcal{O}_S} \mathcal{O}_Y) \simeq \mathcal{H}om_{\mathcal{O}_E}(\Omega_{E/S}^1, J\mathcal{O}_E) \otimes_{\mathcal{O}_S} \mathcal{O}_Y.$$

Let us denote by $\pi: E \rightarrow S$ and $g: Y \rightarrow S$ the structural morphisms; applying 0.13 to the diagram

$$\begin{array}{ccc} E & \xleftarrow{g'} & E \times_S Y \\ \pi \downarrow & & \downarrow \pi' \\ S & \xleftarrow{g} & Y, \end{array}$$

and to the $\mathcal{O}_E$ -module $\mathcal{F} = \mathcal{H}om_{\mathcal{O}_E}(\Omega_{E/S}^1, J\mathcal{O}_E)$, one obtains

$$\Gamma(E \times_S Y, g'^*\mathcal{F}) = \Gamma(Y, \pi'_*g'^*\mathcal{F}) = \Gamma(Y, g^*\pi_*\mathcal{F}) = W(\pi_*\mathcal{F})(Y).$$

One has therefore shown that, if E is of finite presentation over S , one has

$$(0.14.1) \quad L'_X = W(\pi_*\mathcal{H}om_{\mathcal{O}_E}(\Omega_{E/S}^1, J\mathcal{O}_E))$$

on the category of S -schemes *flat* over S . Let us note moreover that the module of which one takes the W is *quasi-coherent*, by EGA I, 9.1.1 and 9.2.1.

⁽³¹⁾Let us denote by L_0 the S_0 -functor

$$W(\pi_{0*}\mathcal{H}om_{\mathcal{O}_{E_0}}(\Omega_{E_0/S_0}^1, J\mathcal{O}_{E_0})).$$

⁽³⁰⁾**N.D.E.** One has set out the first part of this remark in detail, and one has added a second part.

⁽³¹⁾**N.D.E.** One has added what follows.

Then, returning to the definition of $L'_X(Y)$ and taking into account the isomorphism

$$J\mathcal{O}_{E \times_S Y} \simeq (J\mathcal{O}_E) \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0},$$

one obtains, by reasoning as above, that

$$L'_X(Y) = L_0(Y_0) = L_0(Y \times_S S_0) = \left(\prod_{S_0/S} L_0 \right)(Y).$$

Thus, on the category of S -schemes *flat* over S , one has:

$$L'_X = \prod_{S_0/S} W(\pi_{0*} \mathcal{H}om_{\mathcal{O}_{E_0}}(\Omega_{E_0/S_0}^1, J\mathcal{O}_{E_0})).$$

It is not evident that the action of X on L'_X defined in 0.12 comes from an action of $X_0 = \text{Aut}_{S_0}(E_0)$ on L_0 ; this is however the case when, moreover, E is *flat* over S .

Indeed, one has in this case canonical isomorphisms:

$$\begin{aligned} J\mathcal{O}_E &\simeq J \otimes_{\mathcal{O}_S} \mathcal{O}_E \simeq J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{E_0}. \\ L_0 &\simeq W(\pi_{0*} \mathcal{H}om_{\mathcal{O}_{E_0}}(\Omega_{E_0/S_0}^1, J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{E_0})), \end{aligned}$$

$$(0.14.2) \quad L'_X = \prod_{S_0/S} W(\pi_{0*} \mathcal{H}om_{\mathcal{O}_{E_0}}(\Omega_{E_0/S_0}^1, J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{E_0})).$$

Then, for every S_0 -scheme T , one has

$$L_0(T) \simeq \text{Hom}_{\mathcal{O}_{E_0 \times_{S_0} T}}(\Omega_{E_0 \times_{S_0} T/T}^1, J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{E_0 \times_{S_0} T})$$

and $\text{Hom}_{S_0}(T, X_0) = \text{Aut}_T(E_0 \times_{S_0} T)$ acts by transport of structure on $L_0(T)$, in a manner functorial in T ; finally, for every S -scheme Y flat over S , the action by transport of structure of $\text{Hom}_S(Y, X) = \text{Aut}_Y(E \times_S Y)$ on $L'_X(Y) = L_0(Y_0)$ factors through $\text{Aut}_{Y_0}(E_0 \times_{S_0} Y_0)$.

Let us finally extract from SGA 1 III the following two propositions.

Proposition 0.15. (SGA 1 III, 6.8)⁽³²⁾ *For every S_J -scheme Y smooth over S_J and affine, there exists an S -scheme X smooth over S such that $X \times_S S_J \simeq Y$, and such an X is unique up to (non-unique) isomorphism.*

Proposition 0.16. (SGA 1 III, 5.5)⁽³³⁾ *Let X be an S -scheme smooth over S . For every affine S -scheme Y , the canonical map*

$$p_X(Y): \text{Hom}_S(Y, X) \longrightarrow \text{Hom}_S(Y, X^+) = \text{Hom}_{S_J}(Y_J, X_J)$$

is surjective.

Corollary 0.17. *Let E be an S -scheme smooth over S and affine; let us write $X = \text{Aut}_S(E)$. For every affine S -scheme Y , the canonical map*

$$\text{Aut}_Y(E \times_S Y) = \text{Hom}_S(Y, X) \longrightarrow \text{Hom}_S(Y, X^+) = \text{Aut}_{Y_J}(E_J \times_{S_J} Y_J)$$

is surjective.

PROOF. Indeed, $Y \times_S E$ is affine over Y , itself affine, hence affine. Applying 0.16, one deduces from this that every S_J -morphism $Y_J \times_{S_J} E_J \longrightarrow E_J$ extends to an S -morphism $Y \times_S E \longrightarrow E$.⁽³⁴⁾ In other terms, every Y_J -endomorphism of $Y_J \times_{S_J} E_J$ lifts to a Y -endomorphism of $Y \times_S E$. Then, 0.7 a) shows that every Y_J -*automorphism* of $Y_J \times_{S_J} E_J$ lifts to a Y -*automorphism* of $Y \times_S E$, which is the announced property. $\square$

⁽³²⁾**N.D.E.** This “global” lifting result uses the “local” lifting result of *loc. cit.*, 4.1, which is stated for S locally noetherian; see EGA IV₄ 18.1.1 for the general case.

⁽³³⁾**N.D.E.** This is an immediate consequence of the definition of (formally) “smooth” adopted in EGA IV₄, 17.3.1 (and 17.1.1).

⁽³⁴⁾**N.D.E.** One has slightly modified the original, so as to be exactly under the hypotheses of 0.7.

1. Extensions and cohomology

1.1. — Let $\mathcal{C}$ be a category stable under fiber products.⁽³⁵⁾ Let S be an object of $\mathcal{C}$, G an S -group (*representable*) and F a commutative group S -functor on which G operates. One has defined in I, 5.1 the cohomology groups $H^n(G, F)$. One recalls that these are the homology groups of a complex denoted $C^*(G, F)$ where, writing $(G/S)^n = G \times_S \cdots \times_S G$ (n factors),

$$C^n(G, F) = \text{Hom}_S((G/S)^n, F).$$

As G and hence⁽³⁵⁾ the $(G/S)^n$ are representable, one also has

$$C^n(G, F) = F((G/S)^n);$$

from this, and from the definition of the boundary operator, one sees that the complex $C^*(G, F)$ depends only on the restriction of F to the full subcategory of $\mathcal{C}_S$ whose objects are the cartesian powers of G over S . Consequently, one has the

Lemma 1.1.1. *Let $\mathcal{C}$ be a category stable under fiber products⁽³⁵⁾, S an object of $\mathcal{C}$, G a representable S -group. Let us denote by $\mathcal{C}(G)$ the full subcategory of $\mathcal{C}_S$ whose objects are the cartesian powers of G over S . Let F and F' be two commutative group S -functors on which G operates. If F and F' have the same restriction to $\mathcal{C}(G)$, one has a canonical isomorphism*

$$H^*(G, F) \xrightarrow{\sim} H^*(G, F').$$

1.1.2. Cohomology and restriction of scalars. — ⁽³⁶⁾ Let us state another comparison result. Let now $T \rightarrow S$ be a morphism of $\mathcal{C}$. If F is a commutative group T -functor, then the functor obtained by “restriction of scalars” (cf. Exp. II, 1)

$$F_1 = \prod_{T/S} F$$

is a commutative group S -functor and one has a morphism of group S -functors

$$u: \prod_{T/S} \text{Aut}_{T\text{-gr.}}(F) \longrightarrow \text{Aut}_{S\text{-gr.}}(F_1).$$

⁽³⁷⁾ Let now G be a group S -functor and let

$$G_T \longrightarrow \text{Aut}_{T\text{-gr.}}(F)$$

be an operation of G_T on F . By definition of the functor $\prod_{T/S}$, one deduces from this a morphism of group S -functors

$$G \longrightarrow \prod_{T/S} \text{Aut}_{T\text{-gr.}}(F),$$

whence, by composition with u , an operation of G on $F_1 = \prod_{T/S} F$.⁽³⁸⁾

⁽³⁵⁾**N.D.E.** One has added the hypothesis that $\mathcal{C}$ be stable under fiber products (which is the case in the applications where $\mathcal{C}$ is the category of schemes (Sch) or that of the functors $(\text{Sch})^\circ \rightarrow (\text{Ens})$). If one omits this hypothesis, one must suppose in the sequel that the fiber products $G \times_S \cdots \times_S G$ are representable.

⁽³⁶⁾**N.D.E.** One has added the title of this paragraph.

⁽³⁷⁾**N.D.E.** Indeed, let $S' \rightarrow S$ and $\alpha \in \text{Aut}_{(T \times_S S')\text{-gr.}}(F_{T \times_S S'})$; for every $U \rightarrow T \times_S S'$, let us denote by $\alpha_U \in \text{Aut}_{\text{gr.}}(F_{T \times_S S'}(U))$ the element defined by α ; then $u(S')(\alpha)$ is the element β of $\text{Aut}_{S\text{-gr.}}(F_1)(S') = \text{Aut}_{S'\text{-gr.}}(\prod_{T \times_S S'/S'} F_{T \times_S S'})$ such that $\beta_{S''} = \alpha_{T \times_S S''}$ for every $S'' \rightarrow S'$.

⁽³⁸⁾**N.D.E.** Explicitly, for every $S' \rightarrow S$, the operation of $G(S')$ on $F_1(S') = F(T \times_S S')$ is given by the operation of $G_T(T \times_S S') = G(T \times_S S')$ on $F(T \times_S S')$ and the morphism of groups $G(S') \rightarrow G(T \times_S S')$ corresponding to the projection $T \times_S S' \rightarrow S'$. One may also say that the operation of G_T on F is given by a morphism $G_T \times_T F \rightarrow F$; applying the functor $\prod_{T/S}$ and denoting by G_1 the S -group $\prod_{T/S} G_T$, one obtains, taking into account II 1.2, a morphism $G_1 \times_S F_1 \rightarrow F_1$ which defines an operation of G_1 on F . The operation of G is then obtained via the natural morphism $G \rightarrow G_1$, which corresponds by adjunction to id_G .

Lemma 1.1.2. *Under the preceding conditions, one has a canonical isomorphism*

$$H^*\left(G, \prod_{T/S} F\right) \simeq H^*(G_T, F).$$

PROOF. Indeed, by the definition of the cohomology, the standard complexes are canonically isomorphic. $\square$

1.2. Lifting of morphisms of groups. — ⁽³⁹⁾ Following the general principles, one sets down the following definition:

Definition 1.2.1. Let $1 \rightarrow M \xrightarrow{u} E \xrightarrow{v} G$ be a sequence of morphisms of $\widehat{\mathcal{C}}$ -groups. One says that it is *exact* if the following equivalent conditions are verified:

(i) for every $S \in \text{Ob } \mathcal{C}$, the sequence of ordinary groups below is exact:

$$1 \rightarrow M(S) \xrightarrow{u(S)} E(S) \xrightarrow{v(S)} G(S)$$

(ii) for every object H of $\widehat{\mathcal{C}}$, the sequence of ordinary groups below is exact:

$$1 \rightarrow \text{Hom}(H, M) \xrightarrow{u(H)} \text{Hom}(H, E) \xrightarrow{v(H)} \text{Hom}(H, G)$$

Taking in particular $H = G$ in (ii), one sees that the set of sections of v (not respecting *a priori* the group structures) is empty or principal homogeneous under $\text{Hom}(G, M)$. Let us suppose it non-empty; let therefore

$$s: G \rightarrow E$$

be a section of v . Then for every $S \in \text{Ob } \mathcal{C}$ and every $x \in G(S)$, the element $s(x)$ of $E(S)$ defines an inner automorphism of E_S which normalizes M_S (more correctly the image of M_S under u_S), hence an automorphism of M_S .

Scholium 1.2.1.1. ⁽⁴⁰⁾ *If M is commutative, one sees “set-theoretically” that this automorphism does not depend on the chosen section, but only on x , and that it depends multiplicatively on it. In summary, to every exact sequence*

$$(E) \quad 1 \rightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

such that M is commutative and that v possesses a section, there is associated a morphism of $\widehat{\mathcal{C}}$ -groups

$$G \rightarrow \text{Aut}_{\widehat{\mathcal{C}}\text{-gr.}}(M)$$

which one calls the operation of G on M defined by the extension (E) .

Definition 1.2.1.2. One has seen in I, 2.3.7 that v possesses a section which is a morphism of $\widehat{\mathcal{C}}$ -groups if and only if the extension (E) is isomorphic (“as an extension”) to the semidirect product of M by G relative to the preceding operation. Such a section of v will be called a *section of the extension (E)* , or simply a *section of (E)* .

If s is a section of (E) and if $m \in \Gamma(M) \simeq \text{Ker}(\Gamma(E) \rightarrow \Gamma(G))$ (for the definition of Γ , see I, 1.2), then the morphism $G \rightarrow E$ defined by⁽⁴¹⁾

$$x \mapsto u(m) s(x) u(m)^{-1}$$

is likewise a section of (E) , said to be *deduced from s by the inner automorphism defined by m (or by $u(m)$)*.

⁽³⁹⁾**N.D.E.** One has added this title.

⁽⁴⁰⁾**N.D.E.** One has added the numbering 1.2.1.1 and 1.2.1.2 in order to bring out the notions which are introduced there.

⁽⁴¹⁾**N.D.E.** In the right-hand term, one has corrected x to $s(x)$.

Lemma 1.2.2. *Let $(E): 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$ be an exact sequence of $\widehat{\mathcal{C}}$ -groups such that M is commutative and that v possesses a section. Let us make G operate on M in the manner defined by (E) .*

- (i) *The extension (E) canonically defines a class $c(E) \in H^2(G, M)$ whose vanishing is necessary and sufficient for the existence of a section of (E) .*
- (ii) *If $c(E) = 0$, the set of sections of (E) is principal homogeneous under the group $Z^1(G, M)$, and the set of sections of (E) modulo the action of the inner automorphisms defined by the elements of $\Gamma(M)$ is principal homogeneous under the group $H^1(G, M)$.*
- (iii) ⁽⁴²⁾ *Let s be a section of (E) ; the set of the conjugates of s by the inner automorphisms defined by $\Gamma(M)$ is in bijection with $\Gamma(M)/H^0(G, M)$.*

PROOF. The proof is carried out exactly as in the case of ordinary groups, the fact that one starts from a section of v ensuring the functoriality of the set-theoretic computations. Let us briefly indicate the principal steps of the proof.

a) To every section s of v one associates the morphism

$$Ds: G \times G \longrightarrow M,$$

defined set-theoretically by

$$u(Ds(x, y)) = s(xy) s(y)^{-1} s(x)^{-1}.$$

One shows that Ds is a 2-cocycle by the following computation.⁽⁴³⁾ By the definition of the differential of the standard complex (I, 5.1), one has:

$$(\partial^2 Ds)(x, y, z) = (s(x) Ds(y, z) s(x)^{-1}) \cdot Ds(x, y)^{-1} \cdot Ds(xy, z)^{-1} \cdot Ds(x, yz);$$

it suffices to substitute the definition of Ds into this formula in order to find (without using any commutativity) $Ds \in Z^2(G, M)$.

b) If s and s' are two sections of v , there exists $f: G \longrightarrow M$ such that $s(x) = f(x)s'(x)$. One then has

$$\begin{aligned} Ds'(x, y) &= f^{-1}(xy) Ds(x, y) s(x) f(y) s(x)^{-1} f(x), \\ &= Ds(x, y) \cdot f^{-1}(xy) \cdot (s(x) f(y) s(x)^{-1}) \cdot f(x), \end{aligned}$$

that is,

$$Ds' = Ds \cdot \partial^1 f.$$

⁽⁴⁴⁾ This shows that the class of Ds in $H^2(G, M)$ does not depend on the chosen section s of v ; it is the class $c(E)$ of the extension (E) .

c) Let s and s' be two sections of v and let $m \in \Gamma(M)$. Then, the equality $s(x) = m^{-1}s'(x)m$ (for every $x \in G(S)$, $S \in \text{Ob } \mathcal{C}$) is equivalent to

$$s(x) = m^{-1}s'(x)m s'(x)^{-1}s'(x), \quad \text{i.e.} \quad s = \partial^0 m \cdot s'.$$

In particular, the stabilizer of s in $\Gamma(M)$ is the subgroup of those $m \in \Gamma(M)$ such that $\partial^0 m = e_M$, i.e. the subgroup $H^0(G, M)$. This already proves (iii).

d) The reasoning is now the usual one:⁽⁴⁵⁾ Let s_0 be an arbitrary section of v ; there exists a section s , necessarily of the form $s = f \cdot s_0$, which is a morphism of groups, i.e. which satisfies $Ds = 0$, if and only if $(Ds_0)^{-1} = \partial^1 f$, i.e. , if and only if the class $c(E)$ is zero. This proves (i).

In this case, the set of sections of (E) is formed of the sections $s' = h \cdot s$, where $h: G \longrightarrow M$ satisfies $\partial^1 h = 0$, i.e. $h \in Z^1(G, M)$. Moreover, by point c), two such sections $h_1 \cdot s$ and $h_2 \cdot s$

⁽⁴²⁾**N.D.E.** One has added this point, the counterpart of 1.2.4 (iii). Let us note on the other hand that the assertion is valid for *every* section of v , cf. the proof.

⁽⁴³⁾**N.D.E.** One has corrected the original, so as to make it compatible with I, 5.1.

⁽⁴⁴⁾**N.D.E.** One has added the sentence which follows.

⁽⁴⁵⁾**N.D.E.** One has set out in detail what follows.

are conjugate under $\Gamma(M)$ if and only if h_1 and h_2 have the same image in $H^1(G, M)$. This proves (ii). $\square$

Let still

$$(E) \quad 1 \longrightarrow M \xrightarrow{u} E \xrightarrow{v} G$$

be an exact sequence of $\widehat{\mathcal{C}}$ -groups with M commutative. Let

$$f: H \longrightarrow G$$

be a morphism of $\widehat{\mathcal{C}}$ -groups. Consider $E_f = H \times_G E$; it is a $\widehat{\mathcal{C}}$ -group and the projection $v_f: E_f \longrightarrow H$ is a morphism of $\widehat{\mathcal{C}}$ -groups. Likewise for $e_f: E_f \longrightarrow E$. On the other hand, if one sends M into E by u and into H by the unit morphism, one defines a morphism of $\widehat{\mathcal{C}}$ -groups $u_f: M \longrightarrow E_f$. One has therefore constructed a commutative diagram of $\widehat{\mathcal{C}}$ -groups:

$$\begin{array}{ccccccc} (E) & 1 & \longrightarrow & M & \xrightarrow{u} & E & \xrightarrow{v} & G \\ & & & \uparrow \text{id} & & \uparrow e_f & & \uparrow f \\ (E_f) & 1 & \longrightarrow & M & \xrightarrow{u_f} & E_f & \xrightarrow{v_f} & H. \end{array}$$

One has immediately:

Lemma 1.2.3.

- (i) *The sequence (E_f) is exact.*
- (ii) *The map $s \mapsto e_f \circ s = f'$ realizes a bijective correspondence between the morphisms*

$$s: H \longrightarrow E_f$$

such that $v_f \circ s = \text{id}$ (that is, the sections of v_f) and the morphisms

$$f': H \longrightarrow E$$

such that $v \circ f' = f$ (that is, the morphisms f' “lifting” f).

- (iii) *In the preceding correspondence, sections of (E_f) and morphisms of groups f' lifting f correspond.*

Applying lemma 1.2.2 to the extension (E_f) and taking 1.2.3 into account, one obtains the following proposition (which formally contains 1.2.2):

Proposition 1.2.4. *Let $(E): 1 \longrightarrow M \longrightarrow E \xrightarrow{v} G$ be an exact sequence of $\widehat{\mathcal{C}}$ -groups with M commutative. Let*

$$f: H \longrightarrow G$$

be a morphism of $\widehat{\mathcal{C}}$ -groups; suppose that it lifts to a morphism (not necessarily of groups) $f': H \longrightarrow E$. Let us make H operate on M by the composite morphism (multiplicative and independent of the choice of f'),

$$H \xrightarrow{f'} E \xrightarrow{\text{int}} \text{Aut}_{\widehat{\mathcal{C}}\text{-gr.}}(M).$$

- (i) *The morphism f defines canonically a class $c(f) \in H^2(H, M)$ whose vanishing is necessary and sufficient for the existence of a morphism of $\widehat{\mathcal{C}}$ -groups*

$$f': H \longrightarrow E$$

lifting f .

- (ii) *If $c(f) = 0$, the set of morphisms of $\widehat{\mathcal{C}}$ -groups f' lifting f , modulo the action of the inner automorphisms defined by the elements of $\Gamma(M)$ (i.e. by the elements m of $\Gamma(E)$ such that $v(m) = e$) is a principal homogeneous space under $H^1(H, M)$.*
- (iii) *If $f': H \longrightarrow E$ is a morphism of groups lifting f , the set of the transforms of f' by the inner automorphisms defined by the elements of $\Gamma(M)$ is isomorphic to $\Gamma(M)/\Gamma(M^H) = \Gamma(M)/H^0(H, M)$.*

1.3. Extensions of group laws. — Consider the following situation: one has a morphism of $\widehat{\mathcal{C}}$

$$(\dagger) \quad p: X \longrightarrow Y$$

and a commutative $\widehat{\mathcal{C}}$ -group M operating on X , such that X is formally principal homogeneous over Y under M_Y .

If $g: Y \longrightarrow Z$ is an arbitrary morphism of $\widehat{\mathcal{C}}$, then $g \circ p: X \longrightarrow Z$ is invariant under M : for each $S \in \text{Ob}(\mathcal{C})$, $(g \circ p)(S)$ is invariant under the action of $M(S)$ operating on $X(S)$. Conversely, we shall suppose the following condition verified for $n = 1, 2, 3, 4$.

$(+)_n$: Every morphism from X^n into M , invariant under the action of M^n operating on X^n , factors uniquely through $p^n: X^n \longrightarrow Y^n$ (where the powers n denote cartesian powers).

Lemma 1.3.1.

- (i) If h is a morphism from Y into M , the automorphism u_h of X defined set-theoretically by $x \mapsto h(p(x)) \cdot x$ preserves the fibers of p and commutes with the operations of M on X ,⁽⁴⁶⁾ i.e. for all $S \in \text{Ob}(\mathcal{C})$ and $x \in X(S)$, $m \in M(S)$, one has

$$p(h(p(x)) \cdot x) = p(x), \quad m \cdot h(p(x)) \cdot x = h(p(m \cdot x)) \cdot m \cdot x.$$

- (ii) This construction realizes a bijective correspondence between morphisms from Y into M and automorphisms of X preserving the fibers of p and commuting with the operations of M .

The first part is clear, since $p(m \cdot x) = p(x)$ and since M is commutative. Conversely, an automorphism u of X preserving the fibers of p is written set-theoretically $x \mapsto g(x)x$, where g is some morphism from X into M . If u commutes with the operations of M , the morphism g is invariant under $M^{(47)}$ and one concludes by condition $(+)_1$.

We now suppose that there are given in addition a group law on Y and an operation of Y on M , that is, a morphism of $\widehat{\mathcal{C}}$ -groups:

$$(\ddagger) \quad f: Y \longrightarrow \text{Aut}_{\widehat{\mathcal{C}\text{-gr.}}}(M).$$

Definition 1.3.2. A composition law on X

$$P: X \times X \longrightarrow X$$

is said to be *admissible* if it satisfies the following two conditions:

- (i) P lifts the group law of Y , i.e. the diagram

$$\begin{array}{ccc} X \times X & \xrightarrow{P} & X \\ (p,p) \downarrow & & \downarrow p \\ Y \times Y & \longrightarrow & Y \end{array}$$

is commutative.

- (ii) For every $S \in \text{Ob}(\mathcal{C})$ and all $x, y \in X(S)$, $m, n \in M(S)$, one has the following relation

$$(++) \quad P(m \cdot x, n \cdot y) = m \cdot f(p(x))(n) \cdot P(x, y).$$

Proposition 1.3.3. In order that a group law $*$ on X be admissible, it is necessary and sufficient that the following four conditions be satisfied:

- (i) $p: X \longrightarrow Y$ is a morphism of groups.
 (ii) The morphism $i: M \longrightarrow X$ defined by $i(m) = m \cdot e_X$ is an isomorphism of groups from M onto $\text{Ker}(p)$, that is: one has set-theoretically $(m \cdot e_X) * (n \cdot e_X) = (mn) \cdot e_X$.

⁽⁴⁶⁾**N.D.E.** Here and in the sequel, one has replaced “commutes with p and with the operations of M ” by “preserves the fibers of p and commutes with the operations of M ”. On the other hand, one has added the equalities which follow.

⁽⁴⁷⁾**N.D.E.** Indeed, the equality $mg(x)x = mu(x) = u(mx) = g(mx)mx$ implies $g(mx) = g(x)$.

- (iii) One has $m \cdot x = (m \cdot e_X) * x = i(m) * x$ for each $m \in M(S)$, $x \in X(S)$.
- (iv) The inner automorphisms of X operate on $\text{Ker}(p)$ according to the set-theoretic formula:

$$x * i(m) * x^{-1} = i(f(p(x))m).$$

The proof is immediate.

Lemma 1.3.4. ⁽⁴⁸⁾ Let h be a morphism $Y \longrightarrow M$ and u_h the automorphism $x \mapsto h(p(x)) \cdot x$ of X (cf. 1.3.1). Let P be an admissible composition law (resp. a group law) on X and let P' be the composition law on X deduced from P by means of u_h , i.e., $P'(x, y) = u_h^{-1}(P(u_h(x), u_h(y)))$. Then:

- (i) P' is an admissible composition law (resp. group law).
- (ii) For all $x, y \in X(S)$ ($S \in \text{Ob}(\mathcal{C})$), set $v = p(x)$ and $w = p(y)$; then

$$P'(x, y) = h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w)) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

PROOF. One has $u_h^{-1} = u_{h^\vee}$, where $h^\vee: Y \longrightarrow M$ is defined by $h^\vee(y) = h(y)^{-1}$. By 1.3.2 (i) and (ii), one has $P(h(v) \cdot x, h(w) \cdot y) = h(v) \cdot f(v)(h(w)) \cdot P(x, y)$ and $p(P(x, y)) = vw$, whence

$$P'(x, y) = h(vw)^{-1} \cdot h(v) \cdot f(v)(h(w)) \cdot P(x, y) = (\partial^1 h)(p(x), p(y)) \cdot P(x, y).$$

It is then immediate that P' satisfies conditions (i) and (ii) of 1.3.2. $\square$

Definition 1.3.5. Two admissible composition laws deduced from one another by the procedure of 1.3.4 are said to be *equivalent*. ⁽⁴⁹⁾

Proposition 1.3.6. Suppose that there exists an admissible composition law on X . Then:

- (i) There exists a class $c \in H^3(Y, M)$ (determined canonically), whose vanishing is necessary and sufficient for the existence of an associative admissible composition law on X .
- (ii) If $c = 0$, the set of associative admissible composition laws (resp. of equivalence classes of associative admissible composition laws) on X is a principal homogeneous space under $Z^2(G, M)$ (resp. $H^2(G, M)$).

The proof is carried out in several steps.

a) Let P be an admissible composition law on X . As P lifts the composition law of Y , which is associative, there exists a unique morphism $a: X^3 \longrightarrow M$ such that

$$(*) \quad P(x, P(y, z)) = a(x, y, z) P(P(x, y), z).$$

By applying conditions 1.3.2 (i) and (ii), one sees at once that a is invariant under the action of M^3 on X^3 , ⁽⁵⁰⁾ whence, by applying hypothesis $(+)_3$, the following result:

- (1) There exists a unique morphism $DP: Y^3 \longrightarrow M$ such that

$$P(x, P(y, z)) = DP(p(x), p(y), p(z)) P(P(x, y), z),$$

and P is associative if and only if $DP = 0$.

b) Let us compute step by step $P(P(P(x, y), z), t)$ by means of the preceding formula. Setting $p(x) = u$, $p(y) = v$, $p(z) = w$, $p(t) = h$, one obtains ⁽⁵¹⁾ the following pentagonal

⁽⁴⁸⁾**N.D.E.** One has set out the statement as well as its proof in detail, with a view to point (e) of the proof of 1.3.6.

⁽⁴⁹⁾**N.D.E.** This is indeed an equivalence relation, since $u_h^{-1} = u_{h^\vee}$.

⁽⁵⁰⁾**N.D.E.** For $x, y, z \in X(S)$, set $u = p(x)$, $v = p(y)$ and $w = p(z)$. Then $P(mx, P(m'y, m''z)) = P(mx, m'f(v)(m'')P(y, z)) = mf(u)(m')f(uv)(m'')P(x, P(y, z))$. On the other hand, as $p(P(x, y)) = uv$, one also has $P(P(mx, m'y), m''z) = P(mf(u)(m')P(x, y), m''z) = mf(u)(m')f(uv)(m'')P(P(x, y), z)$, and the comparison of these equalities with $(*)$ gives $a(mx, m'y, m''z) = a(x, y, z)$.

⁽⁵¹⁾**N.D.E.** One has added the diagram which follows.

diagram, where an arrow $a \xrightarrow{m} b$ means that $b = m \cdot a$:

$$\begin{array}{ccc}
 & P(x, P(y, P(z, t))) & \\
 DP(u, v, wh) \swarrow & & \searrow f(u)(DP(v, w, h)) \\
 P(P(x, y), P(z, t)) & & P(x, P(P(y, z), t)) \\
 DP(uv, w, h) \downarrow & & \downarrow DP(u, vw, h) \\
 P(P(P(x, y), z), t) & \xleftarrow{DP(u, v, w)} & P(P(x, P(y, z)), t)
 \end{array}$$

hence one finds

$$\begin{aligned}
 & DP(u, v, w) \cdot DP(u, vw, h) \cdot f(u)DP(v, w, h) \\
 & \cdot DP(u, v, wh)^{-1} \cdot DP(uv, w, h)^{-1} = e_M,
 \end{aligned}$$

i.e. , $\partial^3 DP(u, v, w, h) = e_M$. As on the other hand the left-hand side of the preceding formula can be written, with the aid of P and of a , as the expression in (x, y, z, t) of some morphism $X^4 \rightarrow M$, it follows from the uniqueness hypothesis in $(+)_4$ that $\partial^3 DP$ and e_M , which factor the same morphism, are equal, hence

(2) DP is a cocycle, i.e. one has $DP \in Z^3(Y, M)$.

c) If P and P' are two admissible composition laws on X , there exists a unique morphism

$$b: X^2 \rightarrow M$$

such that $P'(x, y) = b(x, y)P(x, y)$. Applying 1.3.2 (ii) to P and to P' , one sees that b is invariant under M^2 , whence, by $(+)_2$:

(3) For every pair of admissible composition laws (P, P') , there exists a unique

$$d(P, P'): Y^2 \rightarrow M$$

such that

$$P'(x, y) = d(P, P')(p(x), p(y)) P(x, y),$$

and the set of admissible composition laws thus becomes a principal homogeneous space under $\text{Hom}(Y^2, M) = C^2(Y, M)$.

d) Under the preceding conditions, one has the formula:

$$(4) \quad DP' - DP = \partial^2 d(P, P').$$

e) P and P' are equivalent if and only if there exists a morphism $h \in C^1(Y, M) = \text{Hom}(Y, M)$ such that $d(P, P') = \partial^1 h$; this follows from the definition of equivalence and from 1.3.4 (ii).

f) It remains only to conclude: one seeks a P' which is associative, i.e. such that $DP' = e_M$. Now DP is a cocycle whose class in $H^3(Y, M)$ does not depend on the admissible composition law P chosen (by (3) and (4)). This class is the obstruction c sought. One will be able to choose a P' satisfying the conditions if and only if $c = 0$; indeed, choosing an arbitrary P , one will have to solve, by (1):

$$0 = DP' = DP + \partial^2 d(P, P'),$$

which is possible by (3) and (4) if and only if $c = 0$. The set of associative P' is a principal homogeneous space under $Z^2(Y, M)$, again by (3) and (4). The set of associative P' up to equivalence is a principal homogeneous space under $H^2(Y, M)$ by (e).

2. Infinitesimal extensions of a morphism of group schemes

Let us resume the notations of $n^\circ 0$. Let Y and X be two group S -functors. Let M be the kernel of the morphism of groups $p_X: X \rightarrow X^+$. One therefore has an exact sequence of group S -functors

$$1 \rightarrow M \rightarrow X \xrightarrow{p_X} X^+.$$

By the definition of X^+ , one has isomorphisms

$$\begin{aligned}\mathrm{Hom}_S(Y, X^+) &\xrightarrow{\sim} \mathrm{Hom}_{S_J}(Y_J, X_J) \\ \mathrm{Hom}_{S\text{-gr.}}(Y, X^+) &\xrightarrow{\sim} \mathrm{Hom}_{S_J\text{-gr.}}(Y_J, X_J),\end{aligned}$$

and the morphism

$$\mathrm{Hom}_S(Y, p_X): \mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_S(Y, X^+)$$

associates with an S -morphism $f: Y \longrightarrow X$, the S -morphism $f^+: Y \longrightarrow X^+$ corresponding, via the preceding isomorphisms, to the S_J -morphism $f_J: Y_J \longrightarrow X_J$ obtained by base change from f . If M is *commutative*, one can apply proposition 1.2.4 to this situation.

2.0. — ⁽⁵²⁾ In the sequel, we shall be interested in the following case: Y is *flat* over S , and X is a group S -functor of one of the following two kinds:

- (a) X is an S -group scheme,
- (b) $X = \mathrm{Aut}_S(E)$ where E is an S -scheme, *of finite presentation over S* .

Let us denote by $(\mathbf{Plats})_S$ the category of S -schemes *flat* over S . In case (a) (resp. (b)), the group S -functor $M = \mathrm{Ker}(X \longrightarrow X^+)$, its restriction L to $(\mathbf{Plats})_S$, and the operations of the inner automorphisms of X on M , have been computed in 0.9, 0.5, and 0.10 (resp. 0.11, 0.14, and 0.12). That is, in case (a), let L_0 be the commutative group S_0 -functor defined by: for every S_0 -scheme T_0 ,

$$\mathrm{Hom}_{S_0}(T_0, L_0) = \mathrm{Hom}_{\mathcal{O}_{T_0}}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}, J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}),$$

on which X_0 operates via its adjoint representation in ω_{X_0/S_0}^1 ; then $L = \prod_{S_0/S} L_0$, i.e. for every S -scheme T , one has $L(T) = L_0(T \times_S S_0)$.

In case (b), let us denote by π the structural morphism $E \longrightarrow S$; then L is the abelian group functor on $(\mathbf{Plats})_S$ defined by

$$\mathrm{Hom}_S(T, L) = \Gamma(T, \pi_*(\mathcal{H}\mathrm{om}_{\mathcal{O}_E}(\Omega_{E/S}^1, J \mathcal{O}_E)) \otimes_{\mathcal{O}_S} \mathcal{O}_T),$$

on which X , considered as a functor on $(\mathbf{Plats})_S$, operates as one has seen in 0.12.

Then, one has an exact sequence of group functors on $(\mathbf{Plats})_S$:

$$(E) \quad 1 \longrightarrow L \longrightarrow X \longrightarrow X^+.$$

On the other hand, Y being supposed *flat* over S , the groups $H^i(Y, M)$ depend, by 1.1.1, only on the restriction L of M to $(\mathbf{Plats})_S$. As $L = \prod_{S_0/S} L_0$ in case (a), then by 1.1.2, one has in this case isomorphisms $H^i(Y, L) \simeq H^i(Y_0, L_0)$.

Then, taking account of what precedes, one deduces from proposition 1.2.4 the⁽⁵³⁾

Theorem 2.1. *Let S be a scheme, I and J two quasi-coherent ideals such that $I \supset J$ and $I \cdot J = 0$, defining the closed subschemes S_0 and S_J , and let:*

- X a group S -functor of type (a) or (b), and L_0, L as above;
- Y an S -group scheme *flat* over S and $f_J: Y_J \longrightarrow X_J$ a morphism of S_J -groups.

Then:

- (i) *In order that f_J lift to a morphism of S -groups $Y \longrightarrow X$, it is necessary and sufficient that the following two conditions be satisfied:*
 - (i₁) *f_J lifts to a morphism of S -functors $Y \longrightarrow X$ (by 1.2.4, this defines an operation of Y on L , which does not depend on the chosen lifting; moreover, in case (a), the operation of Y_0 on L_0 thus obtained comes from the morphism $f_0: Y_0 \longrightarrow X_0$ and from the “adjoint action” of X_0 on L_0);*

⁽⁵²⁾**N.D.E.** One has added the numbering 2.0, for later references.

⁽⁵³⁾**N.D.E.** One has placed higher up the definitions which in the original figured in the statement of theorem 2.1 and formed the essential part of page 113 of the original.

- (i₂) A certain obstruction $c(f_J)$, defined canonically by f_J , is zero, where $c(f_J)$ is a class in $H^2(Y, L) (\simeq H^2(Y_0, L_0)$ in case (a)).
- (ii) If the conditions of (i) are satisfied, the set E of morphisms of group S -functors $Y \rightarrow X$ extending f_J is a principal homogeneous space under $Z^1(Y, L) (\simeq Z^1(Y_0, L_0)$ in case (a)), and E modulo the action of the inner automorphisms of X defined by the sections of X over S inducing the unit section of X_J over S_J , is a principal homogeneous space under $H^1(Y, L) (\simeq H^1(Y_0, L_0)$ in case (a)).
- (iii) If $f: Y \rightarrow X$ is a morphism of group S -functors extending f_J , the set of the morphisms $Y \rightarrow X$ transforms of f by the inner automorphisms defined by the sections of X over S inducing the unit section of X_J over S_J , is isomorphic to $\Gamma(L)/H^0(Y, L) (\simeq \Gamma(L_0)/H^0(Y_0, L_0)$ in case (a)).

Remark 2.1.1. ⁽⁵⁴⁾ If $f, f': Y \rightarrow X$ are morphisms of group S -functors extending f_J , one therefore obtains a cocycle $d(f, f') \in Z^1(Y, L) (\simeq Z^1(Y_0, L_0)$ in case (a)), such that

$$(*) \quad f' = d(f, f') \cdot f.$$

⁽⁵⁵⁾ One will denote by $d(f, f')$ the image of $d(f, f')$ in $H^1(Y, L) (\simeq H^1(Y_0, L_0)$ in case (a)).

Remark 2.2. One keeps the preceding notations; in particular, Y is flat over S . In case (b), L is, by (0.14.1), the restriction to $(\mathbf{Plats})_S$ of the functor

$$\mathbf{W}(\pi_*(\mathcal{H}\text{om}_{\mathcal{O}_E}(\Omega_{E/S}^1, J\mathcal{O}_E))),$$

where $\pi: E \rightarrow S$ is the structural morphism. In case (a), suppose moreover that X is locally of finite presentation over S ; then by (0.6.1), L is the restriction to $(\mathbf{Plats})_S$ of the functor

$$\coprod_{S_0/S} \mathbf{W}(\mathcal{H}\text{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, J)).$$

In both cases, the module of which one takes the $\mathbf{W}$ is quasi-coherent, by EGA I, 9.1.1. Suppose moreover that Y is affine over S ⁽⁵⁶⁾. Then, by I, 5.3, one obtains:

- (a) $H^i(Y, L) = H^i(Y_0, L_0) = H^i(Y_0, \mathcal{H}\text{om}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, J)),$
 (b) $H^i(Y, L) = H^i(Y, \pi_*(\mathcal{H}\text{om}_{\mathcal{O}_E}(\Omega_{E/S}^1, J\mathcal{O}_E))).$

Remark 2.3. 1) By 0.16 and 0.17, condition (i₁) is automatically satisfied when Y is an affine scheme and

$$(*) \quad \begin{cases} \text{in case (a), } X \text{ is smooth over } S; \\ \text{in case (b), } E \text{ is smooth and affine over } S. \end{cases}$$

2) Moreover, under these conditions (Y being still supposed flat over S , cf. 2.0), one can write in case (a), by 2.2 a) and (0.6.2),

$$H^i(Y, L) = H^i(Y_0, L_0) = H^i(Y_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} J),$$

⁽⁵⁷⁾ and in case (b), by (0.14.2), 1.1.2 and I, 5.3,

$$H^i(Y, L) = H^i(Y_0, \pi_{0*} \mathcal{H}\text{om}_{\mathcal{O}_{E_0}}(\Omega_{E_0/S_0}^1, J \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{E_0})).$$

⁽⁵⁴⁾**N.D.E.** One has added this remark, analogous to 4.5.1, in order to introduce the notations $d(f, f')$ and $d(f, f')$, used in 4.38; consequently, one has also added in 2.1 (ii) above, the part concerning E itself.

⁽⁵⁵⁾**N.D.E.** One has conformed to the sign conventions of the original. One could have chosen to write $f' = d(f', f) \cdot f$, but then, in order to have in 4.27 the equality $d^1(d(i, i')) = d(Y, i'(Y))$ when i, i' are two immersions $Y \hookrightarrow X$, one would have had to change the sign of the class $d(Y, Y')$ introduced in 4.5.1, and that would have led to changes of sign in the formulas of 4.8, 4.14, 4.17. One has preferred to keep the signs given in the original (all correct!).

⁽⁵⁶⁾**N.D.E.** One has added this hypothesis, as well as the reference to I, 5.3.

⁽⁵⁷⁾**N.D.E.** One has added what follows, cf. the N.D.E. (31) in 0.14.

Let us now state a certain number of corollaries concerning the case where Y is a diagonalizable S -group (I, 4.4); one then knows (*loc. cit.* 5.3.3) that if S is affine, $H^n(Y, \mathcal{F}) = 0$ for $n > 0$ and every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{F}$. First a particular case:

Corollary 2.4. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal. Let Y be a diagonalizable S -group and let:*

- (a) *X an S -group locally of finite presentation over S ,*
- (b) *$X = \text{Aut}_S(E)$ where E is an S -scheme locally of finite presentation.*

Let $f: Y \rightarrow X$ be a morphism of S -groups such that the morphism $f_0: Y_0 \rightarrow X_0$ obtained by base change is the unit morphism. Then f is the unit morphism.

Indeed, the question is local on S and (in (b)) on E . One may therefore suppose S affine and (in (b)) E of finite presentation over S . Introducing now the closed subschemes S_n of S defined by the powers of the ideal defining S_0 , one is reduced to the case where S_0 is defined by an ideal of square zero, and in this case the stated assertion follows from the theorem, via 2.2.

In the case where one does not necessarily suppose that f_0 is the unit morphism, one has:

Corollary 2.5. *Let S and S_0 be as in 2.4. Suppose moreover S affine. Let Y be a diagonalizable S -group, X a group S -functor and $f_0: Y_0 \rightarrow X_0$ a morphism of group S_0 -functors.*

- (i) ⁽⁵⁸⁾
- (ii) *Suppose that one has one of the following two properties:*
 - (a) *X is an S -group smooth over S ;*
 - (b) *$X = \text{Aut}_S(E)$ where E is smooth and affine over S .*

Then f_0 extends to a morphism of S -groups $Y \rightarrow X$; two such extensions are conjugate by an inner automorphism of X defined by a section of X over S inducing the unit section of X_0 over S_0 .

PROOF. Let us introduce the S_n as above.⁽⁵⁹⁾ For (ii), let us first note that a scheme smooth over S is necessarily locally of finite presentation over S ; hence, in case (b), E , being smooth and affine over S , is necessarily of finite presentation over S , i.e. one is indeed under hypothesis (b) of 2.0.

Then, under the hypotheses of (ii), condition (i_1) of 2.1 is automatically satisfied by 0.16 and 0.17; moreover every section of X_{S_n} over S_n lifts to a section of $X_{S_{n+1}}$ over S_{n+1} , by the definition of “smooth over S ” in case (a), and by 0.17 in case (b). Consequently, if f and f' are two liftings of f_0 , one may suppose step by step that $f_n = f'_n$, by lifting the inner automorphism whose existence is asserted by part (ii) of the theorem, which completes the proof.

Arguing in the same way, one obtains, taking Remark 2.3 into account:

Corollary 2.6. *Let S be a scheme, $\mathcal{I}$ a nilpotent ideal defining the closed subscheme S_0 , Y an S -group flat over S and affine, X an S -group smooth over S .*

- (i) *If, for every $n \geq 0$, one has $H^2(Y_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$, every morphism of S_0 -groups $f_0: Y_0 \rightarrow X_0$ lifts to a morphism of S -groups $f: Y \rightarrow X$.*

⁽⁵⁸⁾**N.D.E.** The original stated: “Suppose that one has one of the following two properties: (a) X is an S -group locally of finite presentation; (b) $X = \text{Aut}_S(E)$ where E is of finite presentation over S . Then f_0 extends to a morphism of S -groups $Y \rightarrow X$ if and only if it extends to a morphism of S -functors $Y \rightarrow X$.” and indicated: “(i) follows step by step from part (ii) of the theorem.”. This proof does not seem sufficient: if for example $I^3 = 0$ and $J = I^2$, and if $f: Y \rightarrow X$ is a morphism of S -functors lifting f_0 , then f_0 lifts to a morphism of S_J -groups $g_J: Y_J \rightarrow X_J$; next, does g_J lift to a morphism of S -functors $g: Y \rightarrow X$? In any event, this assertion (i) is not used in the sequel, where X is everywhere supposed smooth over S .

⁽⁵⁹⁾**N.D.E.** In what follows, one has slightly modified the original and set it out in detail.

- (ii) If, for every $n \geq 0$, one has $H^1(Y_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$, two such liftings are conjugate by an inner automorphism of X defined by a section of X over S inducing the unit section of X_0 over S_0 .

Now one has the following lemma:

Lemma 2.7. *Let S be an affine scheme, G an affine S -group, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_S$ -module, $\mathcal{L}$ a locally free $\mathcal{O}_S$ -module. Suppose that one has an operation of G on $\mathcal{F}$ in the sense of Exposé I, which defines an operation of G on $\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{L}$ ⁽⁶⁰⁾. Denote by Λ the ring of S , and by L the Λ -module defining $\mathcal{L}$ (which is therefore a projective module). One has a canonical isomorphism*

$$H^*(G, \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{L}) \simeq H^*(G, \mathcal{F}) \otimes_{\Lambda} L.$$

PROOF. ⁽⁶¹⁾ Indeed, denote by $\mathcal{A}$ the $\mathcal{O}_S$ -algebra $\mathcal{A}(G)$ and consider the complex $\mathcal{C}$ of quasi-coherent $\mathcal{O}_S$ -modules:

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A} \longrightarrow \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{A} \longrightarrow \cdots$$

By I, 5.3, $H^*(G, \mathcal{F})$ (resp. $H^*(G, \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{L})$) is the cohomology of the complex $\Gamma(S, \mathcal{C})$ (resp. $\Gamma(S, \mathcal{C} \otimes_{\mathcal{O}_S} \mathcal{L})$). Now, as S is affine, one has (cf. EGA I, 1.3.12)

$$\Gamma(S, \mathcal{C} \otimes_{\mathcal{O}_S} \mathcal{L}) \simeq \Gamma(S, \mathcal{C}) \otimes_{\Lambda} L.$$

As L is a projective Λ -module (hence flat), one also has $H^*(\Gamma(S, \mathcal{C}) \otimes_{\Lambda} L) \simeq H^*(\Gamma(S, \mathcal{C})) \otimes_{\Lambda} L$, whence the announced result.

Using the lemma, one transforms 2.6 into:

Corollary 2.8. *Let S be an affine scheme, $\mathcal{I}$ a nilpotent ideal on S defining the closed subscheme S_0 . Suppose that the $\mathcal{I}^{n+1}/\mathcal{I}^{n+2}$ are locally free over S_0 . Let Y be an S -group flat over S and affine, X an S -group smooth over S , and $f_0: Y_0 \rightarrow X_0$ a morphism of S -groups.*

- (i) *If $H^2(Y_0, \text{Lie}(X_0/S_0)) = 0$, f_0 lifts to a morphism of S -groups $Y \rightarrow X$.*
- (ii) *If $H^1(Y_0, \text{Lie}(X_0/S_0)) = 0$, two such liftings are conjugate by an inner automorphism of X defined by a section of X over S inducing the unit section of X_0 over S_0 .*

In particular, setting $Y = X$:

Corollary 2.9. *Let S and S_0 be as above. Let X be an S -group smooth over S and affine.*

- (i) *If $H^1(X_0, \text{Lie}(X_0/S_0)) = 0$, every endomorphism of X over S inducing the identity on X_0 is the inner automorphism defined by a section of X over S inducing the unit section of X_0 over S_0 .*
- (ii) *If*

$$H^2(X_0, \text{Lie}(X_0/S_0)) = 0,$$

every S_0 -automorphism of X_0 extends to an S -automorphism of X .⁽⁶²⁾

Remark 2.10. The assertions concerning the H^1 have converses by the theorem. Let us point out the following as an example: if $S = \mathbb{I}_{S_0}$ is the scheme of dual numbers over S_0 (II, 2.1) and if X is a flat S -group such that every automorphism of X over S inducing the identity on S_0 is the inner automorphism defined by a section of X over S inducing the unit section of X_0 over S_0 , then $H^1(X_0, \text{Lie}(X_0/S_0)) = 0$.⁽⁶³⁾

⁽⁶⁰⁾N.D.E. Where $\mathcal{L}$ is equipped with the trivial action of G .

⁽⁶¹⁾N.D.E. One has set out in detail (and simplified) the proof of the original (the latter moreover invoked the isomorphisms $\mathcal{H}^n(\mathcal{C} \otimes_{\mathcal{O}_S} \mathcal{L}) \simeq \mathcal{H}^n(\mathcal{C}) \otimes_{\mathcal{O}_S} \mathcal{L}$ and $H^n(\Gamma(-)) \simeq \Gamma(\mathcal{H}^n(-))$, where the $\mathcal{H}^n(\mathcal{C})$ denote the cohomology sheaves of the complex $\mathcal{C}$).

⁽⁶²⁾N.D.E. Indeed, let f_0 be an S_0 -automorphism of X_0 and let g_0 be its inverse. By 2.8 (i), f_0 (resp. g_0) lifts to an S -endomorphism f (resp. g) of X . Then $g \circ f$ and $f \circ g$ are endomorphisms of X inducing the identity on X_0 ; they are therefore, by 0.7, S -automorphisms of X , and the same is therefore true of f and g .

⁽⁶³⁾N.D.E. This is used in XXIV, 1.13.

Corollary 2.11. *Let $S, \mathcal{I}$ and $\mathcal{J}$ be as in 2.1. Let Y be an S -group scheme flat over S , X an S -group scheme, $f: Y \rightarrow X$ a morphism of S -groups. The set of morphisms from Y to X deduced from f by conjugation by elements $x \in X(S)$ inducing the unit of $X(S_{\mathcal{J}})$ is isomorphic to the quotient*

$$E = \text{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) / \text{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J})^{\text{ad}(Y_0)},$$

where the second group is formed of the $\mathcal{O}_{S_0}$ -morphisms $\omega_{X_0/S_0}^1 \rightarrow \mathcal{J}$ which, by every base change $S' \rightarrow S_0$, give morphisms $\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{S'} \rightarrow \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{S'}$ invariant under the action of $Y_0(S')$ on the first factor.

PROOF. By 2.1 (iii), one knows that the set sought is isomorphic to $\Gamma(L_0)/H^0(Y_0, L_0)$. Now $\Gamma(L_0) = \text{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J})$ and $H^0(Y_0, L_0)$ is evidently none other than $\Gamma(L_0)^{\text{ad}(Y_0)}$ in the sense of the preceding statement.

Corollary 2.12. *Under the conditions of 2.11, suppose moreover ω_{X_0/S_0}^1 locally free of finite rank. Then*

$$E \simeq \Gamma(S_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}) / H^0(Y_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

PROOF. ⁽⁶⁴⁾ Indeed, if ω_{X_0/S_0}^1 is locally free of finite rank, one has $\text{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) \simeq \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}$.

Corollary 2.13. *Suppose moreover Y_0 diagonalizable. Then*

$$E \simeq \Gamma(S_0, \text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}) / \Gamma(S_0, \text{Lie}(X_0/S_0)^{\text{ad}(Y_0)} \otimes_{\mathcal{O}_{S_0}} \mathcal{J})$$

where $\text{Lie}(X_0/S_0)^{\text{ad}(Y_0)}$ may be constructed as the factor of the decomposition of I , 4.7.3, corresponding to the zero character of Y_0 .

PROOF. Indeed, if $Y_0 \simeq D_{S_0}(M)$, one has by *loc. cit.* a decomposition into a direct sum:

$$\text{Lie}(X_0/S_0) = \text{Lie}(X_0/S_0)_0 \oplus \bigoplus_{\substack{m \in M \\ m \neq 0}} \text{Lie}(X_0/S_0)_m.$$

On tensoring by $\mathcal{J}$, one finds an analogous decomposition for $\text{Lie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}$, whence the relation

$$H^0(Y_0, \text{Lie}(X_0/S_0) \otimes \mathcal{J}) \simeq \Gamma(S_0, \text{Lie}(X_0/S_0)_0 \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

Corollary 2.14. *Suppose moreover S_0 affine. Then*

$$E \simeq \Gamma\left(S_0, [\text{Lie}(X_0/S_0) / \text{Lie}(X_0/S_0)^{\text{ad}(Y_0)}] \otimes_{\mathcal{O}_{S_0}} \mathcal{J}\right).$$

3. Infinitesimal extensions of a group scheme

Still in the notations of no. 0 ($S, \mathcal{I}, \mathcal{J}$, etc.), let us give ourselves an S -scheme X and suppose $X_{\mathcal{J}}$ endowed with a group structure. We propose to find the S -group structures on X inducing on $X_{\mathcal{J}}$ the given structure.

From now on, we suppose X flat over S . Let $\mathcal{C}$ be the category of S -schemes flat over S . One therefore has $X \in \text{Ob } \mathcal{C}$. We shall denote by Y , resp. M , the functor on $\mathcal{C}$ defined by X^+ , resp. L'_X . The canonical morphism $p_X: X \rightarrow X^+$ defines a morphism of $\widehat{\mathcal{C}}$

$$p: X \rightarrow Y$$

⁽⁶⁴⁾**N.D.E.** One has added the sentence which follows.

and the operation of L'_X on X in $(\widehat{\text{Sch}})_{/S}$ defines an operation of M on X in $\widehat{\mathcal{C}}$. One checks at once that X does indeed thus become formally principal homogeneous under M_Y over Y (cf. 0.2 (i) and 0.4).

The operation of X^+ on L'_X defined in 0.8 (denoted Ad in *loc. cit.*) defines an operation, denoted f , of Y on M . One knows, on the other hand (0.5), that

$$\text{Hom}_{\widehat{\mathcal{C}}}(Z, M) \simeq \text{Hom}_{S_0}(Z_0, L_0), \quad Z \in \text{Ob } \mathcal{C},$$

where L_0 is the functor defined in 0.5.

Lemma 3.1.

- (i) *The condition $(+)_n$ of 1.3 is satisfied for every positive integer n .*
- (ii) *If one lets the S_0 -group X_0 act on the S_0 -functor L_0 via its adjoint representation, one has a canonical isomorphism*

$$H^*(X_0, L_0) \simeq H^*(Y, M),$$

(the first cohomology is computed in $(\text{Sch})_{/S_0}$, the second in $\mathcal{C}$).

PROOF. The two parts of the lemma result from the relation:

$$\begin{aligned} \text{Hom}_{\widehat{\mathcal{C}}}(Y, M) &\simeq \text{Hom}_{(\widehat{\text{Sch}})_{/S_0}}(X^+ \times_S S_0, L_0) \\ &\simeq \text{Hom}_{S_0}(X_0, L_0) \\ &\simeq \text{Hom}_{\widehat{\mathcal{C}}}(X, M), \end{aligned}$$

which comes at once from the definition of M as a $\prod_{S_0/S}$. This relation being more generally satisfied on replacing X, Y by X^n, Y^n , one deduces from it that every morphism $X^n \rightarrow M$ factors uniquely through Y^n , which entails $(+)_n$. One also deduces from it the relation $C^*(Y, M) = C^*(X_0, L_0)$, which entails (ii).

We may therefore apply the constructions of 1.3. In particular:

Lemma 3.2. *Let $P: X \times_S X \rightarrow X$ be a morphism. For P to induce the group law of $X_{\mathcal{J}}$, it is necessary and sufficient that P be an admissible composition law (cf. 1.3.2) on X .*

PROOF. Indeed, for P to induce the group law of $X_{\mathcal{J}}$, it is necessary and sufficient that P lift the group law of X^+ , or again that of Y . One therefore only has to show that every morphism P lifting the group law of $X_{\mathcal{J}}$ satisfies the identity $(++)$ of 1.3.2 (ii), which is exactly what one saw in 0.8.

Proposition 3.3. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal. Let X be an S -scheme flat, and quasi-compact or locally of finite presentation over S . Let $P: X \times_S X \rightarrow X$ be a composition law on X . For P to be a group law, it is necessary and sufficient that the following two conditions be satisfied:*

- (i) *P is associative.*
- (ii) *P induces on $X_0 = X \times_X S_0$ a group law.*

PROOF. These conditions are evidently necessary. Let us show that they are sufficient. Suppose first that $X \rightarrow S$ has a section. As $X(S')$ is then nonempty for each $S' \rightarrow S$, it suffices⁽⁶⁵⁾ to show that, for every $x \in X(S')$, the translations *on the left and on the right* by x are isomorphisms of $X_{S'}$.⁽⁶⁶⁾

⁽⁶⁵⁾**N.D.E.** One has corrected the original, by removing the inadequate reference to an exercise of Bourbaki on semigroups (cf. [BAI], § I.2, Exercises 9 to 13) and by indicating the role of the translations on the left and on the right; see the following N.D.E.

⁽⁶⁶⁾**N.D.E.** Let E be a nonempty set equipped with an associative composition law, such that every left translation ℓ_x is bijective; let us fix $x_0 \in E$. There exists a unique $e \in E$ such that $x_0 \cdot e = x_0$; then $x_0 \cdot e \cdot x = x_0 \cdot x$ entails $e \cdot x = x$, for every $x \in E$. On the other hand, for every x there exists a unique x' such that $x \cdot x' = e$. Suppose moreover that there exists $b \in E$ such that the right translation r_b is injective. Then,

One may evidently suppose $S' = S$; the translation t in question induces on X_0 a translation t_0 of X_0 , which is therefore an automorphism since X_0 is a group. One concludes by flatness (SGA 1 III 4.2).⁽⁶⁷⁾

No longer assuming now that X has a section over S , suppose that there exists an $S' \rightarrow S$ such that $X_{S'}$ has a section over S' . Then $X_{S'}$ is an S' -group by what one has just seen; let us consider its unit section e' . The inverse image of e' under $\text{pr}_i: S' \times_S S' \rightarrow S'$ ($i = 1, 2$) is the unit section of $X_{S''}$ for the group law inverse image of $P_{S'}$ under pr_i . But as P is “defined over S ”, these two group laws coincide, hence also their unit section. One therefore has $\text{pr}_1^*(e') = \text{pr}_2^*(e')$.

If $S' \rightarrow S$ is a descent morphism (cf. Exp. IV no. 2), there will exist a section of X giving e' by extension of the base, and one will be done. As X_X has a section over X (the diagonal section), one sees that it now suffices to prove that $X \rightarrow S$ is a descent morphism. Now it is flat and surjective, and quasi-compact or locally of finite presentation, hence covering for (fpqc), hence a descent morphism (Exp. IV, no. 6).

Remark. In fact the hypothesis that $X \rightarrow S$ be quasi-compact or locally of finite presentation is superfluous, by virtue of the following result, which the reader will prove as an exercise on Exposé IV:

Under the conditions of the text on S and S_0 , if $X \rightarrow S$ is a flat morphism and $X_0 \rightarrow S_0$ a morphism covering for (fpqc), then $X \rightarrow S$ is a descent morphism.

Lemma 3.4. *For two admissible composition laws on X to be equivalent (cf. 1.3.5), it is necessary and sufficient that they be deduced from one another by an automorphism of X over S inducing the identity on $X_{\mathcal{J}}$.*

PROOF. Indeed, the morphisms constructed in 1.3.1 are exactly those of the preceding statement (by 0.7).⁽⁶⁸⁾

Taking all the preceding results into account, Proposition 1.3.6 gives:

Theorem 3.5. *Let S be a scheme, $\mathcal{I}$ and $\mathcal{J}$ two ideals on S such that $\mathcal{I} \supset \mathcal{J}$, $\mathcal{I} \cdot \mathcal{J} = 0$, and S_0 and $S_{\mathcal{J}}$ the closed subschemes of S which they define. Let X be an S -scheme flat over S (and locally of finite presentation or quasi-compact over S), and X_0 and $X_{\mathcal{J}}$ the schemes obtained by base change. Suppose $X_{\mathcal{J}}$ endowed with a structure of $S_{\mathcal{J}}$ -group, and denote by L_0 the S_0 -functor in abelian groups defined by the formula*

$$\text{Hom}_{S_0}(T, L_0) = \text{Hom}_{\mathcal{O}_T}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T)$$

on which X_0 acts via its adjoint representation.

- (i) *For there to exist an S -group structure on X inducing the given structure on $X_{\mathcal{J}}$, it is necessary and sufficient that the following conditions be satisfied:*

for every x , the equality $x \cdot e \cdot b = x \cdot b$ gives $x \cdot e = x$ (i.e. e is a neutral element), and $x \cdot x' \cdot x = x = x \cdot e$ entails $x' \cdot x = e$, i.e. x' is the inverse of x on the left and on the right, hence E is a group.

Note that the hypothesis “ r_b injective” is necessary: on every set E one may define a composition law by $x \cdot y = y$, for all $x, y \in E$; then every left translation is the identity (whence the associativity of the law), but for every y one has $r_y(E) = \{y\}$, hence E is not a group if $|E| > 1$.

⁽⁶⁷⁾**N.D.E.** Since X and X_0 have the same underlying topological space and t_0 is an automorphism, t is a homeomorphism, hence an *affine* morphism, cf. Exp. VI_B, 2.9.1 or EGA IV₄, 18.12.7.1. It therefore suffices to see that if J is a nilpotent ideal of a ring Λ , and $\phi: A \rightarrow B$ a morphism of Λ -algebras, with B flat over Λ , such that $\phi \otimes_{\Lambda} (\Lambda/J)$ is bijective, then ϕ is bijective. By the “nilpotent Nakayama lemma”, ϕ is surjective; moreover, B being flat over Λ , one also has $\text{Ker}(\phi) \otimes_{\Lambda} (\Lambda/J) = 0$, whence $\text{Ker}(\phi) = 0$, hence ϕ is bijective.

⁽⁶⁸⁾**N.D.E.** Indeed, by the proof of 0.7, the S -endomorphisms of X inducing the identity on $X_{\mathcal{J}}$ are the automorphisms $m \cdot \text{id}_X$, for m running through $M(X) = \text{Hom}_S(X, M)$ (for every $S' \rightarrow S$ and $x \in X(S')$, one has $(m \cdot \text{id}_X)(x) = m(x) \cdot x$). Now, by the proof of 3.1, each $m: X \rightarrow M$ factors uniquely as a morphism h from $Y = X^+$ to M , and therefore $m \cdot \text{id}_X$ is the automorphism u_h introduced in 1.3.1. The lemma then follows from the definition of the equivalence, cf. 1.3.4 and 1.3.5.

- (i₁) *There exists a morphism of S -schemes $X \times_S X \rightarrow X$ inducing the group law of $X_{\mathcal{J}}$.*
- (i₂) *A certain obstruction class belonging to $H^3(X_0, L_0)$ (defined canonically by the datum of X and of the group law of $X_{\mathcal{J}}$) is zero.*
- (ii) *If the conditions of (i) are satisfied, the set E of group laws on X inducing the given law of $X_{\mathcal{J}}$ is a principal homogeneous set under $Z^2(X_0, L_0)$, and E modulo the S -automorphisms of X inducing the identity on $X_{\mathcal{J}}$, is a principal homogeneous set under $H^2(X_0, L_0)$.*

PROOF. ⁽⁶⁹⁾ Indeed, every morphism of S -schemes $f: X \times_S X \rightarrow X$ inducing the group law of $X_{\mathcal{J}}$ is, by 3.2, an *admissible* composition law on X ; then, by 1.3.6 (i), the existence of an *associative* admissible composition law $P: X \times_S X \rightarrow X$ is equivalent to the vanishing of a certain class $c(f) \in H^3(X_0, L_0)$, and in this case, by 3.3, P is a group law. This proves (i), and (ii) then follows from 3.3 and 1.3.6 (ii).

Remark 3.5.1. ⁽⁷⁰⁾ If μ, μ' are group laws on X inducing the given law of $X_{\mathcal{J}}$, one therefore obtains a cocycle $\delta(\mu, \mu') \in Z^2(X_0, L_0)$, the sign convention chosen being that $\mu' = \delta(\mu, \mu') \cdot \mu$, i.e. for every $S' \rightarrow S$ and $x, y \in X(S')$,

$$\mu'(x, y) = \delta(\mu, \mu')(x_0, y_0) \cdot \mu(x, y).$$

(71)

One will denote by $\delta(\mu, \mu')$ the image of $\delta(\mu, \mu')$ in $H^2(X_0, L_0)$. Finally, if X equipped with the group law μ (resp. μ') is denoted simply by X (resp. X'), one will write $\delta(X, X')$ instead of $\delta(\mu, \mu')$, and likewise for $\delta(X, X')$.

Remark 3.6. Let X_J be an S_J -scheme *smooth* over S_J and *affine*. By 0.15, there exists, up to isomorphism, a unique S -scheme X , *smooth* over S , and reducing to X_J . If X_J is equipped with a structure of S_J -group, it follows from 0.16 that condition (i₁) is automatically satisfied. Moreover, by 0.6 the definition of L_0 simplifies and one obtains:

Corollary 3.7. *Let $S, \mathcal{I}$ and $\mathcal{J}$ be as in 3.1. Let X_J be an S_J -group smooth over S_J and affine.*

- (i) *The set of S -groups smooth over S and reducing to X_J , up to isomorphism (inducing the identity on X_J), is empty or a principal homogeneous set under the group*

$$H^2(X_0, \mathcal{L}ie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

- (ii) *There exists an S -group smooth over S reducing to X_J if and only if a certain obstruction in*

$$H^3(X_0, \mathcal{L}ie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J})$$

is zero.

One deduces from this, as usual, the following corollaries:

Corollary 3.8. *Let S be a scheme and S_0 a closed subscheme defined by a nilpotent ideal $\mathcal{I}$. Let X_0 be an S_0 -group smooth over S and affine.*

- (i) *If $H^2(X_0, \mathcal{L}ie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$ for every $n \geq 0$, two S -groups smooth over S and reducing to X_0 are isomorphic (by an isomorphism inducing the identity on X_0).*

⁽⁶⁹⁾N.D.E. One has added what follows.

⁽⁷⁰⁾N.D.E. One has added this remark, analogous to 4.5.1, in order to introduce the notation $\delta(\mu, \mu')$ (or $\delta(X, X')$), used in 4.38; consequently, one has also added in 3.5 (ii) above the part concerning E itself.

⁽⁷¹⁾N.D.E. One has conformed to the sign conventions of the original, so as to have in 4.38 (5) the equality $\partial^1 d(X, X') = \delta(X, X')$ (see also N.D.E. (54)).

- (ii) If $H^3(X_0, \mathcal{L}ie(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$ for every $n \geq 0$, there exists an S -group smooth over S , reducing to X_0 .

Corollary 3.9. *Let S be an affine scheme and S_0 a closed subscheme defined by a nilpotent ideal $\mathcal{I}$. Suppose the $\mathcal{I}^{n+1}/\mathcal{I}^{n+2}$ are locally free over S_0 . Let X_0 be an S_0 -group smooth and affine over S_0 .*

- (i) *If $H^2(X_0, \mathcal{L}ie(X_0/S_0)) = 0$, two S -groups smooth over S and reducing to X_0 are isomorphic.*
 (ii) *If $H^3(X_0, \mathcal{L}ie(X_0/S_0)) = 0$, there exists an S -group smooth over S reducing to X_0 .*

Corollary 3.10. *Let S_0 be a scheme and $S = \mathbf{I}_{S_0}$ the scheme of dual numbers over S_0 . Let X_0 be an S_0 -group smooth over S_0 . In order that every S -group Y , smooth over S , such that Y_0 is S_0 -isomorphic to X_0 , be S -isomorphic to $X = X_0 \times_{S_0} S$, it is necessary and sufficient that $H^2(X_0, \mathcal{L}ie(X_0/S_0)) = 0$.⁽⁷²⁾*

Indeed, by virtue of 3.5, the set of classes, up to an isomorphism of S -groups “inducing the identity on X_0 ”, of such groups Y , is in bijection with $H^2(X_0, \mathcal{L}ie(X_0/S_0))$, hence the set of classes, up to an arbitrary isomorphism of S -groups, is in bijection with

$$H^2(X_0, \mathcal{L}ie(X_0/S_0))/\Gamma_0,$$

where

$$\Gamma_0 = \text{Aut}_{S_0\text{-gr.}}(X_0)$$

(which operates in an evident way on the H^2). The conclusion follows at once from this.⁽⁷³⁾

4. Infinitesimal extensions of closed subgroups

Let us first state a result valid in an arbitrary *abelian category*.

Lemma 4.1. *Let $0 \longrightarrow A' \xrightarrow{i} A \xrightarrow{p} A'' \longrightarrow 0$ be an exact sequence, $\phi: A' \longrightarrow Q$ a morphism and $\pi: A'' \longrightarrow P$ an epimorphism with kernel C . Let E be the set (up to isomorphism) of the quadruples (B, f, g, h) such that the sequence*

$$0 \longrightarrow Q \xrightarrow{f} B \xrightarrow{g} P \longrightarrow 0$$

is exact and the diagram below is commutative:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A' & \xrightarrow{i} & A & \xrightarrow{p} & A'' \longrightarrow 0 \\ & & \downarrow \phi & & \downarrow h & & \downarrow \pi \\ 0 & \longrightarrow & Q & \xrightarrow{f} & B & \xrightarrow{g} & P \longrightarrow 0. \end{array}$$

- (i) *In order that E be non-empty, it is necessary and sufficient that the image in $\text{Ext}^1(C, Q)$ of the element A of $\text{Ext}^1(A'', A')$ be zero.*
 (ii) *Under these conditions, E is a principal homogeneous set under the abelian group $\text{Hom}(C, Q)$.*

⁽⁷²⁾**N.D.E.** This is used in XXIV, 1.13.

⁽⁷³⁾**N.D.E.** Indeed, $\text{Aut}_{S_0\text{-gr.}}(X_0)$ operates by group automorphisms on the abelian group $H^2(X_0, \mathcal{L}ie(X_0/S_0))$, hence the orbit of 0 is the singleton $\{0\}$; consequently the quotient set is a singleton if and only if $H^2(X_0, \mathcal{L}ie(X_0/S_0)) = \{0\}$.

Let us introduce the amalgamated sum $B' = A \amalg^{A'} Q$. One then has a commutative diagram in which the rows are exact:⁽⁷⁴⁾

$$\begin{array}{ccccccc} 0 & \longrightarrow & A' & \xrightarrow{i} & A & \xrightarrow{p} & A'' \longrightarrow 0 \\ & & \downarrow \phi & & \downarrow & & \downarrow \\ 0 & \longrightarrow & Q & \xrightarrow{j} & B' & \longrightarrow & A'' \longrightarrow 0, \end{array}$$

and it is clear that the category of solutions of the problem posed is canonically isomorphic to the category of solutions of the corresponding problem for the sequence

$$0 \longrightarrow Q \longrightarrow B' \longrightarrow A'' \longrightarrow 0$$

and the morphisms id_Q and $\pi: A'' \rightarrow P$.⁽⁷⁵⁾ In this case, the set E is in bijection with the set of subobjects N of B' such that $B' \rightarrow A''$ induces an isomorphism of N with the kernel C of $A'' \rightarrow P$, that is, the set of morphisms $\varepsilon: C \rightarrow B'$ lifting the canonical morphism $C \rightarrow A''$. The abelian group $G = \text{Hom}(C, Q)$ acts on E by $g \cdot \varepsilon = g + \varepsilon$ (addition in $\text{Hom}(C, B')$), and if $E \neq \emptyset$ this makes E into a principal homogeneous set under G .

One deduces from this:

Proposition 4.2. ⁽⁷⁶⁾ *Let S be a scheme, S_J the closed subscheme defined by a quasi-coherent ideal $\mathcal{J}$ of square zero, X an S -scheme, $\mathcal{F}$ an $\mathcal{O}_X$ -module, $X_J = X \times_S S_J$, $\mathcal{F}_J = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S_J}$, and $\mathcal{G}_J = \mathcal{F}_J / \mathcal{H}_J$ a quotient module of $\mathcal{F}_J$. Suppose given a morphism of $\mathcal{O}_{X_J}$ -modules*

$$f: \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{G}_J \longrightarrow \mathcal{Q}.$$

Let $\mathcal{E}$ be the sheaf of sets on X defined as follows: for each open U of X , $\mathcal{E}(U)$ is the set of quotient modules $\mathcal{G}$ of $\mathcal{F}|_U$, such that $\mathcal{G}/\mathcal{J}\mathcal{G} = \mathcal{G}_J|_U$ and such that there exists an isomorphism

$$h: \mathcal{J}\mathcal{G} \xrightarrow{\sim} \mathcal{Q}|_U$$

making the following diagram commutative:

$$\begin{array}{ccc} \mathcal{J} \otimes_{\mathcal{O}_{S_J}} (\mathcal{G}_J|_U) & \xrightarrow{f|_U} & \mathcal{Q}|_U \\ & \searrow \text{can.} \nearrow h & \\ & \mathcal{J}\mathcal{G} & \end{array}$$

(h is then unique, since $\mathcal{J} \otimes_{\mathcal{O}_{S_J}} (\mathcal{G}_J|_U) \rightarrow \mathcal{J}\mathcal{G}$ is an epimorphism). Then $\mathcal{E}$ is a formally principal homogeneous sheaf under the sheaf of commutative groups

$$\mathcal{A} = \text{Hom}_{\mathcal{O}_X}(\mathcal{H}_J, \mathcal{Q}) = \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{H}_J, \mathcal{Q}).$$

PROOF. If $\mathcal{E}(U) = \emptyset$ there is nothing to prove; one may therefore suppose that $\mathcal{E}(U)$ contains an element $\tilde{\mathcal{G}}$. Then, in the diagram below, h is an isomorphism and all the arrows are epimorphisms:

$$\begin{array}{ccccc} \mathcal{J} \otimes_{\mathcal{O}_{S_J}} (\mathcal{F}_J|_U) & \longrightarrow & \mathcal{J} \otimes_{\mathcal{O}_{S_J}} (\mathcal{G}_J|_U) & \xrightarrow{f|_U} & \mathcal{Q}|_U \\ \text{can.} \downarrow & & \text{can.} \downarrow & \nearrow \sim h & \\ \mathcal{J}\mathcal{F}|_U & \longrightarrow & \mathcal{J}\tilde{\mathcal{G}} & & \end{array}$$

⁽⁷⁴⁾**N.D.E.** One has $\text{Coker}(j) = B' \amalg^Q 0 = A \amalg^{A'} 0 = A''$, and one sees that $\text{Ker}(j) \simeq \text{Ker}(i) = 0$ by arguing “as if C were a category of modules”; for a proof solely in terms of arrows, see for example [Fr64], Th. 2.5.4 (*).

⁽⁷⁵⁾**N.D.E.** In what follows, one has replaced A by B' , and set out the end of the argument.

⁽⁷⁶⁾**N.D.E.** One has rewritten the statement so as to lie exactly in the setting of the application which is made of it in 4.3; on the other hand, one has set out the proof, following the indications given by M. Demazure.

Thus, the morphism $\mathcal{J} \otimes_{\mathcal{O}_{S_J}} (\mathcal{F}_J|_U) \longrightarrow \mathcal{Q}|_U$ induces an epimorphism (necessarily unique) $\phi: \mathcal{J}\mathcal{F}|_U \longrightarrow \mathcal{Q}|_U$, and if $\mathcal{G}$ is an $\mathcal{O}_U$ -module such that $\mathcal{G}/\mathcal{J}\mathcal{G} = \mathcal{G}_J|_U$ and one has a commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{J}\mathcal{F}|_U & \longrightarrow & \mathcal{F}|_U & \longrightarrow & \mathcal{F}_J|_U \longrightarrow 0 \\ & & \phi \downarrow & & \downarrow & & \pi \downarrow \\ 0 & \longrightarrow & \mathcal{Q}|_U & \longrightarrow & \mathcal{G} & \xrightarrow{p_J} & \mathcal{G}_J|_U \longrightarrow 0 \end{array}$$

(where p_J is the projection $\mathcal{G} \longrightarrow \mathcal{G}/\mathcal{J}\mathcal{G} = \mathcal{G}_J|_U$, so that $\mathcal{Q}|_U = \text{Ker}(p_J) = \mathcal{J}\mathcal{G}$), then one may identify $\mathcal{G}$ with a quotient module of $\mathcal{F}|_U$. Consequently, by 4.1 (ii), the set $\mathcal{E}(U)$ is a principal homogeneous set under the abelian group

$$\text{Hom}_{\mathcal{O}_X}(\mathcal{H}_J, \mathcal{Q})(U) = \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{H}_J, \mathcal{Q})(U).$$

□

Proposition 4.3. (TDTE IV 5.1) *Let S be a scheme, S_J the closed subscheme defined by a quasi-coherent ideal $\mathcal{J}$ of square zero, X an S -scheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module, $X_J = X \times_S S_J$, $\mathcal{F}_J = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S_J}$. Let $\mathcal{G}_J = \mathcal{F}_J/\mathcal{H}_J$ be a quasi-coherent quotient module of $\mathcal{F}_J$, flat over S_J .*

For every open U of X , let $\mathcal{E}(U)$ be the set of quasi-coherent quotient modules $\mathcal{G}^{(77)}$ of $\mathcal{F}|_U$, flat over S and such that $\mathcal{G}/\mathcal{J}\mathcal{G} \simeq \mathcal{G}_J|_U$. Then the $\mathcal{E}(U)$ form a sheaf of sets $\mathcal{E}$ on X , which is formally principal homogeneous under the sheaf of commutative groups

$$\mathcal{A} = \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{H}_J, \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{G}_J).$$

PROOF. Let us denote by $\pi: X \longrightarrow S$ the structural morphism. Let U be an open of X and $\mathcal{G}$ an $\mathcal{O}_U$ -module flat over S and such that $\mathcal{G}/\mathcal{J}\mathcal{G} \simeq \mathcal{G}_J|_U$. Then, for every $x \in U$, $\mathcal{G}_x$ is a module flat over the local ring $\mathcal{O}_{S,s}$ (where $s = \pi(x)$), and therefore the morphism

$$\mathcal{J}_s \otimes_{\mathcal{O}_{S,s}} (\mathcal{G}/\mathcal{J}\mathcal{G})_x = \mathcal{J}_s \otimes_{\mathcal{O}_{S,s}} \mathcal{G}_x \longrightarrow (\mathcal{J}\mathcal{G})_x$$

is bijective; one therefore has an exact sequence

$$0 \longrightarrow \mathcal{J} \otimes_{\mathcal{O}_S} (\mathcal{G}_J|_U) \longrightarrow \mathcal{G} \longrightarrow \mathcal{G}_J|_U \longrightarrow 0$$

and since $\mathcal{J} \otimes_{\mathcal{O}_S} (\mathcal{G}_J|_U)$ and $\mathcal{G}_J|_U$ are quasi-coherent $\mathcal{O}_U$ -modules, so is $\mathcal{G}$ (cf. EGA III, 1.4.17).

Conversely, since one has supposed $\mathcal{G}_J$ flat over S_J , if $\mathcal{G}$ is a quasi-coherent $\mathcal{O}_U$ -module such that $\mathcal{G}/\mathcal{J}\mathcal{G} \simeq \mathcal{G}_J$ and the morphism $\mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{G}_J \longrightarrow \mathcal{J}\mathcal{G}$ is bijective, then $\mathcal{G}$ is flat over S , by the “fundamental criterion of flatness” (cf. SGA 1 IV, 5.5⁽⁷⁸⁾).

Consequently, the set $\mathcal{E}(U)$ considered here coincides with the set considered in 4.2, taking for f the identity morphism of $\mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{G}_J$, and the conclusion therefore follows from 4.2. □

⁽⁷⁹⁾One keeps the preceding notations. Let Y_J be a closed subscheme of X_J , defined by a quasi-coherent ideal $\mathcal{I}_{Y_J}$. One supposes Y_J flat over S_J . Then, applying 4.3 to $\mathcal{F} = \mathcal{O}_X$ and $\mathcal{G}_J = \mathcal{O}_{Y_J} = \mathcal{O}_{X_J}/\mathcal{I}_{Y_J}$, one obtains the following corollary.

Corollary 4.3.1. *Let S , S_J , $\mathcal{J}$, X , X_J , Y_J and $\mathcal{I}_{Y_J}$ be as above; one supposes Y_J flat over S_J . Let us denote by $\mathcal{A}_J$ the sheaf of commutative groups*

$$\text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{I}_{Y_J}, \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J})$$

⁽⁷⁷⁾**N.D.E.** In order to lighten the statement, one has added here the hypothesis that $\mathcal{G}$ be quasi-coherent, and deferred to the proof the remark that this hypothesis is automatically verified; one has set out the proof accordingly.

⁽⁷⁸⁾**N.D.E.** See also [BAC], § III.5, th. 1.

⁽⁷⁹⁾**N.D.E.** One has set out what follows and added corollary 4.3.1. On the other hand, one recalls that “pseudo-torsor” is synonymous with “formally principal homogeneous”.

on X_J and $\mathcal{A} = i_*(\mathcal{A}_J)$, where i is the immersion $X_J \hookrightarrow X$.

For every open U of X , let $\mathcal{E}(U)$ be the set of closed subschemes Y of U , flat over S , such that $Y \times_S S_J = Y_J \cap U$. Then $\mathcal{E}$ is an $\mathcal{A}$ -pseudo-torsor.

If moreover a Y exists locally (i.e., if every $x \in X$ has an open neighborhood U such that $\mathcal{E}(U) \neq \emptyset$), then $\mathcal{E}$ is an $\mathcal{A}$ -torsor. Now one knows (see for example EGA IV₄, 16.5.15) that the $\mathcal{A}$ -torsors on X are parametrized by the group $H^1(X, \mathcal{A}) = H^1(X_J, \mathcal{A}_J)$, and that $\mathcal{E}$ possesses a global section (i.e., $\mathcal{E}(X) \neq \emptyset$) if and only if the cohomology class corresponding to $\mathcal{E}$ is zero. One therefore obtains the:

Corollary 4.4. *Let $S, S_J, \mathcal{J}, X, X_J, Y_J$ and $\mathcal{I}_{Y_J}$ be as above; one supposes Y_J flat over S_J . Let E be the set of closed subschemes Y of X , flat over S , such that $Y \times_S S_J = Y_J$.*

(i) *The set E is empty or a principal homogeneous set under the abelian group*

$$H^0(X, \mathcal{A}) = H^0(X_J, \mathcal{A}_J) = \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{I}_{Y_J}, \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J}).$$

(ii) *In order that E be non-empty, it is necessary and sufficient that the following two conditions be satisfied:*

(a) *There exists locally on X a solution of the problem.*

(b) *A certain obstruction is zero, which lies in*

$$H^1(X_J, \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{I}_{Y_J}, \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J})).$$

Complement 4.4.1. ⁽⁸⁰⁾Let us keep the notations of 4.4 and suppose that E contains an element Y . Let us denote by $\mathcal{I}_Y$ the ideal of $\mathcal{O}_X$ defining Y , and by $\mathcal{I}_{Y_J}$ its image in $\mathcal{O}_{X_J}$. Then, as one has seen in the proof of 4.2, one has a commutative diagram

$$\begin{array}{ccc} \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{X_J} & \longrightarrow & \mathcal{J} \mathcal{O}_X \\ \downarrow & \nearrow & \downarrow \\ \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J} & \xrightarrow{\sim} & \mathcal{J} \mathcal{O}_Y \end{array}$$

hence an epimorphism of $\mathcal{O}_X$ -modules $\phi: \mathcal{J} \mathcal{O}_X \rightarrow \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J}$; let us denote by $\mathcal{K}$ its kernel. Then, for every element Y' of E , the morphism $\mathcal{O}_X \rightarrow \mathcal{O}_{Y'}$ factors through $\mathcal{O}_X/\mathcal{K}$ (which is the amalgamated sum B' of the proof of lemma 4.1) and, denoting by $\mathcal{I}_{Y'}$ the ideal of Y' in $\mathcal{O}_X$, one has a commutative diagram:

$$\begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ & & \mathcal{I}_{Y'}/\mathcal{K} & \xrightarrow{\sim} & \mathcal{I}_{Y_J} & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & (\mathcal{J} \mathcal{O}_X)/\mathcal{K} & \longrightarrow & \mathcal{O}_X/\mathcal{K} & \longrightarrow & \mathcal{O}_{X_J} \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J} & \longrightarrow & \mathcal{O}_{Y'} & \longrightarrow & \mathcal{O}_{Y_J} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

Thus, replacing X by the closed subscheme defined by $\mathcal{K}$, one reduces to $\mathcal{K} = 0$. Then, giving Y' is equivalent to giving the sub- $\mathcal{O}_X$ -module $\mathcal{I}_{Y'}$ of $\mathcal{O}_X$, mapping bijectively onto $\mathcal{I}_{Y_J}$ by the projection $p: \mathcal{O}_X \rightarrow \mathcal{O}_{X_J}$; let us denote by $f': \mathcal{I}_{Y_J} \xrightarrow{\sim} \mathcal{I}_{Y'}$ (resp. $f: \mathcal{I}_{Y_J} \xrightarrow{\sim} \mathcal{I}$) the inverse isomorphism. Then $f' - f$ is an element of

$$\text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{I}_{Y_J}, \mathcal{J} \otimes_{\mathcal{O}_{S_J}} \mathcal{O}_{Y_J}) = \text{Hom}_{\mathcal{O}_{X_J}}(\mathcal{I}_{Y_J}, \mathcal{J} \mathcal{O}_Y)$$

⁽⁸⁰⁾**N.D.E.** One has added this complement, useful for proving point (ii) of proposition 4.8.

which one will denote by $d(Y', Y)$. (Note that $d(Y, Y') = -d(Y', Y)$.)

For our fixed Y and variable Y' , let us consider the morphism

$$\mathcal{I}_{Y'} \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y = \mathcal{O}_X / \mathcal{I}_Y ;$$

since the composite with $\mathcal{O}_Y \longrightarrow \mathcal{O}_{Y_{\mathcal{J}}}$ is zero, one knows that it takes values in $\mathcal{J}\mathcal{O}_Y = \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}}$. More precisely, if V is an open of X , x' a section of $\mathcal{I}_{Y'}$ over V and $x_{\mathcal{J}}$ its image in $\Gamma(V, \mathcal{I}_{Y_{\mathcal{J}}})$, then

$$x' = f'(x_{\mathcal{J}}) = f(x_{\mathcal{J}}) + (f' - f)(x_{\mathcal{J}}) = f(x_{\mathcal{J}}) + d(Y', Y)(x_{\mathcal{J}}).$$

Consequently: the morphism $\mathcal{I}_{Y'} \longrightarrow \mathcal{J}\mathcal{O}_Y$ is given by $d(Y', Y)$.

4.5.0. — ⁽⁸¹⁾ Let us keep the notations of 4.3.1 and 4.4 and carry out a certain number of transformations: $\mathcal{I}_{Y_{\mathcal{J}}} / \mathcal{I}_{Y_{\mathcal{J}}}^2$ is a quasi-coherent $\mathcal{O}_{X_{\mathcal{J}}}$ -module annihilated by $\mathcal{I}_{Y_{\mathcal{J}}}$, hence is the direct image of a quasi-coherent $\mathcal{O}_{Y_{\mathcal{J}}}$ -module denoted $\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}$, and called the *conormal sheaf* to $Y_{\mathcal{J}}$ in $X_{\mathcal{J}}$. ⁽⁸²⁾ As $\mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}}$ is annihilated by $\mathcal{I}_{Y_{\mathcal{J}}}$, the sheaf of commutative groups $\mathcal{A}_{\mathcal{J}}$ of 4.3.1 is identified with

$$\mathcal{H}\text{om}_{\mathcal{O}_{Y_{\mathcal{J}}}}(\mathcal{I}_{Y_{\mathcal{J}}} / \mathcal{I}_{Y_{\mathcal{J}}}^2, \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}}) = \mathcal{H}\text{om}_{\mathcal{O}_{Y_{\mathcal{J}}}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}, \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}}),$$

whence, for every $i \geq 0$,

$$\mathcal{H}^i(X_{\mathcal{J}}, \mathcal{A}_{\mathcal{J}}) = \mathcal{H}^i(Y_{\mathcal{J}}, \mathcal{H}\text{om}_{\mathcal{O}_{Y_{\mathcal{J}}}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}, \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}})) .$$

⁽⁸³⁾ One may then drop the hypothesis “ Y closed”, as follows. Let us first note that every open $U_{\mathcal{J}}$ of $X_{\mathcal{J}}$ arises by base change from the open subscheme U of X having the same underlying topological space as $U_{\mathcal{J}}$. Let now $Y_{\mathcal{J}}$ be a closed subscheme of $U_{\mathcal{J}}$, flat over $S_{\mathcal{J}}$, and $\mathcal{I}_{Y_{\mathcal{J}}}$ the quasi-coherent ideal of $\mathcal{O}_{U_{\mathcal{J}}}$ defining $Y_{\mathcal{J}}$. If $Y_{\mathcal{J}}$ lifts to a subscheme Y of X , then Y , having the same underlying topological space as $Y_{\mathcal{J}}$, is a closed subscheme of U ; consequently, the obstruction to lifting $Y_{\mathcal{J}}$ to a subscheme, flat over S , of X or of U is “the same”, and it lies in

$$\mathcal{H}^1(Y_{\mathcal{J}}, \mathcal{H}\text{om}_{\mathcal{O}_{Y_{\mathcal{J}}}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}, \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}})) .$$

Finally, let us return to the notations of no. 0: let $\mathcal{I}$ be a quasi-coherent ideal of $\mathcal{O}_S$ such that $\mathcal{J} \subset \mathcal{I}$ and $\mathcal{I}\mathcal{J} = 0$, and let S_0 be the closed subscheme of $S_{\mathcal{J}}$ defined by $\mathcal{I}$. For every S -scheme Z , one denotes $Z_{\mathcal{J}} = Z \times_S S_{\mathcal{J}}$ and $Z_0 = Z \times_S S_0$. Then, as $\mathcal{J}$ is annihilated by $\mathcal{I}$, one has, with the notations of 4.4,

$$\mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}} = \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}$$

$$\mathcal{H}\text{om}_{\mathcal{O}_{Y_{\mathcal{J}}}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}, \mathcal{J} \otimes_{\mathcal{O}_{S_{\mathcal{J}}}} \mathcal{O}_{Y_{\mathcal{J}}}) = \mathcal{H}\text{om}_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}),$$

etc. One therefore obtains:

Proposition 4.5. *Let S be a scheme, S_0 and $S_{\mathcal{J}}$ the closed subschemes defined by the quasi-coherent ideals $\mathcal{I}$ and $\mathcal{J}$, such that $\mathcal{I} \supset \mathcal{J}$ and $\mathcal{I} \cdot \mathcal{J} = 0$. Let X be an S -scheme and $Y_{\mathcal{J}}$ a subscheme of $X_{\mathcal{J}}$, flat over $S_{\mathcal{J}}$. Let $\mathcal{A}_0$ be the $\mathcal{O}_{Y_0}$ -module defined by*

$$\mathcal{A}_0 = \mathcal{H}\text{om}_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}).$$

- (i) *In order that there exist a subscheme Y of X , reducing to $Y_{\mathcal{J}}$, flat over S , it is necessary and sufficient that the following conditions be satisfied:*
 - (a) *Such a Y exists locally on X .*

⁽⁸¹⁾**N.D.E.** One has added the numbering 4.5.0 in order to mark the return to the original.

⁽⁸²⁾**N.D.E.** One has corrected the following sentence.

⁽⁸³⁾**N.D.E.** One has set out what follows in more detail.

- (b) *A certain obstruction in $R^1\Gamma(Y_0, \mathcal{A}_0)$ is zero.*⁽⁸⁴⁾
 (ii) *Under these conditions, the set of the Y satisfying the required conditions is a principal homogeneous space under the commutative group $\Gamma(Y_0, \mathcal{A}_0)$.*

Remark 4.5.1.⁽⁸⁵⁾ It follows from 4.5(ii) that there is given, for every pair (Y, Y') of subschemes⁽⁸⁶⁾ of X , flat over S and reducing to $Y_{\mathcal{J}}$, a “deviation”

$$d(Y', Y) \in \Gamma(Y_0, \mathcal{A}_0) = \text{Hom}_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0});$$

the sign convention adopted in 4.4.1 being that $d(Y', Y)$ corresponds to the morphism of $\mathcal{O}_X$ -modules

$$\mathcal{I}_{Y'} \hookrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y$$

(which takes values in $\mathcal{J}\mathcal{O}_Y \simeq \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}$ and factors through $\mathcal{I}_{Y'_{\mathcal{J}}} = \mathcal{I}_{Y_{\mathcal{J}}}$ and then through $\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}$).

Remark 4.6.⁽⁸⁷⁾ If X is flat over S and if $Y_{\mathcal{J}}$ is *locally a complete intersection* in $X_{\mathcal{J}}$, then condition (a) is always satisfied, and every Y flat over S lifting $Y_{\mathcal{J}}$ is then locally a complete intersection in X . If moreover Y_0 is *affine*, condition (b) is likewise satisfied.

Definition 4.6.1. (cf. SGA 6, VII 1.1) Let B be a commutative ring, $f: E \rightarrow B$ a B -linear morphism, where E is a free B -module of finite rank d , and I the ideal $f(E)$ (if one chooses a basis of E , f is given by a d -tuple $(f_1, \dots, f_d)$ of elements of B , and I is the ideal generated by the f_i). The *Koszul complex* $K_{\bullet}(f)$ is the graded B -module $\bigwedge_B^{\bullet} E$, equipped with the differential (of degree -1)

$$x_1 \wedge \dots \wedge x_i \longmapsto \sum_{j=1}^i (-1)^{j-1} f(x_j) x_1 \wedge \dots \wedge \widehat{x_j} \wedge \dots \wedge x_i.$$

One therefore has an augmented chain complex (B/I being in degree -1):

$$\dots \longrightarrow \bigwedge^2 E \longrightarrow E \xrightarrow{f} B \longrightarrow B/I \longrightarrow 0$$

which by definition is exact in degree 0, since $I = f(E)$. One says that f is *regular* if $K_{\bullet}(f)$ is acyclic in degrees > 0 , i.e. if the augmented complex above is a *resolution* of $C = B/I$.

In this case, the proof of SGA 6, VII 1.2.b) shows that the C -modules I^n/I^{n+1} ($n \in \mathbf{N}$) are free, I/I^2 being of rank d .

Definition 4.6.2. (cf. SGA 6, VII 1.4) Let X be a scheme, Y a subscheme, U an open of X such that Y is a closed subscheme of U , defined by the quasi-coherent ideal $\mathcal{I}_Y$.

One says that Y is *locally a complete intersection* in X if $Y \hookrightarrow X$ is a *regular immersion* in the sense of SGA 6, VII 1.4, i.e. if for every $y \in Y$ there exist an affine open neighbourhood V of y in U , a finite free $\mathcal{O}_V$ -module $\mathcal{E}$, and a regular morphism $f: \mathcal{E} \rightarrow \mathcal{O}_V$ with image $\mathcal{I}_Y|_V$, i.e. such that $K_{\bullet}(f)$ is a resolution of $\mathcal{O}_{Y \cap V}$.

This implies that the immersion $Y \hookrightarrow X$ is *locally of finite presentation*, and, by 4.6.1, that the conormal sheaf $\mathcal{N}_{Y/X} = \mathcal{I}_Y/\mathcal{I}_Y^2$ is a *finite locally free* $\mathcal{O}_Y$ -module.

⁽⁸⁴⁾**N.D.E.** Here, one has denoted by $R^1\Gamma(Y_0, \mathcal{A}_0)$ the “coherent” cohomology group $H^1(Y_0, \mathcal{A}_0)$ of the $\mathcal{O}_{Y_0}$ -module $\mathcal{A}_0$, in order to distinguish it from the “Hochschild” cohomology groups $H^i(Y_0, \mathcal{M}_0)$ (Y_0 an S_0 -group, $\mathcal{M}_0$ an $\mathcal{O}_{S_0}$ -module) which will be considered starting from 4.16.

⁽⁸⁵⁾**N.D.E.** One has placed this remark here; it replaces Remark 4.7 of the original.

⁽⁸⁶⁾**N.D.E.** One has corrected “closed subschemes” to “subschemes”.

⁽⁸⁷⁾**N.D.E.** One has kept, for the record, Remark 4.6 of the original, in which the definition of “locally a complete intersection” does not appear. One has added afterwards the “right” definition, taken from SGA 6, VII 1.4 (which *replaces* that of EGA IV₄, 16.9.2), and the proof of the three results stated in the remark.

Lemma 4.6.3. ⁽⁸⁸⁾ *Let A be a ring, J an ideal of A of square zero, $\bar{A} = A/J$, B a flat A -algebra, E a free B -module of finite rank, $f: E \rightarrow B$ a morphism of B -modules. One assumes that the morphism $g: \bar{E} = E \otimes_A \bar{A} \rightarrow \bar{B} = B \otimes_A \bar{A}$ induced by f is regular and that $\bar{B}/g(\bar{E})$ is flat over $\bar{A}$.*

Then f is regular and $B/f(E)$ is flat over A .

PROOF. Set $C = B/f(E)$ and $\bar{C} = C \otimes_A \bar{A} = \bar{B}/g(\bar{E})$. First, the $\bigwedge_B^i(E)$ are free B -modules, hence flat A -modules, since B is flat over A . As $\bigwedge_B^\bullet E \otimes_A \bar{A} \simeq \bigwedge_{\bar{A}}^\bullet \bar{E}$, one therefore obtains an exact sequence of complexes

$$0 \rightarrow J \otimes_A \bigwedge_B^\bullet E \rightarrow \bigwedge_B^\bullet E \rightarrow \bigwedge_{\bar{A}}^\bullet \bar{E} \rightarrow 0.$$

Moreover, as $J^2 = 0$, one has $J \otimes_A M = J \otimes_{\bar{A}} \bar{A} \otimes_A M$ for every A -module M . Denoting by $--\rightarrow$ the augmentation arrows, and by d the rank of E , one therefore obtains the following double complex, in which the rows are exact:

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & J \otimes_A \bigwedge_B^d E & \longrightarrow & \bigwedge_B^d E & \longrightarrow & \bigwedge_{\bar{A}}^d \bar{E} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & \vdots & & \vdots & & \vdots \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & J \otimes_A E & \longrightarrow & E & \longrightarrow & \bar{E} \longrightarrow 0 \\ & & \downarrow \text{id} \otimes g & & \downarrow f & & \downarrow g \\ 0 & \longrightarrow & J \otimes_A B & \longrightarrow & B & \longrightarrow & \bar{B} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & J \otimes_A \bar{C} & \longrightarrow & C & \longrightarrow & \bar{C} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}.$$

Moreover, the right-hand and left-hand columns are exact, since $K_\bullet(g)$ is a resolution of $\bar{C}$ and since the latter is flat over $\bar{A}$. Thus, considering the long exact sequence in homology associated with the exact sequence of non-augmented complexes, one obtains that $K_\bullet(f)$ is acyclic in degrees > 0 , and that in degree 0 one has an exact sequence

$$0 \rightarrow J \otimes_A \bar{C} \rightarrow C \rightarrow \bar{C} \rightarrow 0.$$

Hence C is flat over A , by the “fundamental criterion of flatness” (cf. [BAC], §III.5, th. 1). $\square$

Lemma 4.6.4. ⁽⁸⁸⁾ *Let A be a commutative ring, J a nilpotent ideal, $N \subset M$ A -modules such that M/N is flat over A . If $x_1, \dots, x_n$ are elements of N whose images generate the image $\bar{N}$ of N in M/JM , then they generate N .*

Indeed, denote by N' the submodule of N generated by the x_i , and $Q = N/N'$. Then the morphism $N' \otimes (A/J) \rightarrow \bar{N}$ is surjective. On the other hand, as M/N is flat over A , the morphism $N \otimes (A/J) \rightarrow \bar{N}$ is bijective. One therefore obtains that $Q \otimes (A/J) = 0$, whence $Q = 0$ by the “nilpotent Nakayama lemma” (one has $Q = JQ = J^2Q = \dots = 0$).

One may now prove the following:

Proposition 4.6.5. ⁽⁸⁸⁾ *Let $S, \mathcal{I}, \mathcal{J}$ and $X, Y_{\mathcal{J}}$ be as in 4.5. Assume moreover X flat over S and $Y_{\mathcal{J}}$ locally a complete intersection in $X_{\mathcal{J}}$.*

⁽⁸⁸⁾**N.D.E.** In order to prove the results stated in Remark 4.6, one has added Lemmas 4.6.3, 4.6.4 and Proposition 4.6.5, as well as Remark 4.6.6.

a) Then, condition (a) of 4.5 (i) is satisfied; moreover, every Y flat over S lifting $Y_{\mathcal{J}}$ is locally a complete intersection in X .

b) If moreover Y_0 is affine, condition (b) of loc. cit. is likewise satisfied.

PROOF. The first assertion of (a) follows from Lemma 4.6.3; the second then results from Lemma 4.6.4. On the other hand, the hypothesis implies (cf. 4.6.2) that $\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}$ is a finite locally free $\mathcal{O}_{Y_{\mathcal{J}}}$ -module, hence the $\mathcal{O}_{Y_0}$ -module

$$\mathcal{A}_0 = \mathcal{H}om_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0})$$

is quasi-coherent (cf. EGA I, 1.3.12), whence $R^1\Gamma(Y_0, \mathcal{A}_0) = 0$ if Y_0 is affine. $\square$

Remark 4.6.6. ⁽⁸⁸⁾ Let us end this paragraph with the following example, which shows that, under the hypotheses of Lemma 4.6.3, if (g_1, g_2) is a regular sequence generating the ideal $I = g(E)$, it does not necessarily lift to a regular sequence in B .

Let k be a field, $A = k[X, Y]$, denote by $k\varepsilon$ the A -module $A/(X, Y)$ (i.e. $P \cdot \varepsilon = P(0, 0)\varepsilon$ for every $P \in A$), and let $A = \bar{A} \oplus k\varepsilon$, where $J = k\varepsilon$ is an ideal of square zero. One has $A/J = \bar{A}$.

The algebra $B = A \otimes_k k[Z, T]$ is free over A , hence flat; one has $\bar{B} = k[X, Y, Z, T]$. Set $g_1 = XZ - YT$ and $g_2 = XZ - 1$. As the polynomial g_1 is irreducible, $\bar{B}/(g_1)$ is an integral domain, and hence (g_1, g_2) is a regular sequence in $\bar{B}$, generating the ideal $I = (XZ - 1, YT - 1)$. Thus

$$\bar{C} = \bar{B}/I = k[X, Y, X^{-1}, Y^{-1}] = \bar{A}[X^{-1}, Y^{-1}]$$

is a flat $\bar{A}$ -algebra (and also a flat A -algebra). But every lifting in B of g_1 is of the form $XY - ZT + \lambda\varepsilon$, where $\lambda \in k[Z, T]$, hence annihilates ε .

4.7. — One has deleted here Remark 4.7, placed in 4.5.1.

Remark 4.8.0. ⁽⁸⁹⁾ Let S be a scheme, S' a closed subscheme, X an S -scheme, Y a sub- S -scheme of X , and $X' = X \times_S S'$, $Y' = Y \times_S S'$. Then one has a *surjective* morphism of $\mathcal{O}_{Y'}$ -modules

$$\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \xrightarrow{\text{surj.}} \mathcal{N}_{Y'/X'}.$$

Indeed, after replacing X by a suitable open, one may assume that Y is closed, defined by an ideal $\mathcal{I}_Y$ of $\mathcal{O}_X$; then the image of $\mathcal{I}_Y$ in $\mathcal{O}_{X'}$ is the ideal $\mathcal{I}_{Y'}$ defining Y' , and one has a surjective morphism of $\mathcal{O}_{Y'}$ -modules

$$\pi: (\mathcal{I}_Y/\mathcal{I}_Y^2) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \xrightarrow{\text{surj.}} \mathcal{I}_{Y'}/\mathcal{I}_{Y'}^2.$$

Assume moreover that $\mathcal{O}_Y = \mathcal{O}_X/\mathcal{I}_Y$ is *flat* over $\mathcal{O}_S$; then the natural morphism

$$\mathcal{I}_Y \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} \longrightarrow \mathcal{I}_{Y'}$$

is bijective (cf. EGA IV₂, 2.1.8). One then has the following commutative diagram with exact rows:

$$\begin{array}{ccccccc} \mathcal{I}_Y^2 \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} & \longrightarrow & \mathcal{I}_Y \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} & \longrightarrow & (\mathcal{I}_Y/\mathcal{I}_Y^2) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} & \longrightarrow & 0 \\ \downarrow \text{surj.} & & \downarrow \wr & & \downarrow \pi \text{ surj.} & & \\ 0 \longrightarrow & \mathcal{I}_{Y'}^2 & \longrightarrow & \mathcal{I}_{Y'} & \longrightarrow & \mathcal{I}_{Y'}/\mathcal{I}_{Y'}^2 & \longrightarrow 0 \end{array}$$

⁽⁸⁹⁾**N.D.E.** One has inserted this remark here, used in the proposition which follows; it appeared in 4.10 of the original.

whence one deduces, by the snake lemma:⁽⁹⁰⁾

$$(4.8.0) \quad \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \xrightarrow{\sim} \mathcal{N}_{Y'/X'} \quad \text{if } Y \text{ is flat over } S.$$

Proposition 4.8. *Let $S, S_0, S_{\mathcal{J}}$ and $\mathcal{I}, \mathcal{J}$ be as in 4.5.⁽⁹¹⁾ Let X be an S -scheme, Y a subscheme of X , and i the immersion $Y \hookrightarrow X$.*

- (i) *For every S -morphism $f: T \rightarrow X$ such that $f_{\mathcal{J}}: T_{\mathcal{J}} \rightarrow X_{\mathcal{J}}$ factors through $Y_{\mathcal{J}}$, one may define an obstruction*

$$(*) \quad c(X, Y, f) \in \mathrm{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T)$$

whose vanishing is equivalent to the existence of a factorization of f through Y .

- (ii) *Let Y' be a second subscheme of X . Assume that $Y'_{\mathcal{J}} = Y_{\mathcal{J}}$ and that Y, Y' are flat over S . One then has isomorphisms (cf. 4.8.0)*

$$\mathcal{J}\mathcal{O}_Y \simeq \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0} \simeq \mathcal{J}\mathcal{O}_{Y'} \quad \text{and} \quad \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_{\mathcal{J}}} \xrightarrow{\sim} \mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}}$$

whence an isomorphism

$$u: \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}, \mathcal{J}\mathcal{O}_Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}, \mathcal{J}\mathcal{O}_{Y'}).$$

Denoting by $i': Y' \rightarrow X$ the canonical immersion and by $d(Y, Y')$ the deviation of 4.5.1, one has:⁽⁹²⁾

$$(**) \quad c(X, Y, i') = u(d(Y, Y')).$$

- (iii) *The canonical morphism $\mathcal{N}_{Y/X} \xrightarrow{D} i^*(\Omega_{X/S}^1)$ (cf. SGA 1 II, formula 4.3)⁽⁹³⁾ induces a morphism*

$$D_0: \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0} \rightarrow \Omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{X_0}} \mathcal{O}_{Y_0}$$

and thus, for every S -morphism $f: T \rightarrow X$ as in (i), a morphism

$$v_{f_0}: \mathrm{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\Omega_{X_0/S_0}^1), \mathcal{J}\mathcal{O}_T) \rightarrow \mathrm{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T),$$

$$a \mapsto a \circ f_0^*(D_0)$$

where above the first group is $\mathrm{Hom}_{X^+}(T, L_X)$, cf. 0.1.5. For $a \in \mathrm{Hom}_{X^+}(T, L_X)$, one has

$$(***) \quad c(X, Y, a \cdot f) - c(X, Y, f) = v_{f_0}(a),$$

where one has denoted by $a \cdot f$ the composite morphism $T \xrightarrow{a \times f} L_X \times_{X^+} X \rightarrow X$.

We shall prove part (i) of the proposition, leaving to the reader the task of (not) verifying the assertions (ii) and (iii); this verification is carried out by reduction to the affine case, then by comparison of the explicit definitions.⁽⁹⁴⁾

Let us therefore prove (i). The morphism $f: T \rightarrow X$ defines a morphism of sheaves of rings $\varphi: \mathcal{O}_X \rightarrow f_*(\mathcal{O}_T)$.⁽⁹⁵⁾ Let U be an open subscheme of X in which Y is closed; as T (resp. $Y_{\mathcal{J}}$) has the same underlying space as $T_{\mathcal{J}}$ (resp. Y), the underlying continuous map

⁽⁹⁰⁾**N.D.E.** In the original, this was indicated in Remark 4.10, under the additional hypothesis that Y' be locally a complete intersection in X' . This hypothesis also appeared, consequently, in the statements 4.12–4.14; it seems in fact superfluous, and one has removed it from the aforementioned statements.

⁽⁹¹⁾**N.D.E.** One has removed the hypothesis that $\mathcal{I}$ be nilpotent, which appears superfluous (cf. the proof).

⁽⁹²⁾**N.D.E.** See also 4.27 further on.

⁽⁹³⁾**N.D.E.** See also EGA IV₄, 16.4.21. Recall that if U is an affine open of X such that $Y \cap U$ is defined by the ideal I of $A = \mathcal{O}_X(U)$, if one denotes by d the differential $A \rightarrow \Gamma(U, \Omega_{X/S}^1)$, and if $x \in I$, then $D(x + I^2)$ is the element $d(x) \otimes 1$ of $\Gamma(U, \Omega_{X/S}^1) \otimes_A (A/I)$.

⁽⁹⁴⁾**N.D.E.** One has carried out these verifications further down.

⁽⁹⁵⁾**N.D.E.** On the one hand, one has removed the hypothesis that $\mathcal{I}$ be nilpotent, i.e. that X_0 have the same underlying topological space as X ; on the other hand, one has set out the following sentence in more detail.

of f sends T into U , and as U is an open of X , φ induces a morphism of sheaves of rings $\mathcal{O}_U = \mathcal{O}_X|_U \rightarrow f_*(\mathcal{O}_T)$, i.e. f factors through U .

Thus, one may restrict oneself to the case where Y is closed, hence defined by a sheaf of ideals $\mathcal{I}_Y$. In order that f factor through Y , it is necessary and sufficient that the composite map $\mathcal{I}_Y \rightarrow \mathcal{O}_X \rightarrow f_*(\mathcal{O}_T)$ be zero. As $f_{\mathcal{J}}$ factors through $Y_{\mathcal{J}}$, the composite map $\mathcal{I}_{Y_{\mathcal{J}}} \rightarrow \mathcal{O}_{X_{\mathcal{J}}} \rightarrow f_*(\mathcal{O}_{T_{\mathcal{J}}})$ is zero. Considering the commutative diagram in which the first row is exact

$$\begin{array}{ccccccc} 0 & \longrightarrow & f_*(\mathcal{J}\mathcal{O}_T) & \longrightarrow & f_*(\mathcal{O}_T) & \longrightarrow & f_*(\mathcal{O}_{T_{\mathcal{J}}}) \\ & & \nwarrow \varphi & & \uparrow \varphi & & \uparrow \varphi_{\mathcal{J}} \\ & & & & \mathcal{O}_X & \longrightarrow & \mathcal{O}_{X_{\mathcal{J}}} \\ & & & & \uparrow & & \uparrow \\ & & & & \mathcal{I}_Y & \longrightarrow & \mathcal{I}_{Y_{\mathcal{J}}} \end{array}$$

one deduces from this that φ sends $\mathcal{I}_Y$ into $f_*(\mathcal{J}\mathcal{O}_T)$.⁽⁹⁶⁾ Since $\mathcal{J}^2 = 0$, it follows that $f_*(\mathcal{J}\mathcal{O}_T)$, viewed as an $\mathcal{O}_X$ -module via φ , is annihilated by $\mathcal{I}_Y$; consequently, φ induces a morphism of $\mathcal{O}_X$ -modules

$$h: i_*(\mathcal{N}_{Y/X}) = \mathcal{I}_Y/\mathcal{I}_Y^2 \rightarrow f_*(\mathcal{J}\mathcal{O}_T).$$

On the other hand, one has cartesian squares

$$\begin{array}{ccccc} T_0 & \xrightarrow{f_0} & X_0 & \xleftarrow{i_0} & Y_0 \\ \tau_{T_0} \downarrow & & \tau_{X_0} \downarrow & & \downarrow \tau_{Y_0} \\ T & \xrightarrow{f} & X & \xleftarrow{i} & Y \end{array}$$

where i_{T_0} etc. are the closed immersions deduced by base change from $S_0 \hookrightarrow S$. As $\mathcal{J}\mathcal{O}_T$ is a quasi-coherent $\mathcal{O}_T$ -module annihilated by $\mathcal{I}$, one has an isomorphism

$$\mathcal{J}\mathcal{O}_T \simeq (\tau_{T_0})_* \tau_{T_0}^*(\mathcal{J}\mathcal{O}_T),$$

whence $f_*(\mathcal{J}\mathcal{O}_T) \simeq (\tau_{X_0})_*(f_0)_* \tau_{T_0}^*(\mathcal{J}\mathcal{O}_T)$. Thus h corresponds, by adjunction, to a morphism of $\mathcal{O}_{T_0}$ -modules

$$h_0: f_0^* \tau_{X_0}^* i_*(\mathcal{N}_{Y/X}) \rightarrow i_{T_0}^*(\mathcal{J}\mathcal{O}_T).$$

Now, $\tau_{X_0}^* i_*(\mathcal{N}_{Y/X}) \simeq (i_0)_* \tau_{Y_0}^*(\mathcal{N}_{Y/X}) = \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}$. Thus, reverting to the abuse of notation $i_{T_0}^*(\mathcal{J}\mathcal{O}_T) = \mathcal{J}\mathcal{O}_T$ constantly used, h_0 is identified with a morphism of $\mathcal{O}_{T_0}$ -modules

$$h_0: f_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}) \rightarrow \mathcal{J}\mathcal{O}_T$$

which is the desired obstruction $c(X, Y, f)$. This proves (i).

When f is the immersion $i': Y \hookrightarrow X$, one sees that $c(X, Y, i')$ arises from the morphism $\mathcal{I}_Y \hookrightarrow \mathcal{O}_X \rightarrow \mathcal{O}_{Y'}$, hence corresponds, by 4.4.1 and 4.5.1, to the class $d(Y, Y')$. This proves (ii).

Let us prove (iii). First, $D: \mathcal{N}_{Y/X} \rightarrow i^*(\Omega_{X/S}^1)$ induces a morphism

$$D_0: \tau_{Y_0}^*(\mathcal{N}_{Y/X}) \rightarrow \tau_{Y_0}^* i^*(\Omega_{X/S}^1) = i_0^* \tau_{X_0}^*(\Omega_{X/S}^1)$$

and, as $X_0 = X \times_S S_0$, one has $\tau_{X_0}^*(\Omega_{X/S}^1) \simeq \Omega_{X_0/S_0}^1$ (cf. EGA IV₄, 16.4.5). One therefore obtains the announced morphism

$$D_0: \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0} \rightarrow \Omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{X_0}} \mathcal{O}_{Y_0}.$$

Finally, one will verify the equality $(***)$ after the remark below.

Remark 4.9. The obstruction $c(X, Y, f)$ is computed locally on T . Let $U = \text{Spec}(C)$ be an affine open of T lying over an affine open $\text{Spec}(A)$ of X , itself lying over an affine open $\text{Spec}(\Lambda)$ of S , let $\mathcal{J} \subset \mathcal{I} \subset \Lambda$ (resp. $\mathcal{I}_Y \subset A$) be the ideals corresponding to $\mathcal{J} \subset \mathcal{I}$ (resp.

⁽⁹⁶⁾**N.D.E.** One has set out what follows in more detail.

to $\mathcal{I}_Y$), let $B = A/\mathcal{I}_Y$ and let $\varphi: A \rightarrow C$ be the morphism of Λ -algebras corresponding to $f: T \rightarrow X$; as $f(T_{\mathcal{J}}) \subset Y_{\mathcal{J}}$ one has $\varphi(\mathcal{I}_Y) \subset \mathcal{J}C$ and hence φ induces a morphism of Λ -algebras $B \rightarrow C/\mathcal{J}C \rightarrow C_0 = C/\mathcal{I}C$. Then the obstruction $c = c(X, Y, f)$ is computed by the following commutative diagram

$$\begin{array}{ccccc} \mathcal{I}_Y & \longrightarrow & A & \xrightarrow{\varphi} & C \\ \downarrow & & & & \uparrow \\ \mathcal{I}_Y/\mathcal{I}_Y^2 & \longrightarrow & \mathcal{I}_Y/\mathcal{I}_Y^2 \otimes_B C_0 & \xrightarrow{c} & \mathcal{J}C \end{array}$$

i.e. it is defined, above the open U , as the unique element of

$$\mathrm{Hom}_{C_0}(\mathcal{I}_Y/\mathcal{I}_Y^2 \otimes_B C_0, \mathcal{J}C) = \mathrm{Hom}_{B_0}(\mathcal{I}_Y/\mathcal{I}_Y^2 \otimes_B B_0, \mathcal{J}C)$$

such that, with the obvious notations, one has $c(x \otimes_B 1) = \varphi(x)$, for every $x \in \mathcal{I}_Y$.

⁽⁹⁷⁾One can now complete the proof of 4.8 (iii). The equality $(***)$ is checked locally on T , so one is reduced to the affine situation described above. Denote by $d_{A/\Lambda}$ the differential $A \rightarrow \Omega_{A/\Lambda}^1$. Then a corresponds, above U , to an element a_U of

$$\mathrm{Hom}_{C_0}(\Omega_{A_0/\Lambda_0}^1 \otimes_{A_0} C_0, \mathcal{J}C) \simeq \mathrm{Hom}_{B_0}(\Omega_{A/\Lambda}^1 \otimes_A B_0, \mathcal{J}C) \simeq \mathrm{Hom}_A(\Omega_{A/\Lambda}^1, \mathcal{J}C).$$

Then, on the one hand, $v_{f_0}(a)$ corresponds above U to the element $a_U \circ D_0$, where D_0 is the morphism of B -modules⁽⁹⁸⁾

$$I_Y/I_Y^2 \rightarrow \Omega_{A/\Lambda}^1 \otimes_A B_0, \quad x + I_Y^2 \mapsto d_{A/\Lambda}(x) \otimes 1.$$

On the other hand (cf. the proof of 0.1.8 and 0.1.9), the morphism of Λ -algebras $\phi': A \rightarrow C$ corresponding to $a \cdot f$ differs from ϕ by the Λ -derivation $A \rightarrow \mathcal{J}C$ associated with a_U , i.e. one has:

$$\phi' = \phi + a_U \circ d_{A/\Lambda} = \phi + a_U \circ (d_{A/\Lambda} \otimes 1).$$

Consequently, writing $c' = c(X, Y, a \cdot f)$, one has for every $x \in I_Y$, writing x for its image in I_Y/I_Y^2 :

$$(c' - c)(x \otimes 1) = a_U(d_{A/\Lambda}(x) \otimes 1) = (a_U \circ D_0)(x) = v_{f_0}(a)(x).$$

This shows that $c' - c = v_{f_0}(a)$.

4.10. — One has deleted remark 4.10 of the original, rendered obsolete by the addition of remark 4.8.0.

4.11. — We now propose to study the following situation. Let $S, S_{\mathcal{J}}$ and S_0 be as in 4.8; one has three S -schemes X, X', T , a subscheme Y of X (resp. Y' of X'), and morphisms $f: T \rightarrow X'$ and $g: X' \rightarrow X$.

$$\begin{array}{ccccc} & & Y' & & Y \\ & & \downarrow i' & & \downarrow i \\ T & \xrightarrow{f} & X' & \xrightarrow{g} & X \end{array}$$

One supposes that by reduction modulo $\mathcal{J}$, this diagram completes to a commutative diagram

$$\begin{array}{ccccc} & & Y'_{\mathcal{J}} & \dashrightarrow & Y_{\mathcal{J}} \\ & & \downarrow i'_{\mathcal{J}} & & \downarrow i_{\mathcal{J}} \\ T_{\mathcal{J}} & \xrightarrow{f_{\mathcal{J}}} & X'_{\mathcal{J}} & \xrightarrow{g_{\mathcal{J}}} & X_{\mathcal{J}} \end{array}$$

⁽⁹⁷⁾**N.D.E.** One has added what follows.

⁽⁹⁸⁾**N.D.E.** Cf. N.D.E. (93).

One therefore has, by 4.8, obstructions:

$$\begin{aligned} c(X, Y, g \circ i') &\in \operatorname{Hom}_{\mathcal{O}_{Y'_0}}(i'^*_0 g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_{Y'}), \\ c(X', Y', f) &\in \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0}), \mathcal{J}\mathcal{O}_T), \\ c(X, Y, g \circ f) &\in \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T), \end{aligned}$$

whose relations one seeks to calculate.⁽⁹⁹⁾

Lemma 4.12. *Suppose Y' flat over S , so that $\mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y'_0} = \mathcal{J}\mathcal{O}_{Y'}$.*

(i) *One has a natural morphism*

$$b_{f_0}: \operatorname{Hom}_{\mathcal{O}_{Y'_0}}(i'^*_0 g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_{Y'}) \longrightarrow \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T).$$

(ii) *One also has a natural morphism, functorial in T* ⁽¹⁰⁰⁾

$$a_{g_0}(f_0): \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0}), \mathcal{J}\mathcal{O}_T) \longrightarrow \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T).$$

PROOF. ⁽¹⁰¹⁾Let us first remark that, X, X', Y, Y' being fixed, giving a T as above is equivalent to giving a morphism $(f, f_{\mathcal{J}}): T \longrightarrow X' \times_{X'_+} Y'_+$. Set $Z = X' \times_{X'_+} Y'_+$ and denote by M and M' the Z -functors defined by: for every $f: T \longrightarrow Z$,

$$\begin{aligned} M(T) &= \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J}\mathcal{O}_T), \\ M'(T) &= \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^*(\mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0}), \mathcal{J}\mathcal{O}_T). \end{aligned}$$

⁽¹⁰²⁾One has in any case a commutative diagram:

$$\begin{array}{ccc} f_0^*(\mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y'_0}) & & \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0} \\ \downarrow & & \downarrow \\ f_0^*(\mathcal{J}\mathcal{O}_{Y'}) & \dashrightarrow & \mathcal{J}\mathcal{O}_T \end{array}$$

and as Y' is flat over S , the left-hand arrow is an isomorphism, so one obtains a morphism of $\mathcal{O}_{T_0}$ -modules $f_0^*(\mathcal{J}\mathcal{O}_{Y'}) \longrightarrow \mathcal{J}\mathcal{O}_T$. The latter induces a morphism of abelian groups

$$\operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), f_0^*(\mathcal{J}\mathcal{O}_{Y'})) \longrightarrow M(T)$$

and, composing with the morphism

$$M(Y') \longrightarrow \operatorname{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), f_0^*(\mathcal{J}\mathcal{O}_{Y'})),$$

induced by f_0^* , one obtains the morphism $b_{f_0}: M(Y') \longrightarrow M(T)$.

Likewise, one has in any case a diagram

$$\begin{array}{ccc} g_0^*(\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}) & \dashrightarrow & \mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0} \\ \downarrow & & \downarrow \\ g_0^*(\mathcal{N}_{Y_0/X_0}) & \longrightarrow & \mathcal{N}_{Y'_0/X'_0} \end{array}$$

⁽⁹⁹⁾**N.D.E.** From 4.17 onwards, one will apply this to the case where X is an S -group, $g: X \times_S X \longrightarrow X$ the multiplication, Y a subscheme of X such that $Y_{\mathcal{J}}$ is a subgroup of $X_{\mathcal{J}}$, $Y' = Y \times_S Y$, and to the two morphisms $Y^3 \longrightarrow X^2$ which send (y_1, y_2, y_3) to $(y_1 y_2, y_3)$, resp. $(y_1, y_2 y_3)$. In this case, the comparison of the obstructions above will show that the obstruction to Y being a subgroup of X lies in a certain cohomology group (of Hochschild) $H^2(Y_0, N_0)$.

⁽¹⁰⁰⁾**N.D.E.** One has deleted the hypothesis “ Y'_0 locally a complete intersection in X'_0 ”, superfluous by 4.8.0; on the other hand, one has added that $a_{g_0}(f_0)$ is “functorial in T ”, this playing a crucial role in the proof of 4.17.

⁽¹⁰¹⁾**N.D.E.** One has spelled out the proof, in order to bring out the “functoriality in T ” of a_{g_0} .

⁽¹⁰²⁾**N.D.E.** The situation will simplify from 4.16 onwards: one will restrict oneself to schemes *flat* over S , Y will be a flat S -group and $Y' = Y \times_S Y$, and one will then obtain S_0 -functors N_0 and N'_0 .

and as Y' is flat over S , the second vertical arrow is an isomorphism, by 4.8.0. One therefore obtains an $\mathcal{O}_{Y'_0}$ -morphism

$$i_0'^* g_0^* (\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}) \longrightarrow \mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0}$$

which induces a morphism $a_{g_0}(\text{id}_{Y'}) : M'(Y') \longrightarrow M(Y')$ and, for every $f : T \longrightarrow Z$, a morphism $a_{g_0}(f) : M'(T) \longrightarrow M(T)$ such that one has a commutative diagram

$$\begin{array}{ccc} M'(Y') & \xrightarrow{a_{g_0}(\text{id}_{Y'})} & M(Y') \\ b'_{f_0} \downarrow & & \downarrow b_{f_0} \\ M'(T) & \xrightarrow{a_{g_0}(f_0)} & M(T) \end{array}$$

(where b'_{f_0} is defined as b_{f_0}).

Q.E.D.

Remark 4.12.1. ⁽¹⁰³⁾Let us denote by M_0 and M'_0 the Y'_0 -functors defined by: for every $f : T_0 \longrightarrow Y'_0$,

$$M_0(T) = \text{Hom}_{\mathcal{O}_{T_0}}(f_0^* g_0^* (\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}), \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}),$$

$$M'_0(T) = \text{Hom}_{\mathcal{O}_{T_0}}(f_0^* (\mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_0}), \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}).$$

Let us remark at once that $Z_0 = Y'_0$ and that on the category of Z -schemes T which are *flat* over S , M and M' coincide, respectively, with the functors $\prod_{S_0/S} M_0$ and $\prod_{S_0/S} M'_0$. In this case, b_{f_0} is simply the morphism

$$f_0^* : M_0(Y'_0) \longrightarrow M(T_0)$$

induced by f_0 , and for every morphism $u : U \longrightarrow T$, setting $h = f \circ u$, one has a commutative diagram

$$\begin{array}{ccc} M'_0(T_0) & \xrightarrow{a_{g_0}(f_0)} & M_0(T_0) \\ u_0^* \downarrow & & \downarrow u_0^* \\ M'_0(U_0) & \xrightarrow{a_{g_0}(h_0)} & M_0(U_0) \end{array}$$

i.e. a_{g_0} becomes a morphism of functors $\prod_{S_0/S} M'_0 \longrightarrow \prod_{S_0/S} M_0$.

Proposition 4.13. *Suppose Y' flat over S . One then has the formula:*

$$c(X, Y, g \circ f) = a_{g_0}(c(X', Y', f)) + b_{f_0}(c(X, Y, g \circ i')).$$

As the definition of the various obstructions and of the morphisms a_{g_0} and b_{f_0} is local, one sees easily that it suffices to check the given formula when the various schemes in question are affine. Let us therefore write $S = \text{Spec}(\Lambda)$, $S_J = \text{Spec}(\Lambda/J)$, $S_0 = \text{Spec}(\Lambda/I)$, $T = \text{Spec}(C)$, $X = \text{Spec}(A)$, $Y = \text{Spec}(A/I_Y) = \text{Spec}(B)$, $X' = \text{Spec}(A')$, $Y' = \text{Spec}(A'/I_{Y'}) = \text{Spec}(B')$.

One therefore has a diagram of rings and ideals⁽¹⁰⁴⁾

$$\begin{array}{ccccc} & & B' & & B \\ & & \uparrow \pi' & & \uparrow \pi \\ C & \xleftarrow{f} & A' & \xleftarrow{g} & A \\ & & \uparrow & & \uparrow \\ & & I_{Y'} & & I_Y. \end{array}$$

⁽¹⁰³⁾**N.D.E.** One has added this remark, used in the proof of 4.17.

⁽¹⁰⁴⁾**N.D.E.** One has kept the notations of the original, writing $f : A' \longrightarrow C$ and $g : A \longrightarrow A'$ for the ring morphisms corresponding to $f : T \longrightarrow X'$ and $g : X' \longrightarrow X$. This explains the formula $c(X, Y, g \circ f)(x \otimes 1) = f(g(x))$, for $x \in I_Y$.

Let us study the various terms of the formula to be proved. In what follows, if $x \in I_Y$ (resp. $u \in I_{Y'}$), one denotes by x (resp. u) its image in I_Y/I_Y^2 (resp. $I_{Y'}/I_{Y'}^2$); on the other hand, if m belongs to a Λ -module M , one denotes by m_0 its image in $M_0 = M/IM$.

One has seen that $c = c(X, Y, g \circ f)$ is the unique C_0 -morphism $I_Y/I_Y^2 \otimes_B C_0 \rightarrow JC$ such that $c(x \otimes 1) = f(g(x))$, for every $x \in I_Y$.

Let us fix $x \in I_Y$; one has $g(x) \in I_{Y'} + JA'$ since $g_J(Y_J') \subset Y_J$. Let us write $g(x) = x' + \sum \lambda_i a'_i$, with $x' \in I_{Y'}$, $\lambda_i \in J$, $a'_i \in A'$. One therefore has

$$(1) \quad c(X, Y, g \circ f)(x \otimes 1) = f(g(x)) = f(x') + \sum \lambda_i f(a'_i).$$

Let us now consider $a_{g_0}(c(X', Y', f))$. By the definitions set down, it is defined by the diagram

$$\begin{array}{ccccc} & & I_{Y'} & \xrightarrow{f} & C \\ & & \downarrow & & \\ I_{Y'_0}/I_{Y'_0}^2 \otimes_{B'_0} C_0 & \xleftarrow{\sim} & I_{Y'}/I_{Y'}^2 \otimes_{B'} C_0 & \xrightarrow{c(X', Y', f)} & JC \\ & \uparrow g_0 & \uparrow & \nearrow a_{g_0}(c(X', Y', f)) & \\ I_{Y_0}/I_{Y_0}^2 \otimes_{B_0} C_0 & \xleftarrow{\sim} & I_Y/I_Y^2 \otimes_B C_0 & & \end{array}$$

One therefore has $a_{g_0}(c(X', Y', f))(x \otimes 1) = f(u)$, where u is an element of $I_{Y'}$ whose image u in $I_{Y'_0}/I_{Y'_0}^2$ satisfies $u_0 \otimes 1 = g_0(x_0) \otimes 1 = g_0(x_0) \otimes 1$. One may therefore take $u = x'$, and one finds

$$(2) \quad a_{g_0}(c(X', Y', f))(x \otimes 1) = f(x').$$

Let us finally consider $b_{f_0}(c(X, Y, g \circ i'))$. By hypothesis, the morphism of Λ_0 -algebras $f_0: A'_0 \rightarrow C_0$ factors through B'_0 and hence, as $J \otimes_{\Lambda_0} B'_0 \xrightarrow{\sim} JB'$ (B' being flat over Λ), one obtains a morphism of B'_0 -modules $\psi: JB' \rightarrow JC$ such that one has a commutative diagram:

$$\begin{array}{ccccc} J \otimes_{\Lambda_0} A'_0 & \xrightarrow{\text{id} \otimes \pi'} & J \otimes_{\Lambda_0} B'_0 & \xrightarrow{\text{id} \otimes f_0} & J \otimes_{\Lambda_0} C_0 \\ \downarrow & & \downarrow \wr & & \downarrow \\ JA' & \xrightarrow{\pi'} & JB' & \xrightarrow{\psi} & JC. \end{array}$$

Denote by $\phi: JB' \otimes_{B'_0} C_0 \rightarrow JC$ the morphism of C_0 -modules deduced from ψ ; then one has, for every $a' \in A'$, $\lambda \in J$,

$$(\dagger) \quad \phi(\lambda \pi'(a') \otimes 1) = \lambda f(a').$$

Then, $b_{f_0}(c(X, Y, g \circ i'))$ is defined by the commutative diagram:

$$\begin{array}{ccc} I_Y & \xrightarrow{\pi' \circ g} & B' \\ \downarrow & & \downarrow \\ I_Y/I_Y^2 \otimes_B B'_0 & \xrightarrow{c(X, Y, g \circ i')} & JB' \\ \downarrow & & \downarrow \\ I_Y/I_Y^2 \otimes_B C_0 & \xrightarrow{\quad} & JB' \otimes_{B'_0} C_0 \\ & \searrow b_{f_0}(c(X, Y, g \circ i')) & \downarrow \phi \\ & & JC. \end{array}$$

One therefore has at once

$$(3) \quad b_{f_0}(c(X, Y, g \circ i'))(x \otimes 1) = \phi\left(\sum \lambda_i \pi'(a'_i) \otimes 1\right) = \sum \lambda_i f(a'_i),$$

the last equality following from $(\dagger)$ above. The comparison of the three explicit results (1), (2), (3) gives the desired formula.

Corollary 4.14. *Let Y, Y' be two subschemes of X , flat, reducing to $Y_{\mathcal{J}}$; suppose Y_0 locally a complete intersection in X_0 . If $f: T \rightarrow X$ is an S -morphism such that $f_{\mathcal{J}}$ factors through $Y_{\mathcal{J}} \rightarrow X_{\mathcal{J}}$, one has the formula*

$$c(X, Y, f) - c(X, Y', f) = b_{f_0}(d(Y, Y')).$$

Indeed, applying the preceding formula to the diagram

$$\begin{array}{ccccc} & & Y' & & Y \\ & & \downarrow i' & & \downarrow i \\ T & \xrightarrow{f} & X & \xrightarrow{\text{id}} & X \end{array}$$

one finds $c(X, Y, f) - c(X, Y', f) = b_{f_0}(c(X, Y, i'))$. Moreover, by 4.8 (ii), one has $c(X, Y, i') = d(Y, Y')$.

Proposition 4.15. *Let X be an S -group smooth over S and Y an S -subgroup flat and locally of finite presentation over S . Then Y is locally a complete intersection (cf. 4.6.2) in X .*

PROOF. ⁽¹⁰⁵⁾One will show that the immersion $Y \rightarrow X$ is regular in the sense of EGA IV₄, 16.9.2, which implies that it is so also in the sense of 4.6.2, by EGA IV₄, 19.5.1 (moreover, by *loc. cit.*, the two definitions are equivalent if S is locally noetherian). Thus, in what follows, one takes “regular immersion” in the sense of EGA IV₄, 16.9.2. As X and Y are flat and locally of finite presentation over S , then, by EGA IV₄, 19.2.4, it suffices to show that, for every $s \in S$, $Y_s \rightarrow X_s$ is a regular immersion. By EGA IV₄, 19.1.5 (ii), one is reduced to checking the assertion on the geometric fibers of S , hence when S is the spectrum of an algebraically closed field k .

Then, by VI_A, 3.2, the quotient X/Y exists and is smooth, the morphism $\pi: X \rightarrow X/Y$ is flat, and one has a cartesian square

$$\begin{array}{ccc} Y & \xrightarrow{f} & X \\ \downarrow & & \downarrow \pi \\ e & \xrightarrow{i} & X/Y \end{array}$$

(where e is the image in X/Y of the unit point of X). Thus, by flat base change (cf. EGA IV₄, 19.1.5 (ii)), it suffices to see that i is a regular immersion, which is immediate since the noetherian local ring $\mathcal{O}_{X/Y, e}$ is smooth, hence its maximal ideal is generated by a regular sequence. $\square$

⁽¹⁰⁵⁾**N.D.E.** One has added to the statement the hypothesis that Y be locally of finite presentation over S , and one has given the proof which follows, more direct than the one sketched in the original. For completeness, let us also spell out the latter. As in the proof given above, one first reduces to the case where $S = \text{Spec}(k)$, k being an algebraically closed field. By EGA IV₄, 16.9.10 and 19.3.2, it suffices to see that, for every $y \in Y$, the completion of the local ring $\mathcal{O}_{Y, y}$ is the quotient of a complete noetherian local ring by a regular sequence. By *loc. cit.*, 19.3.3, the set of $y \in Y$ satisfying this property is an open subset U of Y ; as Y is of finite type over k , it suffices to show that U contains every closed point. As Y is a k -group it suffices, by a translation argument, to show that the property holds for the completion of $\mathcal{O}_{Y, e}$, i.e. for the “formal group” $\hat{Y}$ corresponding to Y (cf. Exp. VII_B). Now, as X is smooth, the affine algebra $A(\hat{X})$ is an algebra of formal power series $k[[X_1, \dots, X_n]]$, and one concludes by means of the structure theorem of Dieudonné, which shows that $A(\hat{Y})$ is isomorphic to a quotient $k[[X_1, \dots, X_{r+s}]]/(X_1^{p^{n_1}}, \dots, X_r^{p^{n_r}})$, where p is the characteristic exponent of k and $r + s \leq n$, cf. VII_B, Remark 5.5.2 (b).

4.16. — ⁽¹⁰⁶⁾Let X be an S -group *smooth* over S ; denote by $\mu: X \times_S X \rightarrow X$ its group law. Let us be given an $S_{\mathcal{J}}$ -subgroup $Y_{\mathcal{J}}$ of $X_{\mathcal{J}}$, *flat and locally of finite presentation* over $S_{\mathcal{J}}$. By 4.15, $Y_{\mathcal{J}}$ is locally a complete intersection in X .

Thus, by 4.6.5, every *flat*⁽¹⁰⁷⁾ S -scheme Y lifting $Y_{\mathcal{J}}$ is locally a complete intersection in X . For such a Y one has, by 4.8.0,

$$(4.16.1) \quad \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0} = \mathcal{N}_{Y_0/X_0} = \mathcal{N}_{Y_{\mathcal{J}}/X_{\mathcal{J}}} \otimes_{\mathcal{O}_{Y_{\mathcal{J}}}} \mathcal{O}_{Y_0}.$$

On the other hand, let us denote by $\varepsilon_0: S_0 \rightarrow Y_0$ the unit section of Y_0 and by $\mathbf{n}_{Y_0/X_0}$ the quasi-coherent $\mathcal{O}_{S_0}$ -module:

$$\mathbf{n}_{Y_0/X_0} = \varepsilon_0^*(\mathcal{N}_{Y_0/X_0}).$$

As Y_0 and X_0 are S_0 -groups, one sees easily that $\mathcal{N}_{Y_0/X_0}$ is invariant under the translations (say on the left) of Y_0 , hence⁽¹⁰⁸⁾ is the inverse image under $Y_0 \rightarrow S_0$ of $\mathbf{n}_{Y_0/X_0}$, i.e. one has

$$(4.16.2) \quad \mathcal{N}_{Y_0/X_0} = \mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}.$$

Taking (4.16.1) and (4.16.2) into account, one deduces on the one hand from 4.5 that the set of sub- S -schemes Y of X , *flat* over S , lifting $Y_{\mathcal{J}}$, is empty or a principal homogeneous space under

$$(4.16.3) \quad \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}),$$

and one deduces on the other hand from 4.8 (i) that, for every such Y and every S -morphism $f: T \rightarrow X$ such that $f_{\mathcal{J}}: T_{\mathcal{J}} \rightarrow X_{\mathcal{J}}$ factors through $Y_{\mathcal{J}}$, the obstruction $c(X, Y, f)$ to f factoring through Y is an element of

$$\mathrm{Hom}_{\mathcal{O}_{T_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0});$$

if moreover T is *flat* over S , this last group equals

$$\mathrm{Hom}_{\mathcal{O}_{T_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}).$$

This leads one to introduce the group functor N_0 below:

Definition 4.16.1. Let N_0 be the commutative group S_0 -functor defined by: for every $Z \in \mathrm{Ob}(\mathrm{Sch})/S_0$,

$$(*) \quad \mathrm{Hom}_{S_0}(Z, N_0) = \mathrm{Hom}_{\mathcal{O}_Z}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_Z, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_Z).$$

Then the set of sub- S -schemes Y of X , *flat* over S , lifting $Y_{\mathcal{J}}$, is empty or a principal homogeneous space under

$$\mathrm{Hom}_{S_0}(Y_0, N_0) = C^1(Y_0, N_0).$$

For each such Y , let us consider the following diagram:

$$\begin{array}{ccc} Y \times_S Y & & Y \\ \downarrow (i,i) & & \downarrow i \\ X \times_S X & \xrightarrow{\mu} & X \end{array}$$

and let us denote by $DY = c(X, Y, \mu \circ (i, i))$ the obstruction to $\mu \circ (i, i)$ factoring through Y , i.e. to Y being stable under the group law of X ; by what precedes, DY is an element of

$$N_0(Y_0 \times_{S_0} Y_0) = C^2(Y_0, N_0).$$

⁽¹⁰⁶⁾**N.D.E.** One has reorganized 4.16 by gathering in it, on the one hand, the hypotheses stated at the end of 4.15 and, on the other hand, the definition of the obstruction DY .

⁽¹⁰⁷⁾**N.D.E.** One has corrected the original by adding “flat”.

⁽¹⁰⁸⁾**N.D.E.** See 4.25 below.

Lemma 4.17. ⁽¹⁰⁹⁾ *Let X be an S -group smooth over S , and $Y_{\mathcal{J}}$ a sub- $S_{\mathcal{J}}$ -group of $X_{\mathcal{J}}$, flat and locally of finite presentation over $S_{\mathcal{J}}$. For each subscheme Y of X , flat over S and lifting $Y_{\mathcal{J}}$, let us consider the obstruction defined in 4.16.1:*

$$DY \in \operatorname{Hom}_{S_0}(Y_0 \times_{S_0} Y_0, N_0) = C^2(Y_0, N_0)$$

(i) *In order that Y be a sub- S -group of X , it is necessary and sufficient that $DY = 0$.*

(ii) *If one lets Y_0 act on N_0 by functoriality, starting from the inner automorphisms of Y_0 , then $DY \in Z^2(Y_0, N_0)$.*

(iii) *If Y and Y' are two subschemes of X , flat over S , lifting $Y_{\mathcal{J}}$ (so that the deviation $d(Y, Y') \in C^1(Y_0, N_0)$ is defined, cf. 4.5.1), one has $DY' - DY = \partial^1 d(Y, Y')$.* ⁽¹¹⁰⁾

Let us prove these various assertions in succession.

4.18. — *Proof of 4.17 (i).* By definition, one has $DY = 0$ if and only if Y is stable under the group law of X . Thus $DY = 0$ if Y is a subgroup of X . Conversely, if $DY = 0$, Y is equipped with the induced law μ^Y , which is associative and reduces modulo $\mathcal{J}$ according to the group law on $X_{\mathcal{J}}$; since Y is flat and locally of finite presentation over S , it follows from 3.3 that μ^Y is a group law.

4.19. — *Proof of 4.17 (ii).* This is done by comparing the two values of $u = c(X, Y, \mu^2 \circ (i, i, i))$ computed in the following two diagrams (D_j) , $j = 1, 2$:

$$(D_j) \quad \begin{array}{ccccc} Y \times_S Y \times_S Y & & Y \times_S Y & & Y \\ \downarrow (i, i, i) & & \downarrow (i, i) & & \downarrow i \\ X \times_S X \times_S X & \xrightarrow{f_j} & X \times_S X & \xrightarrow{\mu} & X \end{array}$$

where $f_1 = (1, \pi)$, $f_2 = (\pi, 1)$, and where one denotes by μ^2 the morphism

$$\mu \circ f_1 = \mu \circ f_2: X \times_S X \times_S X \longrightarrow X.$$

⁽¹¹¹⁾ Set $\mu^Y = \mu \circ (i, i)$, $f_j^Y = f_j \circ (i, i, i)$ and $\mu^{2,Y} = \mu^2 \circ (i, i, i)$. For $j = 1, 2$, let us denote by a_j and b_j the morphisms

$$a_j = a_{\mu_0}((f_j^Y)_0) \quad \text{and} \quad b_j = b_{(f_j^Y)_0},$$

associated with the pair of morphisms (f_j^Y, μ) by Lemma 4.12; one therefore has:

$$(\dagger) \quad \begin{cases} \operatorname{Hom}_{\mathcal{O}_{Y_0^3}}((f_j^Y)_0^*(\mathcal{N}_{Y_0 \times Y_0/X_0 \times X_0}), \mathcal{J} \mathcal{O}_{Y_0^3}) \xrightarrow{a_j} \operatorname{Hom}_{\mathcal{O}_{Y_0^3}}((\mu^{2,Y})_0^*(\mathcal{N}_{Y_0/X_0}), \mathcal{J} \mathcal{O}_{Y_0^3}) \\ \operatorname{Hom}_{\mathcal{O}_{Y_0^2}}((\mu^Y)_0^*(\mathcal{N}_{Y_0/X_0}), \mathcal{J} \mathcal{O}_{Y_0^2}) \xrightarrow{b_j} \operatorname{Hom}_{\mathcal{O}_{Y_0^3}}((\mu^{2,Y})_0^*(\mathcal{N}_{Y_0/X_0}), \mathcal{J} \mathcal{O}_{Y_0^3}). \end{cases}$$

Since $\mathcal{N}_{Y_0 \times Y_0/X_0 \times X_0} \simeq \operatorname{pr}_1^* \mathcal{N}_{Y_0/X_0} \oplus \operatorname{pr}_2^* \mathcal{N}_{Y_0/X_0}$ (since X_0 and Y_0 are flat over S_0), and $\mathcal{N}_{Y_0/X_0} \simeq \mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}$, one has:

$$(f_j^Y)_0^*(\mathcal{N}_{Y_0 \times Y_0/X_0 \times X_0}) \simeq (\mathbf{n}_{Y_0/X_0} \oplus \mathbf{n}_{Y_0/X_0}) \otimes \mathcal{O}_{Y_0^3}$$

and, likewise,

$$(\mu^{2,Y})_0^*(\mathcal{N}_{Y_0/X_0}) \simeq \mathbf{n}_{Y_0/X_0} \otimes \mathcal{O}_{Y_0^3} \quad \text{and} \quad (\mu^Y)_0^*(\mathcal{N}_{Y_0/X_0}) \simeq \mathbf{n}_{Y_0/X_0} \otimes \mathcal{O}_{Y_0^2}.$$

Moreover, since Y_0^2 and Y_0^3 are flat over S_0 , $(\dagger)$ is rewritten in the following form:

$$(\ddagger) \quad \begin{cases} a_j: \operatorname{Hom}_{S_0}(Y_0^3, N_0 \oplus N_0) \longrightarrow \operatorname{Hom}_{S_0}(Y_0^3, N_0) \\ b_j: \operatorname{Hom}_{S_0}(Y_0^2, N_0) \longrightarrow \operatorname{Hom}_{S_0}(Y_0^3, N_0). \end{cases}$$

⁽¹⁰⁹⁾**N.D.E.** One has modified 4.17 and 4.18, taking into account the additions made in 4.16.

⁽¹¹⁰⁾**N.D.E.** In the original, one finds $DY' - DY = -\partial d(Y, Y')$, but ∂ there is the opposite of the differential ∂^1 defined in I, 5.1.

⁽¹¹¹⁾**N.D.E.** One has slightly modified the notations, and spelled out the beginning of the argument.

Applying 4.13 twice to $c(X, Y, \mu^{2,Y}) = u$, one obtains:

$$a_1(c(X^2, Y^2, f_1)) + b_1(c(X, Y, \mu^Y)) = u = a_2(c(X^2, Y^2, f_2)) + b_2(c(X, Y, \mu^Y)).$$

Now, $c(X, Y, \mu^Y) = DY$ and, since $f_1 = (1, \mu)$ and $f_2 = (\mu, 1)$, one has, with evident notations:

$$c(X^2, Y^2, f_1) = (0, DY) \quad \text{and} \quad c(X^2, Y^2, f_2) = (DY, 0).$$

Thus one obtains:

$$u = a_1((0, DY)) + b_1(DY) = a_2((DY, 0)) + b_2(DY).$$

The first thing that one notices is that b_j is none other than $\text{Hom}_{S_0}((f_j^Y)_0, N_0)$, that is, the morphism deduced from $(f_j^Y)_0$ by functoriality.

The identity above therefore becomes:

$$a_1((0, DY)) - \text{Hom}((\mu, 1), N_0)(DY) + \text{Hom}((1, \mu), N_0)(DY) - a_2((DY, 0)) = 0.$$

One recognizes the two middle terms: they are the parts “ $DY(xy, z)$ ” and “ $DY(x, yz)$ ” of the formula of the 2-coboundary. It therefore only remains to identify the two other terms.

We must first compute the map a_j . Now it arises, by inverse image under $(f_j^Y)_0$, from the morphism of $\mathcal{O}_{Y_0^2}$ -modules

$$P: \mathbf{n}_{Y_0/X_0} \otimes \mathcal{O}_{Y_0^2} \longrightarrow (\mathbf{n}_{Y_0/X_0} \oplus \mathbf{n}_{Y_0/X_0}) \otimes \mathcal{O}_{Y_0^2}$$

induced by the product in Y_0 . Now this morphism is described in the following way: let us consider the vector bundle $\mathbf{V} = \mathbf{V}(\mathbf{n}_{Y_0/X_0})$; P gives by duality a morphism

$$\mathbf{V}(P): \mathbf{V} \times_{S_0} \mathbf{V} \times_{S_0} Y_0 \times_{S_0} Y_0 \longrightarrow \mathbf{V} \times_{S_0} Y_0 \times_{S_0} Y_0$$

which is expressed set-theoretically by

$$\mathbf{V}(P)(u, v, a, b) = (u + \text{Ad}(a)v, ab, b).$$

⁽¹¹²⁾ This is proved exactly like the corresponding fact for Lie algebras, that is, for the module ω_{Y_0/S_0}^1 . One first observes that $\mathbf{V}$ is equipped, by functoriality in Y_0 , with a group structure in the category of vector bundles over S_0 ; by virtue of the lemma already used for Lie algebras (Exposé II, 3.10), this structure coincides with the group structure underlying its structure of $\mathcal{O}_S$ -module. One then sees that $\mathbf{V}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}) = \mathbf{V}(\mathcal{N}_{Y_0/X_0})$ is likewise equipped with a structure of S_0 -group which is none other than the semidirect product of that of $\mathbf{V}$ by that of Y_0 . It only remains to identify the operations of Y_0 on $\mathbf{V}$ in order to establish the desired formula.

Let us now compute the two remaining terms. Let us first consider $a_1((0, DY))$. One computes it by the diagram (where $\mathbf{n}$ denotes $\mathbf{n}_{Y_0/X_0}$):

$$\begin{array}{ccc} \mathbf{n} \otimes \mathcal{O}_{Y_0^2} & \xrightarrow{P} & (\mathbf{n} + \mathbf{n}) \otimes \mathcal{O}_{Y_0^2} \\ \downarrow (f_1^Y)_0^* & & \downarrow (f_1^Y)_0^* \\ \mathbf{n} \otimes \mathcal{O}_{Y_0^3} & \searrow a_1((0, DY)) & (\mathbf{n} + \mathbf{n}) \otimes \mathcal{O}_{Y_0^3} \\ & & \downarrow (0, DY) \\ & & \mathcal{J} \otimes \mathcal{O}_{Y_0^3}. \end{array}$$

Considering now the vector bundles defined by these various modules as so many schemes over S_0 , and taking the points with values in anything whatsoever, one has, denoting by

⁽¹¹²⁾**N.D.E.** One has replaced a, b by ab, b , in order to make it plain that $\mathbf{V}(P)$ arises by inverse image on Y_0^2 from the multiplication morphism $\mathbf{V}_{Y_0} \times_{S_0} \mathbf{V}_{Y_0} \longrightarrow \mathbf{V}_{Y_0}$.

(u, x, y, z) a point of $\mathbf{V}(\mathcal{J}) \times Y_0^3$:

$$\begin{array}{ccc}
 (\mathrm{Ad}(x) DY_{y,z}(u), x, yz) & \longleftarrow & (0 + DY_{y,z}(u), x, yz) \\
 \uparrow & & \uparrow \\
 & & (0 + DY_{y,z}(u), x, y, z) \\
 & & \uparrow \\
 (\mathrm{Ad}(x) DY_{y,z}(u), x, y, z) & \longleftarrow & (u, x, y, z)
 \end{array}$$

One has therefore obtained $a_1((0, DY))(x, y, z) = \mathrm{Ad}(x) DY(y, z)$, which is indeed the first term of the coboundary. One would likewise have $a_2((DY, 0))(x, y, z) = DY(x, y)$, whence⁽¹¹³⁾

$$0 = \mathrm{Ad}(x) DY(y, z) - DY(xy, z) + DY(x, yz) - DY(x, y) = (\partial^2 DY)(x, y, z).$$

4.20. — *Proof of 4.17 (iii).*⁽¹¹⁴⁾ This is done by comparing the two values of $v = c(X, Y, \mu \circ (i', i'))$ computed in the following two diagrams

$$\begin{array}{ccc}
 (*) & & \\
 & Y' & Y \\
 & \downarrow i' & \downarrow i \\
 Y' \times_S Y \xrightarrow{\mu \circ (i', i')} & X & X \\
 \\
 (\dagger) & & \\
 & Y \times_S Y & Y \\
 & \downarrow (i, i) & \downarrow i \\
 Y' \times_S Y' \xrightarrow{(i', i')} & X \times_S X \xrightarrow{\mu} & X.
 \end{array}$$

Let us denote $f = \mu \circ (i', i')$; then $(*)$ gives

$$(1) \quad v = DY' + f_0^*(c(X, Y, i')).$$

Now $Y'_0 = Y_0$ and f_0 is the multiplication $Y_0^2 \rightarrow Y_0$; one deduces from this that

$$(2) \quad f_0^*(c(X, Y, i'))(x_0, y_0) = c(X, Y, i')(x_0 y_0).$$

Set $c = c(X, Y, i')$; via the identification $N'_0 \simeq N_0 \oplus N_0$, $c(X \times_S X, Y \times_S Y, (i', i'))$ is identified with the pair (c, c) . Then, denoting $h = (i', i')$, $(\dagger)$ gives

$$(3) \quad v = h_0^*(DY) + a_{\mu_0}(c, c).$$

Now h_0 is the identity map of Y_0^2 , whence $h_0^*(DY) = DY$. Finally, by the computation of a_{μ_0} carried out previously, one has, for every $S' \rightarrow S$ and $x_0, y_0 \in Y_0(S'_0)$,

$$(4) \quad a_{\mu_0}(c, c)(x_0, y_0) = c(x_0) + \mathrm{Ad}(x_0)(c(y_0)).$$

One therefore obtains:

$$\begin{aligned}
 (DY' - DY)(x_0, y_0) &= \mathrm{Ad}(x_0)(c(X, Y, i')(y_0)) - c(X, Y, i')(x_0 y_0) + c(X, Y, i')(x_0) \\
 &= (\partial^1 c(X, Y, i'))(x_0, y_0).
 \end{aligned}$$

Since $c(X, Y, i') = d(Y, Y')$ (cf. 4.8 (ii)), this shows that $DY' - DY = \partial^1 d(Y, Y')$.

Theorem 4.21. *Let S be a scheme, $\mathcal{I}$ and $\mathcal{J}$ two ideals⁽¹¹⁵⁾ on S such that $\mathcal{I} \supset \mathcal{J}$ and $\mathcal{I} \cdot \mathcal{J} = 0$. Let X be an S -group smooth over S , and $Y_{\mathcal{J}}$ a sub- $S_{\mathcal{J}}$ -group of $X_{\mathcal{J}}$, flat and*

⁽¹¹³⁾**N.D.E.** One has changed the signs, in order to make them compatible with I 5.1.

⁽¹¹⁴⁾**N.D.E.** One has spelled out the original in what follows.

⁽¹¹⁵⁾**N.D.E.** One has removed the hypothesis that $\mathcal{I}$ be nilpotent, which seems superfluous.

locally of finite presentation over $S_{\mathcal{J}}$. Let us consider the commutative S_0 -group functor N_0 defined by

$$\mathrm{Hom}_{S_0}(T, N_0) = \mathrm{Hom}_{\mathcal{O}_T}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T), \quad T \in \mathrm{Ob}(\mathrm{Sch})_{/S_0},$$

on which Y_0 acts by means of the inner automorphisms of X_0 .

(i) In order that there exist a sub- S -group of X , flat over S , which reduces to $Y_{\mathcal{J}}$, it is necessary and sufficient that the following two conditions be satisfied:

(i₁) There exists a subscheme Y of X , flat over S , lifting $Y_{\mathcal{J}}$ (a condition automatically satisfied if Y_0 is affine, cf. 4.6.5).

(i₂) A certain canonical obstruction, an element of $H^2(Y_0, N_0)$, is zero.

(ii) If the conditions of (i) are satisfied, the set of sub- S -groups Y of X , flat over S and reducing to $Y_{\mathcal{J}}$, is a principal homogeneous set under the group $Z^1(Y_0, N_0)$.⁽¹¹⁶⁾

Indeed, condition (i₁) is necessary. Let us assume it satisfied, and let Y be flat over S and lifting $Y_{\mathcal{J}}$. We must seek a Y' flat over S , also lifting $Y_{\mathcal{J}}$, such that $DY' = 0$,⁽¹¹⁷⁾ cf. 4.17 (i). By 4.17 (iii), this amounts to seeking a $d(Y', Y) \in C^1(Y_0, N_0)$ such that $DY = \partial^1 d(Y', Y)$.⁽¹¹⁸⁾

Let $c \in H^2(Y_0, N_0)$ be the image class of DY , which is a cocycle by 4.17 (ii). It does not depend on the choice of Y by 4.17 (iii), and its vanishing is necessary and sufficient for the existence of a $d(Y', Y)$ satisfying the preceding equation. This proves (i).

If one has now chosen Y such that $DY = 0$, the equation to be solved reads $\partial^1 d(Y', Y) = 0$, which proves (ii).

Remark 4.22. One keeps the notations of 4.21. By 4.15, Y_0 is locally a complete intersection in X_0 , so $\mathcal{N}_{Y_0/X_0}$ is a finite locally free $\mathcal{O}_{Y_0}$ -module, and consequently $\mathbf{n}_{Y_0/X_0} = \varepsilon_0^*(\mathcal{N}_{Y_0/X_0})$ is a finite locally free $\mathcal{O}_{S_0}$ -module. Thus, denoting $\mathbf{n}_{Y_0/X_0}^\vee = \mathcal{H}\mathrm{om}_{\mathcal{O}_{Y_0}}(\mathbf{n}_{Y_0/X_0}, \mathcal{O}_{Y_0})$, one has

$$\mathrm{Hom}_{\mathcal{O}_T}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T) \simeq \mathbf{n}_{Y_0/X_0}^\vee \otimes_{\mathcal{O}_{S_0}} \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T$$

for every $T \rightarrow S_0$.⁽¹¹⁹⁾ Consequently, the S_0 -functor N_0 is isomorphic to the functor

$$\mathbf{W}(\mathbf{n}_{Y_0/X_0}^\vee \otimes_{\mathcal{O}_{S_0}} \mathcal{J}) \simeq \mathbf{W}(\mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathcal{J})).$$

There result the isomorphisms:⁽¹²⁰⁾

$$H^2(Y_0, N_0) \simeq H^2(Y_0, \mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathcal{J})) \simeq H^2(Y_0, \mathbf{n}_{Y_0/X_0}^\vee \otimes_{\mathcal{O}_{S_0}} \mathcal{J}),$$

$$Z^1(Y_0, N_0) \simeq Z^1(Y_0, \mathcal{H}\mathrm{om}_{\mathcal{O}_{S_0}}(\mathbf{n}_{Y_0/X_0}, \mathcal{J})) \simeq Z^1(Y_0, \mathbf{n}_{Y_0/X_0}^\vee \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

4.23. — Still under the hypotheses of 4.21, we shall now study how the set of the Y lifting $Y_{\mathcal{J}}$ behaves with respect to conjugation by sections of X . If x is a section of X over S inducing the unit section of $X_{\mathcal{J}}$, the inner automorphism $\mathrm{Int}(x)$ defined by x transforms flat subgroups of X lifting $Y_{\mathcal{J}}$ into flat subgroups of X lifting $Y_{\mathcal{J}}$. Now, under the conditions of 4.21 (ii), the set of these subgroups is principal homogeneous under $Z^1(Y_0, N_0)$; we shall see that there then exists a subgroup Γ of $B^1(Y_0, N_0)$ ⁽¹²¹⁾ such that two subgroups of X , flat over S and lifting $Y_{\mathcal{J}}$, are conjugate (by $x \in X(S)$ inducing the unit element of $X(S_{\mathcal{J}})$) if and only if their “difference” in $Z^1(Y_0, N_0)$ is an element of Γ . In the most favorable cases, we shall

⁽¹¹⁶⁾**N.D.E.** The question whether the preceding set, modulo conjugation by the $x \in X(S)$ inducing the unit element of $X(S_{\mathcal{J}})$, is principal homogeneous under $H^1(Y_0, N_0)$, occupies nos. 4.23 to 4.36.

⁽¹¹⁷⁾**N.D.E.** One has corrected $\partial DY'$ to DY' .

⁽¹¹⁸⁾**N.D.E.** Cf. N.D.E. (110).

⁽¹¹⁹⁾**N.D.E.** One has spelled out what precedes; this shows that the isomorphism which follows is valid without a flatness hypothesis. On the other hand, since 4.16, one has restricted oneself to the S -schemes $f: T \rightarrow S$ flat over S , in order to ensure that the group $\mathrm{Hom}_{\mathcal{O}_{T_0}}(\mathbf{n}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0}, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{T_0})$, in which the obstruction $c(X, Y, f)$ lies, coincides with $N_0(T_0)$ (cf. the end of 4.16).

⁽¹²⁰⁾**N.D.E.** with the notations of I 5.3, assuming Y_0 affine over S_0 .

⁽¹²¹⁾**N.D.E.** One has replaced $Z^1(Y_0, N_0)$ by $B^1(Y_0, N_0)$, since the proof shows that Γ is a subgroup of $B^1(Y_0, N_0)$, cf. 4.27–4.29.

show that Γ equals $B^1(Y_0, N_0)$, so that the set of flat subgroups of X lifting $Y_{\mathcal{J}}$, modulo conjugation by the $x \in X(S)$ inducing the unit element of $X(S_{\mathcal{J}})$, is empty or principal homogeneous under $H^1(Y_0, N_0)$ (cf. 4.29 and 4.36).

4.24. — One keeps the notations of 4.21. Let Y be a flat subgroup of X , reducing to $Y_{\mathcal{J}}$. Let us recall that we introduced in 0.5 the functor L_{0X} (resp. L_{0Y}) defined by the following identity with respect to the variable S_0 -scheme T :

$$\mathrm{Hom}_{S_0}(T, L_{0X}) = \mathrm{Hom}_{\mathcal{O}_T}(\omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T, \mathcal{J} \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_T)$$

(resp. likewise upon replacing X by Y), as well as the functor $L'_X = \prod_{S_0/S} L_{0X}$.

Now one has:

Lemma 4.25. *There exists a canonical exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules*

$$(+)\quad \mathbf{n}_{Y_0/X_0} \xrightarrow{d} \omega_{X_0/S_0}^1 \longrightarrow \omega_{Y_0/S_0}^1 \longrightarrow 0$$

having the following properties:

- (i) By inverse image on Y_0 , d gives the morphism D_0 of 4.8 (iii).
- (ii) If X_0 and Y_0 are smooth over S_0 , then d is injective. Since the two ω^1 are then locally free of finite type, the same holds for $\mathbf{n}_{Y_0/X_0}$, and the sequence is locally split.

PROOF. ⁽¹²²⁾ Let π_0 denote the morphism $Y_0 \rightarrow S_0$. By SGA 1 II, formula (4.3) (see also EGA IV₄, 16.4.21), one has a canonical exact sequence of $\mathcal{O}_{Y_0}$ -modules

$$(\dagger)\quad \mathcal{N}_{Y_0/X_0} \xrightarrow{D_0} \Omega_{X_0/S_0}^1 \otimes_{\mathcal{O}_{X_0}} \mathcal{O}_{Y_0} \longrightarrow \Omega_{Y_0/S_0}^1 \longrightarrow 0.$$

Since this sequence is formed of $(Y_0 \times_S Y_0)$ -equivariant modules and morphisms, its inverse image $(+)$ by ε_0^* is an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules, and $(\dagger)$ is the inverse image of $(+)$ by π_0^* (cf. Exp. I, § 6.8). This proves (i).

Suppose moreover that X_0 and Y_0 are smooth over S_0 . Then, by SGA 1 II 4.10 (see also EGA IV₄, 17.2.3 (i) and 17.2.5), D is injective and the sequence $(\dagger)$ is formed of $\mathcal{O}_{Y_0}$ -modules locally free of finite type (hence is locally split). By the equivalence of categories I, 6.8.1, d is likewise injective, and thus the sequence $(+)$ has the properties indicated. $\square$

4.26. — ⁽¹²³⁾ For every S_0 -scheme $f: T \rightarrow S_0$, $(+)$ gives an exact sequence of $Y_0(T)$ - $\mathcal{O}(T)$ -modules

$$\begin{aligned} 0 \longrightarrow \mathrm{Hom}_{\mathcal{O}_T}(f^*(\omega_{Y_0/S_0}^1), f^*(\mathcal{J})) &\longrightarrow \mathrm{Hom}_{\mathcal{O}_T}(f^*(\omega_{X_0/S_0}^1), f^*(\mathcal{J})) \\ &\longrightarrow \mathrm{Hom}_{\mathcal{O}_T}(f^*(\mathbf{n}_{Y_0/X_0}), f^*(\mathcal{J})), \end{aligned}$$

one therefore has an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules:

$$(4.26.1)\quad 0 \longrightarrow L_{0Y} \longrightarrow L_{0X} \xrightarrow{d} N_0.$$

One deduces from this an exact sequence of complexes of abelian groups:

$$0 \longrightarrow C^*(Y_0, L_{0Y}) \longrightarrow C^*(Y_0, L_{0X}) \xrightarrow{d^*} C^*(Y_0, N_0),$$

and in particular, a commutative diagram with exact rows

$$\begin{array}{ccccc} 0 \longrightarrow C^0(Y_0, L_{0Y}) & \longrightarrow & C^0(Y_0, L_{0X}) & \xrightarrow{d^0} & C^0(Y_0, N_0) \\ & \downarrow \partial & \downarrow \partial & & \downarrow \partial \\ 0 \longrightarrow C^1(Y_0, L_{0Y}) & \longrightarrow & C^1(Y_0, L_{0X}) & \xrightarrow{d^1} & C^1(Y_0, N_0). \end{array}$$

Let us note that $C^0(Y_0, L_{0Y})$ (resp. $C^0(Y_0, L_{0X})$) is none other than $\mathrm{Hom}_{S_0}(S_0, L_{0Y}) = \mathrm{Hom}_S(S, L'_Y)$ (resp. $\cdots$), i.e. (cf. 0.9) the group of sections of Y (resp. of X) over S inducing

⁽¹²²⁾**N.D.E.** One has spelled out the proof, taking into account the additions made in Exp. I, § 6.8.

⁽¹²³⁾**N.D.E.** One has spelled out the original, in order to make it plain that one has an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules.

the unit section of $X_{\mathcal{J}}$. Let us also note that d^1 is none other than the morphism $v_{i_{Y_0}}$ of 4.8 (iii), where $i_{Y_0}: Y_0 \rightarrow X_0$ is the canonical immersion.⁽¹²⁴⁾

Lemma 4.27. *Under the conditions of 4.21 for $S, \mathcal{I}, \mathcal{J}$ and X , let Y be a subgroup of X , flat over S and lifting $Y_{\mathcal{J}}$. Let $i: Y \hookrightarrow X$ denote the canonical immersion.⁽¹²⁵⁾*

(i) *Let $i': Y \rightarrow X$ be a morphism of S -schemes lifting i_0 (so that i' is also an immersion), let $Y' = i'(Y)$ and let $d(i, i')$ be the element of $C^1(Y_0, L_{0X})$ such that $i' = d(i, i') \cdot i$ (cf. 1.2.4). Then the deviation $d(Y, Y') \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by the formula:*

$$d(Y, Y') = d^1(d(i, i')).$$

(ii) *Let $x \in C^0(Y_0, L_{0X})$ be a section of X over S inducing the unit section of $X_{\mathcal{J}}$ over $S_{\mathcal{J}}$. Then the deviation $d(Y, \text{Int}(x)Y) \in C^1(Y_0, N_0)$ (cf. 4.5.1) is given by the formula:*

$$-d(Y, \text{Int}(x)Y) = d^1 \partial x = \partial d^0 x.$$

Indeed, Y' is the image of Y by the composite morphism:⁽¹²⁶⁾

$$Y \xrightarrow{(d(i, i'), i)} L'_X \times_S X \rightarrow X,$$

which is denoted $d(i, i') \cdot i$ in 4.8 (iii); by *loc. cit.* and the equality $v_{i_0} = d^1$, one therefore has:

$$c(X, Y', d(i, i') \cdot i) - c(X, Y', i) = v_{i_0}(d(i, i')) = d^1(d(i, i')).$$

But $d(i, i') \cdot i = i'$ factors through Y' by definition, so the first term is zero; moreover, by 4.8 (ii), one has $c(X, Y', i) = d(Y', Y) = -d(Y, Y')$. Thus $d(Y, Y') = d^1(d(i, i'))$, which proves (i).

Now let x be as in (ii). By the formula

$$xyx^{-1} = xyx^{-1}y^{-1}y = (x - \text{Ad}(y)x)y = (-\partial x)(y) \cdot y,$$

one sees that Y' is the image of Y by the immersion $i' = (-\partial x) \cdot i_Y$. Thus, by (i), one obtains

$$-d(Y, \text{Int}(x)Y) = d^1 \partial x = \partial d^0 x.$$

Corollary 4.28. *In order that two subgroups Y and Y' of G , flat over S and lifting $Y_{\mathcal{J}}$, be conjugate by a section of X over S inducing the unit section of $X_{\mathcal{J}}$, it is necessary and sufficient that $d(Y, Y') \in \partial d^0 C^0(Y_0, L_{0X}) \subset \partial C^0(Y_0, N_0) = B^1(Y_0, N_0)$.*

Corollary 4.29. *If d^0 is surjective, Y and Y' as above are conjugate by a section of X over S inducing the unit section of $X_{\mathcal{J}}$ if and only if $d(Y, Y') \in B^1(Y_0, N_0)$.*

Corollary 4.30. *Let Y be as in 4.27; the set of conjugates of Y by sections of X over S inducing the unit section of $X_{\mathcal{J}}$ is isomorphic to:*

$$d^1 \partial C^0(Y_0, L_{0X}) = C^0(Y_0, L_{0X}) / \text{Ker } d^1 \partial.$$

Let us now note that $C^0(Y_0/L_{0X})/\text{Ker } d^1 \partial$ is computed solely by means of the left-hand square of the commutative diagram of 4.26. It follows in particular that one can also compute it in any diagram of the same type having the same left-hand square. Let us consider in particular the functor L_{0X}/L_{0Y} over S_0 defined by

$$\text{Hom}_{S_0}(T, L_{0X}/L_{0Y}) = \text{Hom}_{S_0}(T, L_{0X}) / \text{Hom}_{S_0}(T, L_{0Y}).$$

⁽¹²⁴⁾**N.D.E.** This follows from the definition of $d: \mathbf{n}_{Y_0/X_0} \rightarrow \Omega_{X_0/S_0}^1$ (cf. 4.25) and from that of $v_{i_{Y_0}}$ (cf. 4.8).

⁽¹²⁵⁾**N.D.E.** One has added point (i) below, which will be useful in 4.35.1 and then in 4.38 (4) and (5).

⁽¹²⁶⁾**N.D.E.** One recalls that $L'_X = \prod_{S_0/S} L_{0X}$.

One has a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & C^0(Y_0, L_{0Y}) & \longrightarrow & C^0(Y_0, L_{0X}) & \longrightarrow & C^0(Y_0, L_{0X}/L_{0Y}) \longrightarrow 0 \\ & & \downarrow \partial & & \downarrow \partial & & \downarrow \partial \\ 0 & \longrightarrow & C^1(Y_0, L_{0Y}) & \longrightarrow & C^1(Y_0, L_{0X}) & \longrightarrow & C^1(Y_0, L_{0X}/L_{0Y}) \longrightarrow 0, \end{array}$$

whence, by the preceding remark:

Corollary 4.31. *Let Y be as in 4.27; the set of conjugates of Y by sections of X over S inducing the unit section of $X_{\mathcal{J}}$ is isomorphic to*

$$E = \partial C^0(Y_0, L_{0X}/L_{0Y}) = C^0(Y_0, L_{0X}/L_{0Y})/H^0(Y_0, L_{0X}/L_{0Y}).$$

Corollary 4.32. *Suppose moreover that S_0 is affine and that ω_{Y_0/S_0}^1 ⁽¹²⁷⁾ is finite locally free. If one denotes $\mathcal{F}_0 = (\mathcal{L}\mathrm{ie}(X_0/S_0)/\mathcal{L}\mathrm{ie}(Y_0/S_0)) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}$, one has $E = \Gamma(S_0, \mathcal{F}_0)/H^0(Y_0, \mathcal{F}_0)$.*

⁽¹²⁸⁾ Indeed, since ω_{Y_0/S_0}^1 is finite locally free, as is ω_{X_0/S_0}^1 (since X is assumed smooth over S), one has, by 0.6:

$$L_{0Y} = \mathbf{W}(\mathcal{L}\mathrm{ie}(Y_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}) \quad \text{and} \quad L_{0X} = \mathbf{W}(\mathcal{L}\mathrm{ie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

On the other hand, by 4.25, one has an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules:

$$0 \longrightarrow \mathcal{K} \longrightarrow \omega_{X_0/S_0}^1 \xrightarrow{\phi} \omega_{Y_0/S_0}^1 \longrightarrow 0$$

(where $\mathcal{K} = \mathrm{Ker}(\phi)$). Since ω_{Y_0/S_0}^1 and ω_{X_0/S_0}^1 are finite locally free, one has a locally split exact sequence:

$$0 \longrightarrow \mathcal{L}\mathrm{ie}(Y_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J} \longrightarrow \mathcal{L}\mathrm{ie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J} \longrightarrow \mathcal{F}_0 \longrightarrow 0$$

It follows that one has an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules:

$$0 \longrightarrow L_{0Y} \longrightarrow L_{0X} \longrightarrow \mathbf{W}(\mathcal{F}_0).$$

By the argument which we used to prove 4.31, we may compute E as the image of the composite map

$$C^0(Y_0, L_{0X}) \xrightarrow{\pi} C^0(Y_0, \mathbf{W}(\mathcal{F}_0)) \xrightarrow{\partial} C^1(Y_0, \mathbf{W}(\mathcal{F}_0)).$$

Now the map π above is the map $\Gamma(S_0, \mathcal{L}\mathrm{ie}(X_0/S_0) \otimes_{\mathcal{O}_{S_0}} \mathcal{J}) \longrightarrow \Gamma(S_0, \mathcal{F}_0)$. Thus, S_0 being affine, π is surjective, and one indeed finds the result announced.

Corollary 4.33. *Let S , $\mathcal{I}$, $\mathcal{J}$ and X be as in 4.21, and let Y be a diagonalizable subgroup of X . Suppose that ω_{Y_0/S_0}^1 is finite locally free and that S_0 is affine.⁽¹²⁹⁾ The set of subgroups of X , conjugates of Y by a section of X over S inducing the unit section of $X_{\mathcal{J}}$, is isomorphic to*

$$E = \Gamma\left(S_0, \mathcal{L}\mathrm{ie}(X_0/S_0)/\mathcal{L}\mathrm{ie}(X_0/S_0)^{\mathrm{ad}(Y_0)}\right) \otimes_{\Gamma(S_0, \mathcal{O}_{S_0})} \Gamma(S_0, \mathcal{J})$$

⁽¹³⁰⁾ that is, to $L_{0X}(Y_0)/H^0(Y_0, L_{0X})$.

Indeed, one writes, by I 4.7.3 (cf. 2.13):

$$\mathcal{L}\mathrm{ie}(X_0/S_0) = \mathcal{L}\mathrm{ie}(X_0/S_0)^{\mathrm{ad}(Y_0)} \oplus \mathcal{R}.$$

⁽¹²⁷⁾**N.D.E.** One has corrected $\mathcal{L}\mathrm{ie}(Y_0/S_0)$ to ω_{Y_0/S_0}^1 .

⁽¹²⁸⁾**N.D.E.** One has spelled out the original in what follows.

⁽¹²⁹⁾**N.D.E.** One has added the hypothesis on ω_{Y_0/S_0}^1 and replaced the hypothesis “ S affine” by “ S_0 affine”.

⁽¹³⁰⁾**N.D.E.** One has added what follows, cf. 4.34.

Since Y_0 is commutative, one has $\mathcal{L}ie(Y_0/S_0) \subset \mathcal{L}ie(X_0/S_0)^{\text{ad}(Y_0)}$, hence

$$\begin{aligned} \mathcal{F}_0 &= (\mathcal{L}ie(X_0/S_0)^{\text{ad}(Y_0)} / \mathcal{L}ie(Y_0/S_0)) \otimes \mathcal{J} \oplus \mathcal{R} \otimes \mathcal{J}, \\ \mathcal{F}_0^{\text{ad}(Y_0)} &= (\mathcal{L}ie(X_0/S_0)^{\text{ad}(Y_0)} / \mathcal{L}ie(Y_0/S_0)) \otimes \mathcal{J}. \end{aligned}$$

By 4.32, one therefore has $E \simeq \Gamma(S_0, \mathcal{R} \otimes \mathcal{J})$. Returning to the definition of $\mathcal{R}$, one is done.

Corollary 4.34. *Let $S, \mathcal{I}, \mathcal{J}$ and X be as in 4.21, and let Y be a diagonalizable subgroup of X . Suppose that ω_{Y_0/S_0}^1 is finite locally free and that S_0 is affine.⁽¹³¹⁾ If $x \in X(S)$ induces the unit section of $X_{\mathcal{J}}$ and normalizes Y , then it centralizes Y .*

This follows immediately from the comparison of the preceding corollary and of 2.14. Indeed, 4.33 shows that the elements of $C^0(Y_0, L_{0X})$ which globally preserve Y are the elements of $H^0(Y_0, L_{0X})$, and one has seen in 2.14 that they are precisely those which act trivially on the canonical immersion $Y \rightarrow X$.

4.35. — Let us return to the general situation of 4.21 and assume $Y_{\mathcal{J}}$ smooth over $S_{\mathcal{J}}$. Then, by 4.25 (ii), one has an exact sequence of Y_0 - $\mathcal{O}_{S_0}$ -modules:

$$(*) \quad 0 \rightarrow \mathcal{L}ie(Y_0/S_0) \rightarrow \mathcal{L}ie(X_0/S_0) \rightarrow \mathbf{n}_{Y_0/X_0}^{\vee} \rightarrow 0$$

and these are finite locally free $\mathcal{O}_{S_0}$ -modules.

On the other hand, by SGA 1, II 4.10, every subscheme Y of X lifting $Y_{\mathcal{J}}$ and *flat* over S will be *smooth* over S .⁽¹³²⁾ Suppose moreover that S_0 and $Y_{\mathcal{J}}$ are *affine*. Then, since $\mathbf{n}_{Y_0/X_0}^{\vee}$ is a locally free $\mathcal{O}_{S_0}$ -module, the sequence $(*)$ remains exact when one applies $\otimes_{\mathcal{O}_{S_0}} \mathcal{J}$ to it, and then takes the inverse image on Y_0^n , and since the Y_0^n are affine, one therefore obtains an exact sequence of complexes of abelian groups:

$$0 \rightarrow C^*(Y_0, L_{0Y}) \rightarrow C^*(Y_0, L_{0X}) \xrightarrow{d^*} C^*(Y_0, N_0) \rightarrow 0$$

and in particular, a commutative diagram with exact rows

$$\begin{array}{ccccccc} 0 & \rightarrow & C^1(Y_0, L_{0Y}) & \rightarrow & C^1(Y_0, L_{0X}) & \xrightarrow{d^1} & C^1(Y_0, N_0) \rightarrow 0 \\ & & \downarrow \partial & & \downarrow \partial & & \downarrow \partial \\ 0 & \rightarrow & C^2(Y_0, L_{0Y}) & \rightarrow & C^2(Y_0, L_{0X}) & \xrightarrow{d^2} & C^2(Y_0, N_0) \rightarrow 0. \end{array}$$

Now let Y, Y' be two subgroups of X lifting $Y_{\mathcal{J}}$ and *flat*, hence *smooth*, over S . Since $Y_{\mathcal{J}}$ is affine, then, by 0.15, Y and Y' are isomorphic as schemes extending $Y_{\mathcal{J}}$, i.e. there exists an isomorphism of S -schemes

$$f: Y \xrightarrow{\sim} Y'$$

inducing the identity on $Y_{\mathcal{J}}$. On the one hand, by 1.2.4, f defines an element a of $C^1(Y_0, L_{0X})$ such that $f(y) = a(y_0)y$, for every $y \in Y(S')$, $S' \rightarrow S$, and by 4.27 (i), one has

$$d^1(a) = d(Y, Y').$$

Moreover, as Y, Y' are subgroups of X , the above element belongs to $Z^1(Y_0, N_0)$ (cf. 4.21). Then ∂a is an element of $Z^2(Y_0, L_{0Y})$ whose image ∂a in $H^2(Y_0, L_{0Y})$ depends only on the class $d(Y, Y') \in H^1(Y_0, N_0)$; this being the definition of the boundary map $\partial^1: H^1(Y_0, N_0) \rightarrow H^2(Y_0, L_{0Y})$, one therefore has:

$$\partial^1(d(Y, Y')) = \partial a.$$

On the other hand, let us transport by f the group structure of Y' and let Y_1 be the group obtained (which therefore has Y as underlying scheme), that is, the group law μ_1 of Y_1 is defined by: for every $S' \rightarrow S$ and $x, y \in Y(S')$,

$$\mu_1(x, y) = f^{-1}(f(x)f(y)).$$

⁽¹³¹⁾ **N.D.E.** Cf. N.D.E. (129).

⁽¹³²⁾ **N.D.E.** One has added what follows and Proposition 4.35.1, implicit in the original, cf. 4.38 (5).

By 3.5.1, Y_1 defines a cocycle $\delta(Y, Y_1) \in Z^2(Y_0, \mathcal{L}ie(Y_0/S_0))$ such that, for every $S' \rightarrow S$ and $x, y \in Y(S')$, one has

$$\delta(Y, Y_1)(x_0, y_0) xy = \mu_1(x, y) = f^{-1}(f(x)f(y)).$$

Let us set $b = \delta(Y, Y_1)$. For every $S' \rightarrow S$ and $x, y \in Y(S')$, one has $(b(x_0, y_0) xy)_0 = x_0 y_0$, and hence one obtains that $f(b(x_0, y_0) xy)$ equals, on the one hand, $a(x_0 y_0) b(x_0 y_0) xy$ and, on the other hand,

$$f(x)f(y) = a(x_0) x a(y_0) y = a(x_0) \text{Ad}(x_0)(a(y_0)) xy.$$

Comparing the two expressions, one obtains that $b(x_0, y_0)$ equals

$$\begin{aligned} a(x_0 y_0)^{-1} a(x_0) \text{Ad}(x_0)(a(y_0)) &= \text{Ad}(x_0)(a(y_0)) - a(x_0 y_0) + a(x_0) \\ &= (\partial a)(x_0, y_0), \end{aligned}$$

i.e. $\delta(Y, Y_1) = \partial a$. One has therefore obtained the

Proposition 4.35.1. ⁽¹³²⁾ *Under the hypotheses of 4.21, suppose in addition that S_0 is affine and that $Y_{\mathcal{J}}$ is smooth over $S_{\mathcal{J}}$ and affine. Let Y, Y' be two subgroups of X lifting $Y_{\mathcal{J}}$ and flat (hence smooth) over S , let $f: Y \xrightarrow{\sim} Y'$ be an isomorphism of S -schemes inducing the identity on $Y_{\mathcal{J}}$, and denote by Y_1 the group obtained by transporting by f the group structure of Y' . Then one has*

$$\partial^1 d(Y, Y') = \delta(Y, Y_1).$$

Proposition 4.36. *Under the hypotheses of 4.21, suppose in addition that $Y_{\mathcal{J}}$ is smooth over $S_{\mathcal{J}}$ and that S_0 is affine. The set of S -subgroups Y of X flat (or smooth) over S , reducing to $Y_{\mathcal{J}}$, modulo conjugation by sections of X over S inducing the unit section of $X_{\mathcal{J}}$, is either empty or a principal homogeneous set under the group*

$$H^1(Y_0, [\mathcal{L}ie(X_0/S_0)/\mathcal{L}ie(Y_0/S_0)] \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

It is enough for us to check that Corollary 4.29 applies, that is, that

$$d^0: \text{Hom}_{\mathcal{O}_{S_0}}(\omega_{X_0/S_0}^1, \mathcal{J}) \rightarrow \text{Hom}_{\mathcal{O}_{S_0}}(\mathfrak{n}_{Y_0/X_0}, \mathcal{J})$$

is surjective. Now this follows from the fact that the sequence (+) of 4.25 (ii) is split, S_0 being affine.⁽¹³³⁾

Let us finally state a corollary common to 4.21 and 4.36, which will, in fact, be the only form in which we shall use in the sequel the general results of this number.⁽¹³⁴⁾

Corollary 4.37. *Let S be a scheme and S_0 the closed subscheme defined by a nilpotent ideal $\mathcal{I}$. Let X be an S -group smooth over S , and Y_0 an S_0 -subgroup of X_0 , flat over S_0 .*

(i) *If S_0 is affine, Y_0 is smooth over S_0 , and if*

$$H^1(Y_0, [\mathcal{L}ie(X_0/S_0)/\mathcal{L}ie(Y_0/S_0)] \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$$

for every $n \geq 0$, two S -subgroups of X , flat (or smooth) over S , reducing to Y_0 , are conjugate by a section of X over S inducing the unit section of X_0 .

(ii) *If Y_0 is affine and of finite presentation and if⁽¹³⁵⁾*

$$H^2(Y_0, \mathfrak{n}_{Y_0/X_0}^{\vee} \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}) = 0$$

for every $n \geq 0$, there exists an S -subgroup of X , flat over S , reducing to Y_0 .

⁽¹³³⁾ **N.D.E.** This also follows from the proof of 4.32.

⁽¹³⁴⁾ **N.D.E.** For example, 4.37 is used in Exposé IX to prove the statements 3.2 bis and 3.6 bis.

⁽¹³⁵⁾ **N.D.E.** One has replaced $\text{Hom}_{\mathcal{O}_{S_0}}(\mathfrak{n}_{Y_0/X_0}, \mathcal{I}^{n+1}/\mathcal{I}^{n+2})$ by $\mathfrak{n}_{Y_0/X_0}^{\vee} \otimes_{\mathcal{O}_{S_0}} \mathcal{I}^{n+1}/\mathcal{I}^{n+2}$, in accordance with Remark 4.22.

4.38. — It remains for us to relate the three constructions that we have made in this exposé. In order to avoid inessential complications, we shall place ourselves in the following situation: S_0 is the spectrum of a field k , S is the spectrum of the dual numbers $D(k)$, G is an S -group smooth over S , and K an S -subgroup, *smooth* over S and *affine*.⁽¹³⁶⁾ Let us denote $\mathfrak{g}_0 = \mathcal{L}ie G_0$ (which here equals $\Gamma(S_0, \mathcal{L}ie G_0) = \text{Lie}(G_0/S_0)(S_0)$) and $\mathfrak{k}_0 = \mathcal{L}ie K_0$. One has an exact sequence of k -vector spaces⁽¹³⁷⁾:

$$0 \longrightarrow \mathfrak{k}_0 \xrightarrow{i} \mathfrak{g}_0 \xrightarrow{d} \mathfrak{n}_{K_0/G_0}^\vee \longrightarrow 0,$$

giving rise to an exact cohomology sequence:

$$\begin{aligned} 0 \longrightarrow H^0(K_0, \mathfrak{k}_0) \xrightarrow{i^0} H^0(K_0, \mathfrak{g}_0) \xrightarrow{d^0} H^0(K_0, \mathfrak{g}_0/\mathfrak{k}_0) \\ \xrightarrow{\partial^0} H^1(K_0, \mathfrak{k}_0) \xrightarrow{i^1} H^1(K_0, \mathfrak{g}_0) \xrightarrow{d^1} H^1(K_0, \mathfrak{n}_{K_0/G_0}^\vee) \xrightarrow{\partial^1} H^2(K_0, \mathfrak{k}_0). \end{aligned}$$

Now these various groups all have a geometric meaning.

- a) $H^0(K_0, \mathfrak{k}_0) = \mathcal{L}ie \text{ Centr}(K_0)$ ⁽¹³⁸⁾ by II 5.2.3.
- b) $H^0(K_0, \mathfrak{g}_0) = \mathcal{L}ie \text{ Centr}_{G_0}(K_0)$ ⁽¹³⁸⁾ (*idem*).
- c) $H^0(K_0, \mathfrak{g}_0/\mathfrak{k}_0) = \mathcal{L}ie \text{ Norm}_{G_0}(K_0)/\mathfrak{k}_0$ ⁽¹³⁸⁾ (*idem*).
- d) $H^1(K_0, \mathfrak{k}_0) = \mathcal{L}ie(\text{Aut}_{S_0\text{-gr.}}(K_0))/\text{Im}(\mathfrak{k}_0)$, where $\text{Im}(\mathfrak{k}_0)$ denotes the image of $\mathfrak{k}_0$ under the morphism $\mathcal{L}ie(\text{Int}_0)$ deduced from $\text{Int}_0: K_0 \longrightarrow \text{Aut}_{S_0\text{-gr.}}(K_0)$. Indeed, it follows from 2.1 (ii), applied to $Y = X = K$ and $f_0 = \text{id}_{K_0}$, that $Z^1(K_0, \mathfrak{k}_0)$ is the group of infinitesimal automorphisms of the S_0 -group K_0 , and that $H^1(K_0, \mathfrak{k}_0)$ is obtained by quotienting by the inner infinitesimal automorphisms, i.e. by the image of $\mathfrak{k}_0$. Moreover, by II 4.2.2, one also has $Z^1(K_0, \mathfrak{k}_0) = \mathcal{L}ie(\text{Aut}_{S_0\text{-gr.}}(K_0))$.⁽¹³⁹⁾
- e) $H^1(K_0, \mathfrak{g}_0)$ is, by 2.1 (ii), the group of deviations between homomorphisms $K \longrightarrow G$ extending the canonical immersion $i_0: K_0 \longrightarrow G_0$, modulo the deviations obtained by the action of the inner automorphisms of G defined by elements of $G(S)$ giving the unit of $G(S_0)$ (that is, elements of $\mathfrak{g}_0$).
- f) $H^1(K_0, \mathfrak{n}_{K_0/G_0}^\vee)$ is, by 4.36, the group of deviations between subgroups K' of G extending K_0 and flat over S (hence *smooth* over S , cf. SGA 1, II 4.10), modulo the deviations obtained by the action of the inner automorphisms of G constructed as previously.
- g) $H^2(K_0, \mathfrak{k}_0)$ is, by 3.5 (ii), the group of deviations between group structures on K extending that of K_0 , modulo the S -automorphisms of K inducing the identity on K_0 .

We now propose to show how one can make explicit the six morphisms of the preceding exact sequence in the geometric interpretation which we have just given.

- 1) i^0 and d^0 are none other than the morphisms obtained by passage to the Lie algebra (then by passage to the quotient, for d^0), starting from the canonical monomorphisms:

$$\text{Centr}(K_0) \longrightarrow \text{Centr}_{G_0}(K_0) \longrightarrow \text{Norm}_{G_0}(K_0).$$

This is indeed what follows immediately from the definition of the identifications (a), (b), and (c).

- 2) One constructs ∂^0 as follows. Let $x \in \mathcal{L}ie \text{ Norm}_{G_0}(K_0)/\mathfrak{k}_0$. Let us lift it to an $x \in \mathcal{L}ie \text{ Norm}_{G_0}(K_0) \subset \text{Norm}_G(K)(S)$. Then $\text{Int}(x)$ defines an automorphism of K inducing the

⁽¹³⁶⁾**N.D.E.** One has slightly modified the original in what follows. In particular, one has replaced X by G and Y by K , and one has denoted by $\mathfrak{g}_0$ and $\mathfrak{k}_0$ their Lie algebras. On the other hand, one has written explicitly $H^i(K_0, \)$ in place of the abbreviation $H^i(\)$ of the original.

⁽¹³⁷⁾**N.D.E.** Equipped with the adjoint action of K_0 .

⁽¹³⁸⁾**N.D.E.** As the formation of centralizers and normalizers commutes with base change (cf. I 2.3.3.1), one has written $\text{Centr}(K_0)$ in place of $\text{Centr}(K)_0$ in the original, and likewise $\text{Centr}_{G_0}(K_0)$ and $\text{Norm}_{G_0}(K_0)$ in place of $\text{Centr}_G(K)_0$ and $\text{Norm}_G(K)_0$.

⁽¹³⁹⁾**N.D.E.** And this is the Lie algebra $\text{Dér}_k(\mathfrak{k}_0)$ of the derivations of $\mathfrak{k}_0$, so $H^1(K_0, \mathfrak{k}_0)$ is the quotient of $\text{Dér}_k(\mathfrak{k}_0)$ by the inner derivations (i.e. by the image of $\text{ad}: \mathfrak{k}_0 \longrightarrow \text{Dér}_k(\mathfrak{k}_0)$).

identity on K_0 , hence an element of $\mathcal{L}\text{ie Aut}_{S_0\text{-gr.}}(K_0)$. Denote by $\text{Int}(x)$ the image of this element in $\mathcal{L}\text{ie Aut}_{S_0\text{-gr.}}(K_0)/\text{Im}(\mathfrak{k}_0)$. Then one has:

$$(*) \quad \partial^0(x) = -\text{Int}(x) = \text{Int}(x^{-1}).$$

Indeed, let us compute the element of $\mathcal{L}\text{ie Aut}_{S_0\text{-gr.}}(K_0)$ defined by $\text{Int}(x)$. By definition it will correspond to an element a of $Z^1(K_0, \mathfrak{k}_0)$ such that

$$xyx^{-1} = a(y_0)y, \quad \text{for every } y \in K(S'), \quad S' \longrightarrow S.$$

But this may also be written $a(y_0) = xyx^{-1}y^{-1} = x - \text{Ad}(y)x = -\partial(x)(y_0)$, whence $a = -\partial(x)$.

⁽¹⁴⁰⁾ On the other hand, the image of $x \in \mathcal{L}\text{ie Norm}_{G_0}(K_0) \subset \mathfrak{g}_0$ under ∂ is an element of $Z^1(K_0, \mathfrak{k}_0)$, whose image $\partial(x)$ in $H^1(K_0, \mathfrak{k}_0)$ depends only on x , and by definition of the boundary map ∂^0 , one has

$$\partial^0(x) = \partial(x);$$

combined with the equality $a = -\partial(x)$, this proves $(*)$.

3) ⁽¹⁴¹⁾ Denote by $i: K \longrightarrow G$ the canonical immersion. Let u be an element of $H^1(K_0, \mathfrak{k}_0)$, the image of an

$$u \in \mathcal{L}\text{ie Aut}_{S_0\text{-gr.}}(K_0) \subset \text{Aut}_{S\text{-gr.}}(K).$$

Then one has:

$$(**) \quad i^1(u) = d(i, i \circ u),$$

where $d(i, i \circ u)$ is the class defined in 2.1.1.

Indeed, u is the image of an element $v \in Z^1(K_0, \mathfrak{k}_0)$ such that $u(y) = v(y_0)y$, and $i^1(u)$ is the image in $H^1(K_0, \mathfrak{g}_0)$ of the cocycle $i \circ v \in Z^1(K_0, \mathfrak{g}_0)$.

Now, as i is a morphism of groups, the equality $u(y) = v(y_0)y$ implies $iu(y) = iv(y_0)i(y)$. It follows that $i \circ v = d(i, i \circ u)$, whence $(**)$.

4) Let $i': K \longrightarrow G$ be a morphism of groups lifting i_0 , let $d(i, i')$ be the class defined in 2.1.1, and let $d(K, i'(K)) \in C^1(K_0, \mathfrak{n}_{K_0/X_0})$ be the deviation defined in 4.5.1; by 4.21, $d(K, i'(K))$ belongs to $Z^1(K_0, \mathfrak{n}_{K_0/X_0})$. Denote by $d(K, i'(K))$ its image in $H^1(K_0, \mathfrak{n}_{K_0/X_0})$. Then, by 4.27 (i), one has:

$$(\dagger) \quad d^1(d(i, i')) = d(K, i'(K)).$$

5) Finally, let K' be a subgroup of G lifting K_0 and flat, hence *smooth*, over S . One has supposed that K_0 is *affine*. Then one knows that K and K' are isomorphic as schemes extending K_0 (cf. 0.15), hence that there exists an isomorphism of S -schemes

$$f: K \xrightarrow{\sim} K'$$

inducing the identity on K_0 . Let us transport by f the group structure of K' and let K_1 be the group obtained (which therefore has K as underlying scheme), that is, the group law μ_1 of K_1 is defined by: for every $S' \longrightarrow S$ and $x, y \in K(S')$,

$$\mu_1(x, y) = f^{-1}(f(x)f(y)).$$

⁽¹⁴²⁾ By 3.5.1, K_1 defines a cocycle $\delta(K, K_1) \in Z^2(K_0, \mathfrak{k}_0)$ such that, for every $S' \longrightarrow S$ and $x, y \in K(S')$, one has

$$\delta(K, K_1)(x_0, y_0)xy = \mu_1(x, y) = f^{-1}(f(x)f(y)).$$

Then, by 4.35.1, one has:

$$(\ddagger) \quad \partial^1 d(K, K') = \delta(K, K_1).$$

⁽¹⁴⁰⁾**N.D.E.** One has added the following sentence.

⁽¹⁴¹⁾**N.D.E.** One has set out the original in more detail in what follows, and in $(**)$ one has corrected $u \circ i$ to $i \circ u$.

⁽¹⁴²⁾**N.D.E.** One has modified the original in what follows, taking into account the additions made in 3.5.1 and 4.35.1.

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⁽¹⁴³⁾**N.D.E.** : additional references cited in this Exposé.